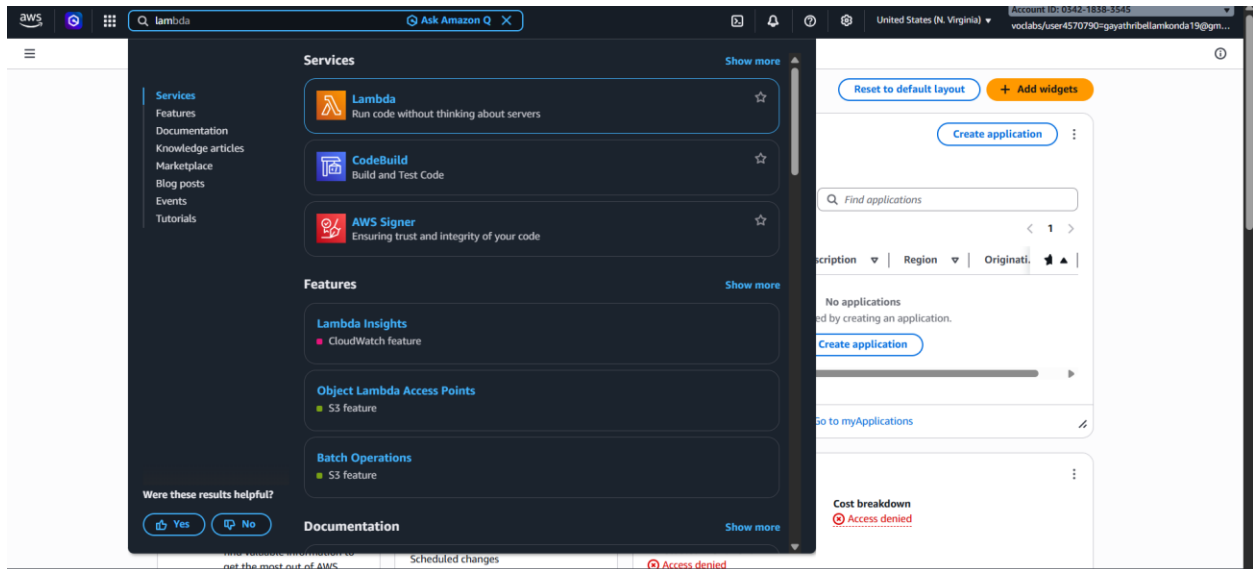


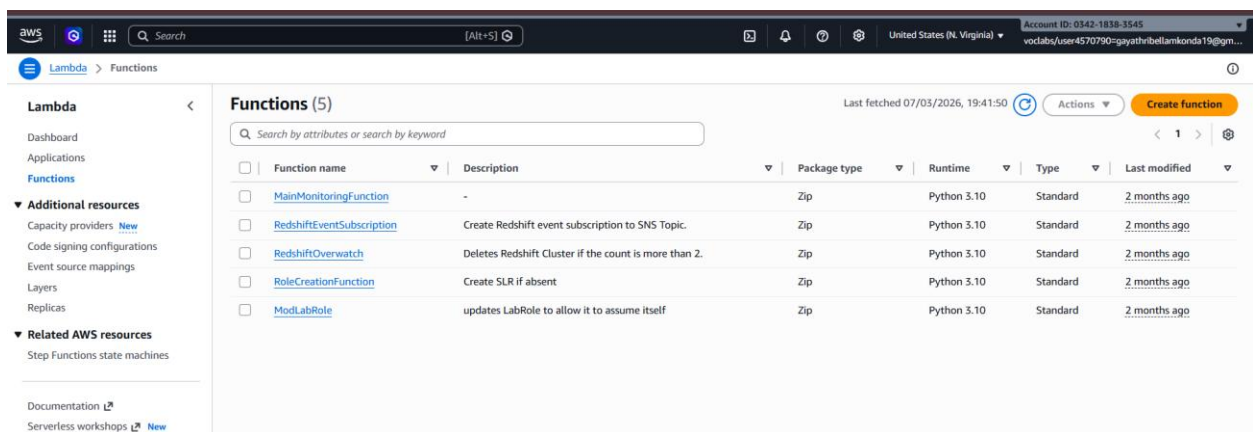
WEEK – 7

AWS Lambda function to trigger whenever an object is uploaded to an S3 bucket and update a DynamoDB table

Open AWS Lambda Console: Navigate to the Lambda service by either searching for "Lambda" in the AWS Management Console

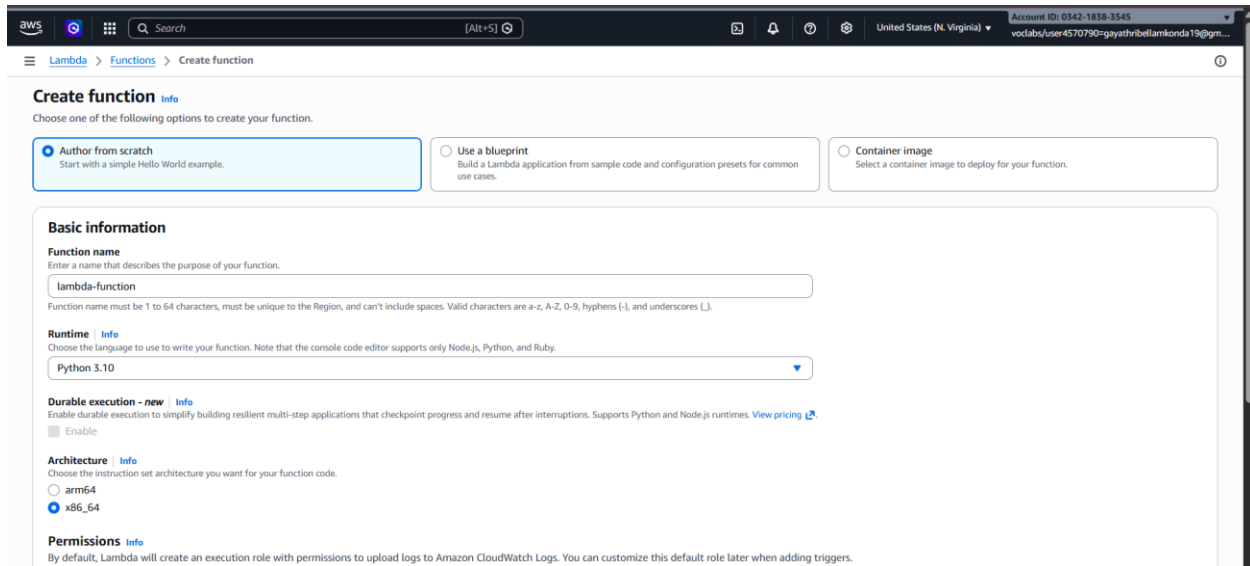


Click on the "Create function" button



- Choose "Author from scratch".

- Enter a name for your function -> lambda-function
- Choose Python 3.10 as the runtime



Create function [Info](#)

Choose one of the following options to create your function.

- ☒ **Author from scratch**
Start with a simple Hello World example.
- ☐ **Use a blueprint**
Build a Lambda application from sample code and configuration presets for common use cases.
- ☐ **Container image**
Select a container image to deploy for your function.

Basic information

Function name
Enter a name that describes the purpose of your function.
lambda-function
Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime [Info](#)
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.
Python 3.10

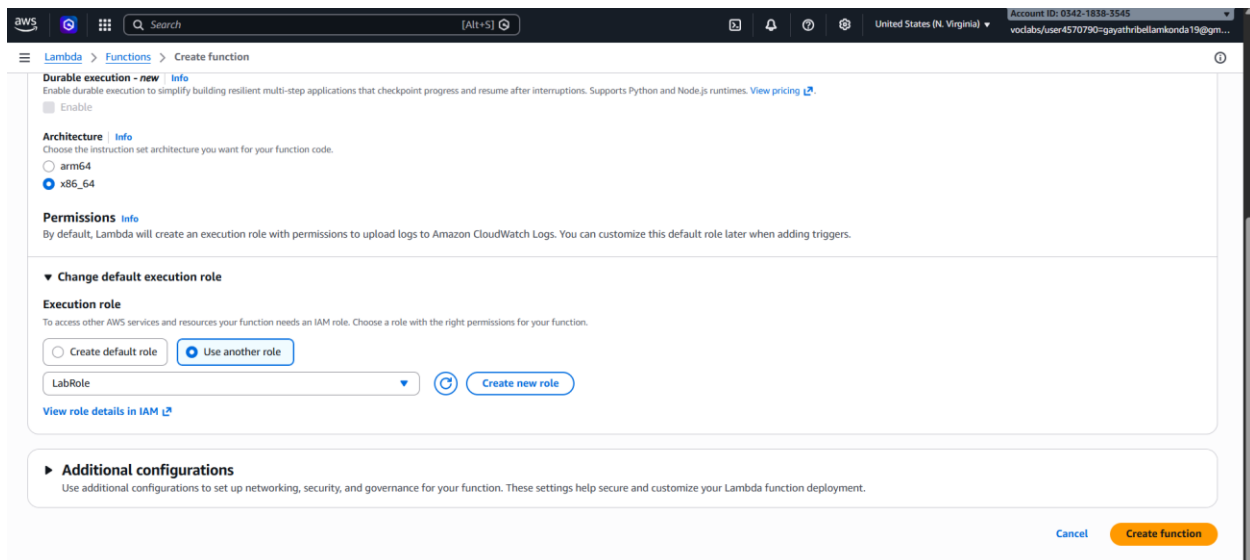
Durable execution - new [Info](#)
Enable durable execution to simplify building resilient multi-step applications that checkpoint progress and resume after interruptions. Supports Python and Node.js runtimes. [View pricing](#).
☐ Enable

Architecture [Info](#)
Choose the instruction set architecture you want for your function code.
☐ arm64
☒ x86_64

Permissions [Info](#)
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

- Under "Permissions" -> choose use another role that has permissions to access S3 and DynamoDB.

Select LabRole -> Click on create function button



Durable execution - new [Info](#)
Enable durable execution to simplify building resilient multi-step applications that checkpoint progress and resume after interruptions. Supports Python and Node.js runtimes. [View pricing](#).
☐ Enable

Architecture [Info](#)
Choose the instruction set architecture you want for your function code.
☐ arm64
☒ x86_64

Permissions [Info](#)
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

▼ Change default execution role

Execution role
To access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

☐ Create default role ☒ Use another role

LabRole [Refresh](#) [Create new role](#)

[View role details in IAM](#)

► Additional configurations
Use additional configurations to set up networking, security, and governance for your function. These settings help secure and customize your Lambda function deployment.

[Cancel](#) [Create function](#)

Test Your Function:

Select create new event -> select template as S3 put

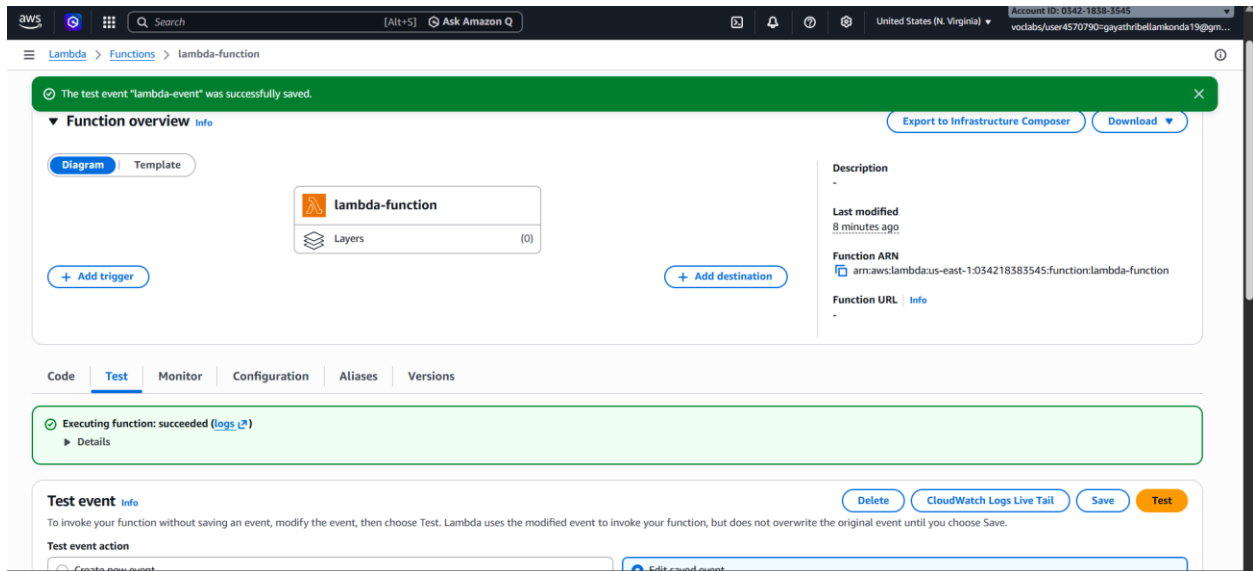
Change the bucket in the code

The screenshot shows the AWS Lambda console interface for testing a function. The 'Test event' tab is active. Under 'Test event action', the 'Create new event' button is selected. The 'Invocation type' is set to 'Synchronous'. The 'Event name' is 'lambda-event'. Under 'Event sharing settings', 'Private' is selected. The 'Template - optional' dropdown is set to 'S3 Put'. Buttons for 'CloudWatch Logs Live Tail', 'Save', and 'Test' are visible at the top right.

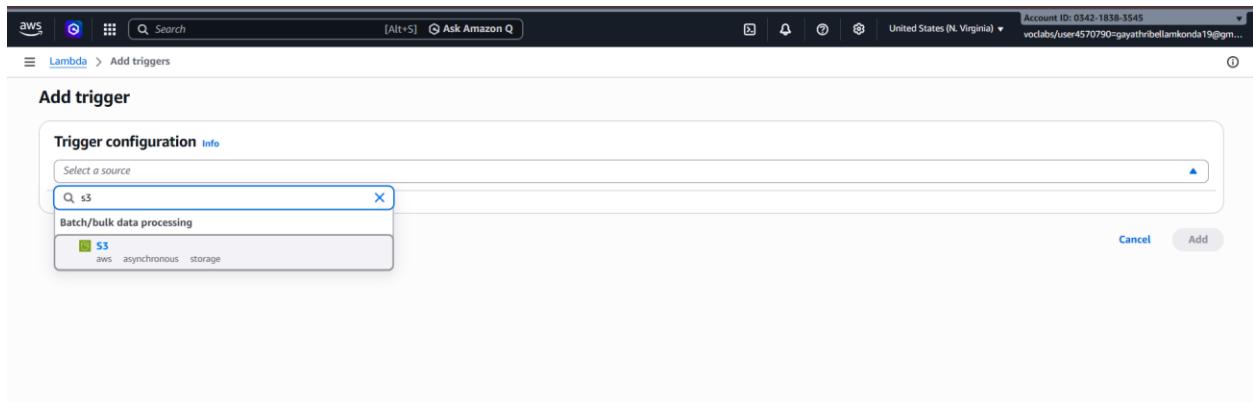
This screenshot shows the 'Event JSON' section of the AWS Lambda console. The 'S3 Put' template is selected. The JSON code is displayed with line numbers 9 through 38. The 'bucket' field in the 's3' object is highlighted, showing the value 'lambda-cc-lab-bucket'. A 'Format JSON' button is located at the top right of the code editor. The bottom right corner of the console shows the time '23:40' and the text 'JSON'.

```
9  {
10   "userIdentity": {
11     "principalId": "EXAMPLE"
12   },
13   "requestParameters": {
14     "sourceIPAddress": "127.0.0.1"
15   },
16   "responseElements": {
17     "x-amz-request-id": "EXAMPLE123456789",
18     "x-amz-id-2": "EXAMPLE123/5678abcdefghijklklamdasawesone/nnopqrstuvwxyzABCDEFGHI"
19   },
20   "s3": {
21     "s3SchemaVersion": "1.0",
22     "configurationId": "testConfigRule",
23     "bucket": {
24       "name": "lambda-cc-lab-bucket",
25       "ownerIdentity": {
26         "principalId": "EXAMPLE"
27       },
28       "arn": "arn:aws:s3:::example-bucket"
29     },
30     "object": {
31       "key": "test%2Fkey",
32       "size": 1024,
33       "eTag": "0123456789abcdef0123456789abcdef",
34       "sequencer": "0A1B2C3D4E5F678901"
35     }
36   }
37 }
38 }
```

Click on save -> click on test – it will execute the code successfully



Click on Add trigger -> Select source as S3



Bucket should be in same region

Select the bucket name -> select I acknowledge -> click on save button

Bucket
Choose or enter the ARN of an S3 bucket that serves as the event source. The bucket must be in the same region as the function.
Q s3/lambda-cc-lab-bucket X C
Bucket region: us-east-1

Event types
Select the events that you want to have trigger the Lambda function. You can optionally set up a prefix or suffix for an event. However, for each bucket, individual events cannot have multiple configurations with overlapping prefixes or suffixes that could match the same object key.
All object create events X

Prefix - optional
Enter a single optional prefix to limit the notifications to objects with keys that start with matching characters. Any special characters must be URL encoded.
e.g. images/

Suffix - optional
Enter a single optional suffix to limit the notifications to objects with keys that end with matching characters. Any special characters must be URL encoded.
e.g. .jpg

Recursive invocation
If your function writes objects to an S3 bucket, ensure that you are using different S3 buckets for input and output. Writing to the same bucket increases the risk of creating a recursive invocation, which can result in increased Lambda usage and increased costs. [Learn more](#)
☒ I acknowledge that using the same S3 bucket for both input and output is not recommended and that this configuration can cause recursive invocations, increased Lambda usage, and increased costs.

Lambda will add the necessary permissions for AWS S3 to invoke your Lambda function from this trigger.
[Info](#)

Cancel Add

Upload any document in the bucket -> click on upload

lambda-cc-lab-bucket Info

Objects Metadata Properties Permissions Metrics Management Access Points

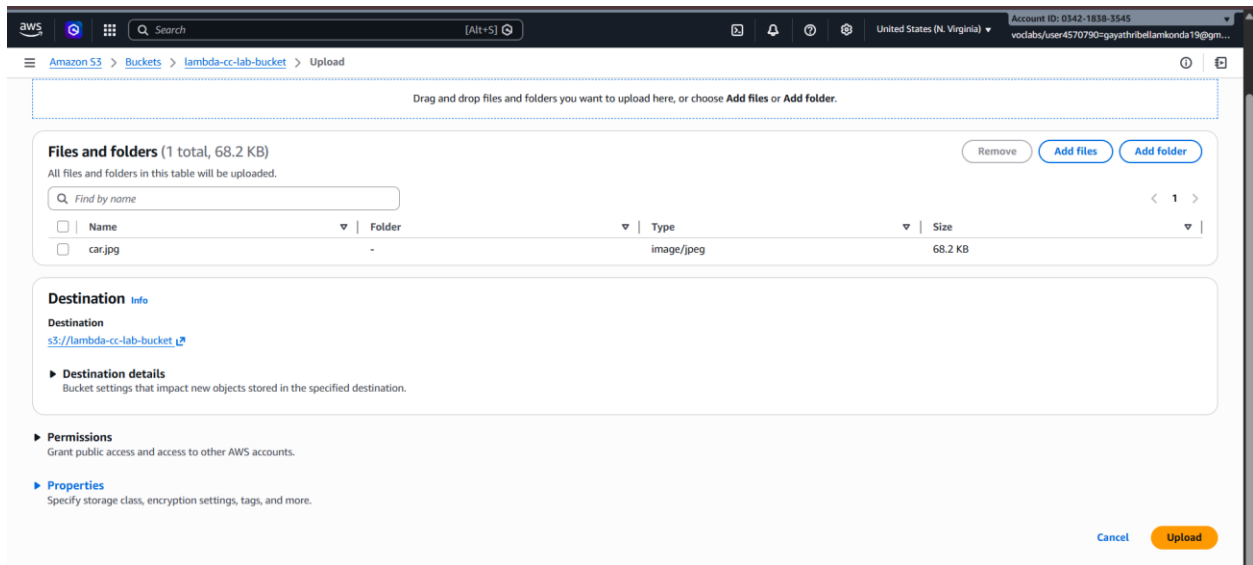
Objects (0) C Copy S3 URI Copy URL Download Open Delete Actions Create folder Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

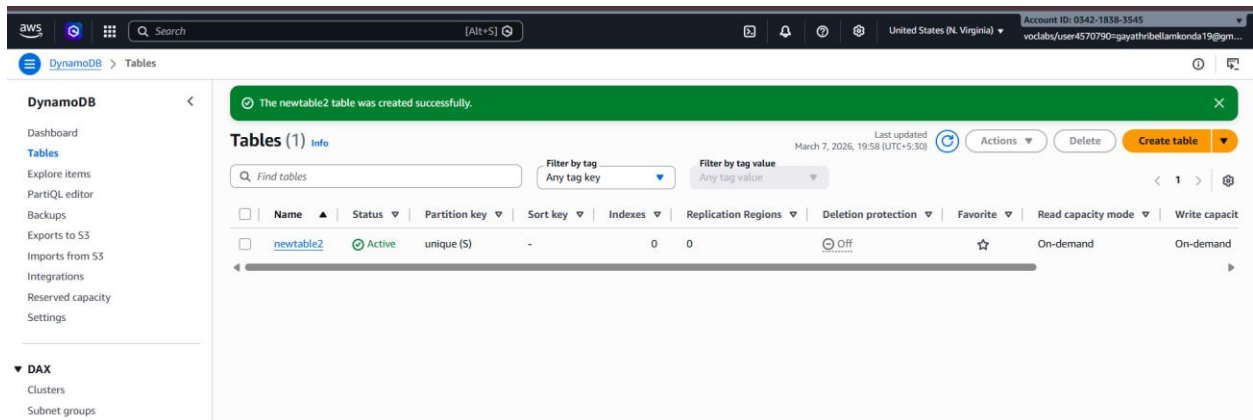
Find objects by prefix

Name	Type	Last modified	Size	Storage class
No objects You don't have any objects in this bucket.				

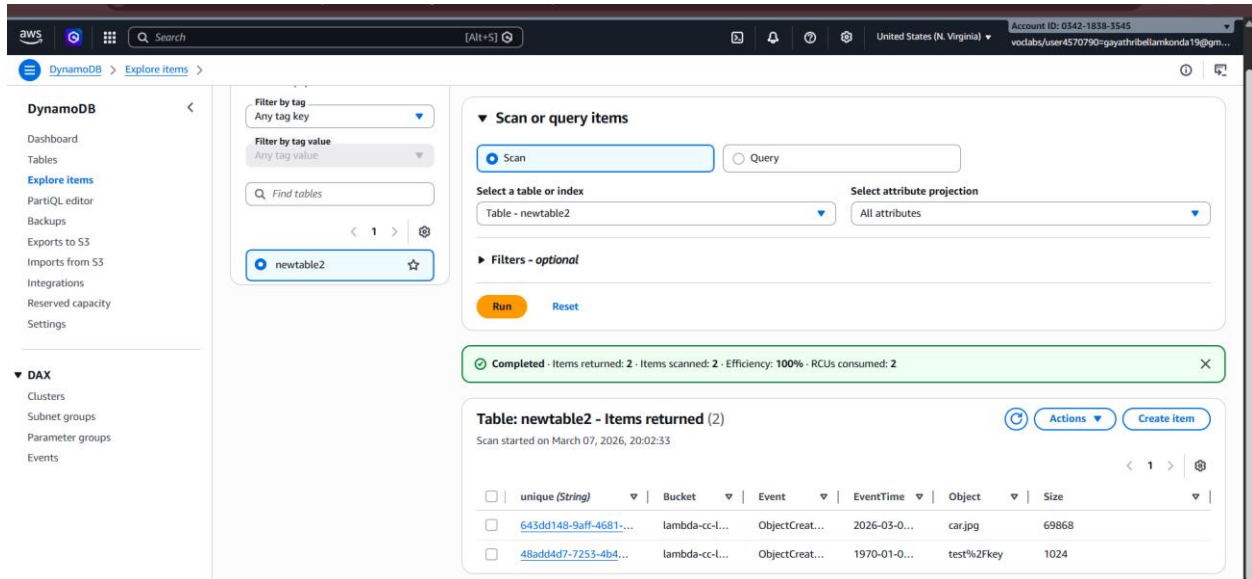
Upload



Go to DynamoDb dashboard -> Click on Explore items



Select the table -> navigate to items returned -> uploaded objects will be visible



Scan or query items

Filter by tag: Any tag key
Filter by tag value: Any tag value
Find tables:

newtable2

Scan or Query: ☒ Scan ☐ Query

Select a table or index: Table - newtable2
Select attribute projection: All attributes

Filters - optional

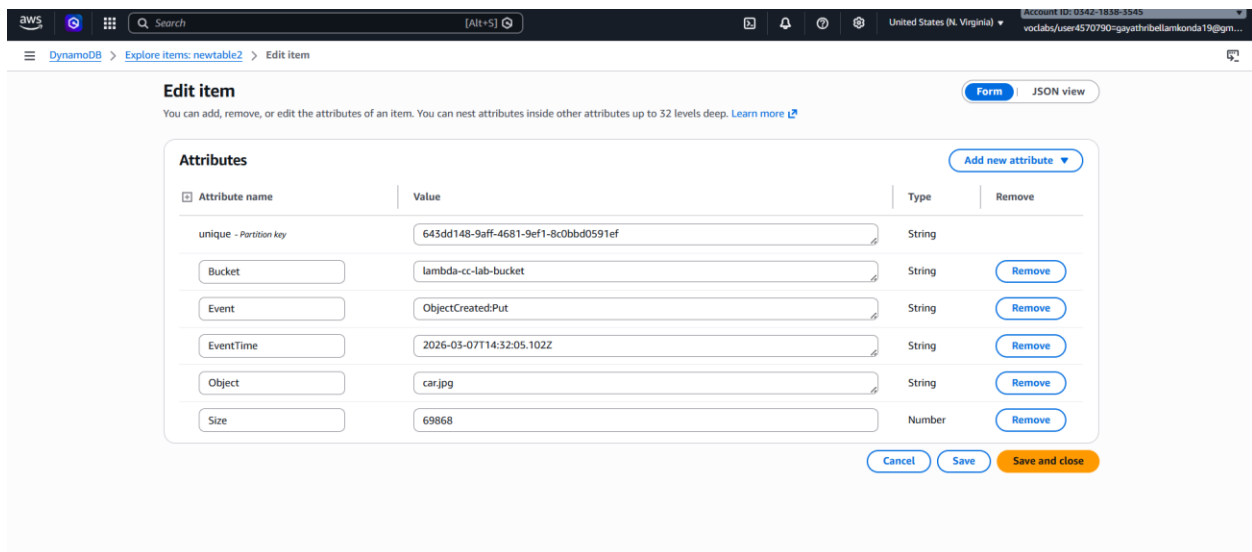
Run Reset

Completed - Items returned: 2 - Items scanned: 2 - Efficiency: 100% - RCUs consumed: 2

Table: newtable2 - Items returned (2)
Scan started on March 07, 2026, 20:02:33

	unique (String)	Bucket	Event	EventTime	Object	Size
☐	643dd148-9aff-4681-...	lambda-cc-l...	ObjectCreat...	2026-03-0...	car.jpg	69868
☐	48add4d7-7253-4b4...	lambda-cc-l...	ObjectCreat...	1970-01-0...	test%2Fkey	1024

The meta data of the object will be stored in the table



Edit item

You can add, remove, or edit the attributes of an item. You can nest attributes inside other attributes up to 32 levels deep. [Learn more](#)

Form | JSON view

Attributes

Attribute name	Value	Type	Remove
unique - Partition key	643dd148-9aff-4681-9ef1-8c0bbd0591ef	String	
Bucket	lambda-cc-lab-bucket	String	Remove
Event	ObjectCreated:Put	String	Remove
EventTime	2026-03-07T14:52:05.102Z	String	Remove
Object	car.jpg	String	Remove
Size	69868	Number	Remove

[Add new attribute](#)

[Cancel](#) [Save](#) [Save and close](#)

In lambda dashboard -> Go to Monitor Tab -> click on view cloudWatch logs

The screenshot shows the AWS Lambda console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and account information. Below the navigation bar, the breadcrumb trail shows 'Lambda > Functions > lambda-function'. The main content area is titled 'lambda-function' and includes tabs for 'Function overview', 'Code', 'Test', 'Monitor' (which is active), 'Configuration', 'Aliases', and 'Versions'. The 'Function overview' tab shows a diagram of the function with layers and a trigger. The 'Monitor' tab displays 'CloudWatch metrics' with a filter set to 'Function' and a 'View CloudWatch logs' button. The 'Monitor' tab also shows 'Alarm recommendations' and a 'Log group' section.

Select the log stream logs will be appeared

The screenshot shows the AWS CloudWatch console interface. The breadcrumb trail is 'CloudWatch > Log management > /aws/lambda/lambda-function'. The main content area is titled 'Log streams' and includes tabs for 'Log streams', 'Tags', 'Data protection', 'Anomaly detection', 'Metric filters', 'Subscription filters', 'Contributor Insights', 'Field indexes', and 'Transformer'. The 'Log streams' tab shows a list of log streams with columns for 'Log stream' and 'Last event time'. A single log stream is visible with the name '2026/03/07/[LATEST]6e117875705343679d1caea7db22e638' and a last event time of '2026-03-07 14:26:01 (UTC)'. The 'Log streams' tab also includes a search bar and a 'Filter log streams or try prefix search' option.

CloudWatch

Log management

aws/lambda/lambda-function

2026/03/07/[\$LATEST]6e117875705343679d1caea7db22e638

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

Clear

1m

30m

1h

12h

Custom

UTC timezone

Display

Timestamp	Message
No older events at this moment. Retry	
2026-03-07T14:26:00.782Z	INIT_START Runtime Version: python:3.10.mainlinev2.v3 Runtime Version ARN: arn:aws:lambda:us-east-1:runtime:a03ff4f729aad44f64166b2291afbcad56a30ba3ab5e1753...
2026-03-07T14:26:01.126Z	START RequestId: 6ab18d68-43f2-4dd2-8b08-f4743f9e5754 Version: \$LATEST
2026-03-07T14:26:03.756Z	[ERROR] ResourceNotFoundException: An error occurred (ResourceNotFoundException) when calling the PutItem operation: Requested resource not found Traceback (no...
2026-03-07T14:26:03.775Z	END RequestId: 6ab18d68-43f2-4dd2-8b08-f4743f9e5754
2026-03-07T14:26:03.775Z	REPORT RequestId: 6ab18d68-43f2-4dd2-8b08-f4743f9e5754 Duration: 2647.76 ms Billed Duration: 2989 ms Memory Size: 128 MB Max Memory Used: 83 MB Init Duration: ...
2026-03-07T14:28:54.895Z	START RequestId: 30ce0207-0d8b-40ed-a262-941810d14a57 Version: \$LATEST
2026-03-07T14:28:55.275Z	END RequestId: 30ce0207-0d8b-40ed-a262-941810d14a57
2026-03-07T14:28:55.275Z	REPORT RequestId: 30ce0207-0d8b-40ed-a262-941810d14a57 Duration: 378.97 ms Billed Duration: 379 ms Memory Size: 128 MB Max Memory Used: 84 MB
2026-03-07T14:32:06.332Z	START RequestId: 01372c4c-b958-4ed8-a922-e11b2b3642aa Version: \$LATEST
2026-03-07T14:32:07.042Z	END RequestId: 01372c4c-b958-4ed8-a922-e11b2b3642aa
2026-03-07T14:32:07.042Z	REPORT RequestId: 01372c4c-b958-4ed8-a922-e11b2b3642aa Duration: 709.30 ms Billed Duration: 710 ms Memory Size: 128 MB Max Memory Used: 84 MB
No newer events at this moment. Auto retry paused . Resume	

CloudWatch

Log management

aws/lambda/lambda-function

2026/03/07/[\$LATEST]6e117875705343679d1caea7db22e638

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

Clear

1m

30m

1h

12h

Custom

UTC timezone

Display

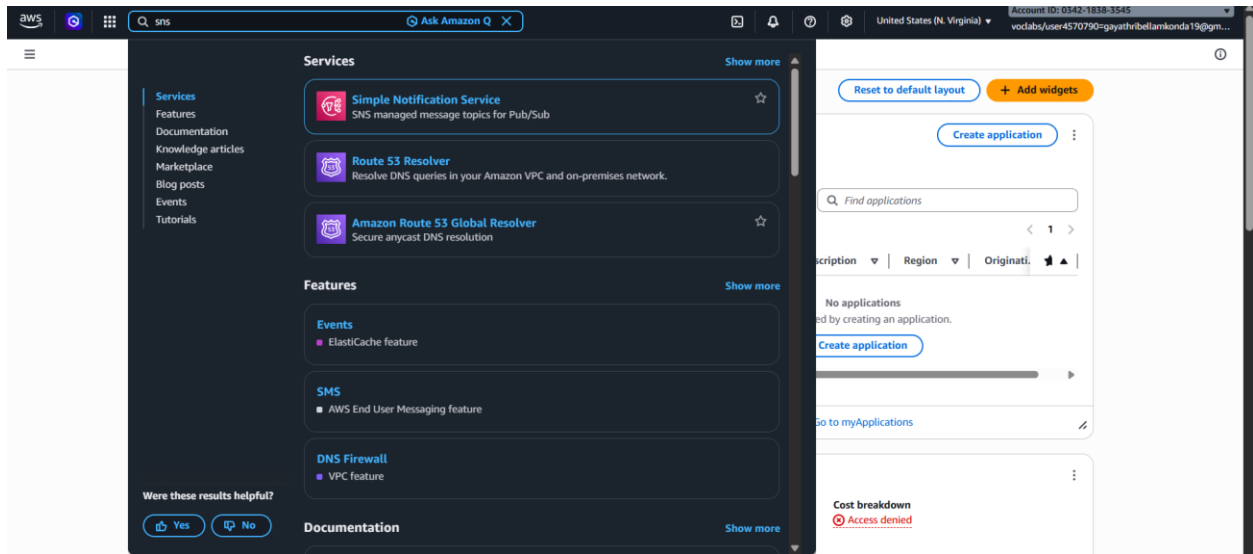
Timestamp	Message
No older events at this moment. Retry	
2026-03-07T14:26:00.782Z	INIT_START Runtime Version: python:3.10.mainlinev2.v3 Runtime Version ARN: arn:aws:lambda:us-east-1:runtime:a03ff4f729aad44f64166b2291afbcad56a30ba3ab5e1753...
2026-03-07T14:26:01.126Z	START RequestId: 6ab18d68-43f2-4dd2-8b08-f4743f9e5754 Version: \$LATEST
2026-03-07T14:26:03.756Z	[ERROR] ResourceNotFoundException: An error occurred (ResourceNotFoundException) when calling the PutItem operation: Requested resource not found Traceback (no...
2026-03-07T14:26:03.775Z	END RequestId: 6ab18d68-43f2-4dd2-8b08-f4743f9e5754
2026-03-07T14:26:03.775Z	REPORT RequestId: 6ab18d68-43f2-4dd2-8b08-f4743f9e5754 Duration: 2647.76 ms Billed Duration: 2989 ms Memory Size: 128 MB Max Memory Used: 83 MB Init Duration: ...
2026-03-07T14:28:54.895Z	START RequestId: 30ce0207-0d8b-40ed-a262-941810d14a57 Version: \$LATEST
2026-03-07T14:28:55.275Z	END RequestId: 30ce0207-0d8b-40ed-a262-941810d14a57
2026-03-07T14:28:55.275Z	REPORT RequestId: 30ce0207-0d8b-40ed-a262-941810d14a57 Duration: 378.97 ms Billed Duration: 379 ms Memory Size: 128 MB Max Memory Used: 84 MB
2026-03-07T14:32:06.332Z	START RequestId: 01372c4c-b958-4ed8-a922-e11b2b3642aa Version: \$LATEST
2026-03-07T14:32:07.042Z	END RequestId: 01372c4c-b958-4ed8-a922-e11b2b3642aa
2026-03-07T14:32:07.042Z	REPORT RequestId: 01372c4c-b958-4ed8-a922-e11b2b3642aa Duration: 709.30 ms Billed Duration: 710 ms Memory Size: 128 MB Max Memory Used: 84 MB
No newer events at this moment. Auto retry paused . Resume	

WEEK-8 -- SNS,SQS & S3-SNS-SQS-LAMBDA SERVICES

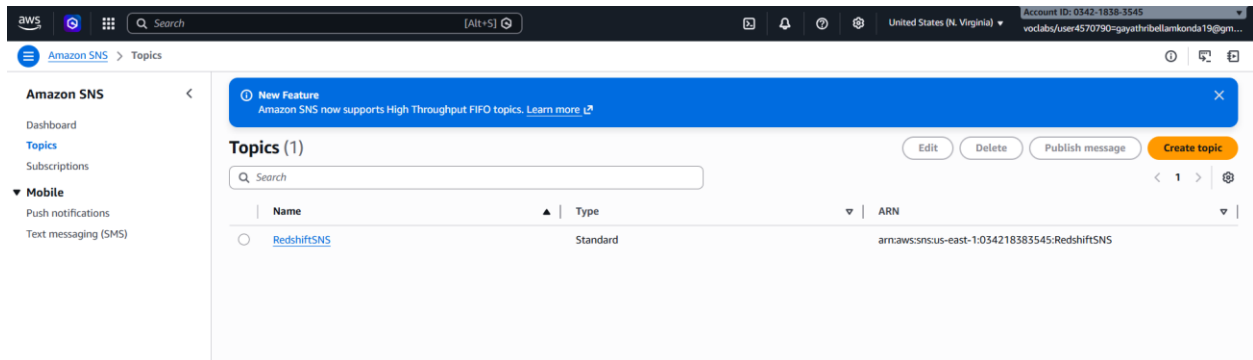
1) Simple Notification Service (SNS)

A) Create SNS Topic and Send Message to Email

Login to AWS Console -> In search bar → Type SNS -> Open Amazon SNS (Simple Notification Service)



In left panel → Click **Topics** -> Click **Create topic**



Type - Select Standard -> Name - MyEmailTopic -> Leave other settings default

Click Create topic

The screenshot shows the 'Create topic' page in the AWS Management Console. The breadcrumb trail is 'Amazon SNS > Topics > Create topic'. The page title is 'Create topic'. Under the 'Details' section, the 'Type' is set to 'Standard' (selected with a blue dot). The 'Name' field contains 'sns-email-topic'. The 'Display name - optional' field contains 'My Topic'. There are expandable sections for 'Encryption - optional' and 'Access policy - optional'. The account ID '034218383545' is visible in the top right corner.

Open created topic -> Go to Subscriptions tab -> Click Create subscription

The screenshot shows the 'sns-email-topic' page in the AWS Management Console, specifically the 'Subscriptions' tab. The breadcrumb trail is 'Amazon SNS > Topics > sns-email-topic'. The left sidebar shows the navigation menu with 'Topics' selected. The 'Details' section shows the topic name 'sns-email-topic', ARN 'arn:aws:sns:us-east-1:034218383545:sns-email-topic', and type 'Standard'. The 'Subscriptions' section shows a table with 0 subscriptions. A 'Create subscription' button is visible in the top right of the subscriptions section.

Protocol - Select Email -> Endpoint - Enter your email address -> Click Create subscription

Amazon SNS > Subscriptions > Create subscription

Details

Topic ARN
arn:aws:sns:us-east-1:034218383545:sns-email-topic

Protocol
The type of endpoint to subscribe
Email

Endpoint
An email address that can receive notifications from Amazon SNS.
gayathribellamkonda19@gmail.com

After your subscription is created, you must confirm it. [Info](#)

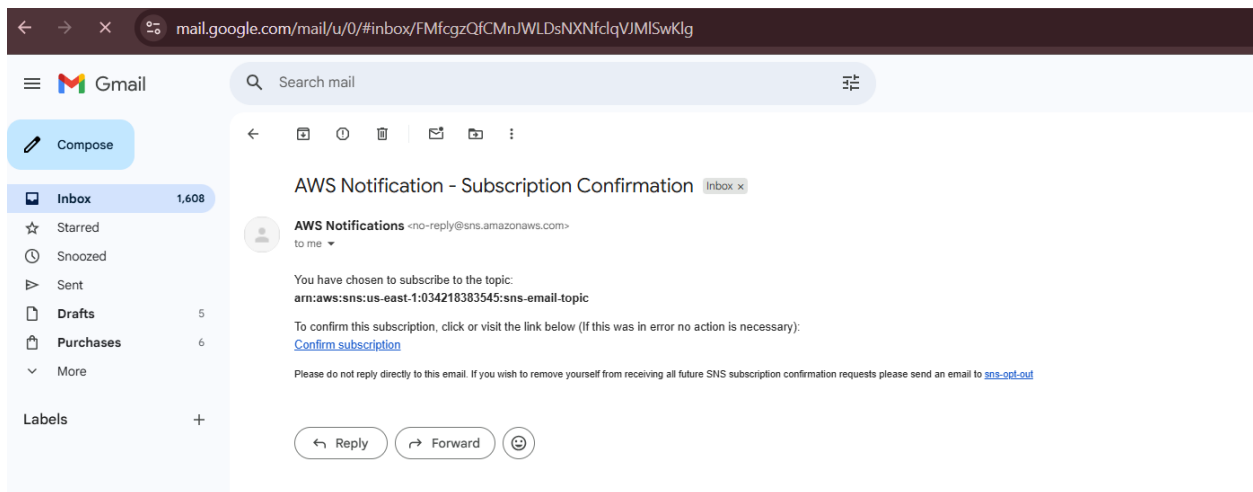
► **Subscription filter policy - optional** [Info](#)
This policy filters the messages that a subscriber receives.

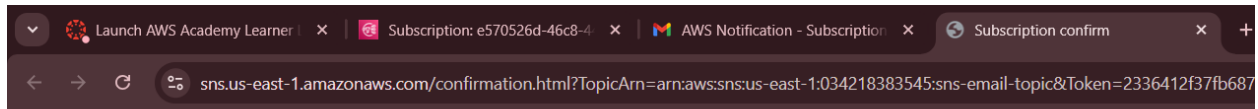
► **Redrive policy (dead-letter queue) - optional** [Info](#)
Send undeliverable messages to a dead-letter queue.

[Cancel](#) [Create subscription](#)

Go to your email inbox -> You will receive a mail from **Amazon SNS**

Click Confirm subscription





Simple Notification Service

Subscription confirmed!

You have successfully subscribed.

Your subscription's id is:

arn:aws:sns:us-east-1:034218383545:sns-email-topic:e570526d-46c8-445d-921e-909ec5579ff2

If it was not your intention to subscribe, [click here to unsubscribe](#).



Amazon S3 > Buckets > sns-bucket-cclab > Create event notification

Create event notification [Info](#)

To enable notifications, you must first add a notification configuration that identifies the events you want Amazon S3 to publish and the destinations where you want Amazon S3 to send the notifications.

General configuration

Event name

upload-notification

Event name can contain up to 255 characters.

Prefix - optional

Limit the notifications to objects with key starting with specified characters.

images/

Suffix - optional

Limit the notifications to objects with key ending with specified characters.

.jpg

Event types

Specify at least one event for which you want to receive notifications. For each group, you can choose an event type for all events, or you can choose one or more individual events.

Object creation

☒ All object create events
s3:ObjectCreated:*

☐ Put
s3:ObjectCreated:Put

☐ Delete

- Destination:
 - Select **SNS topic**
 - Choose your topic (S3UploadNotification)

Click **Save changes**

aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) Account ID: 0342-1838-3545 vodlabs/user4570790@gayathribellamkonda19@gmail...

Amazon S3 > Buckets > sns-bucket-cclab > Create event notification

Destination

Before Amazon S3 can publish messages to a destination, you must grant the Amazon S3 principal the necessary permissions to call the relevant API to publish messages to an SNS topic, an SQS queue, or a Lambda function. [Learn more](#)

Destination

Choose a destination to publish the event. [Learn more](#)

☐ Lambda function
Run a Lambda function script based on S3 events.

☒ SNS topic
Fanout messages to systems for parallel processing or directly to people.

☐ SQS queue
Send notifications to an SQS queue to be read by a server.

Specify SNS topic

☒ Choose from your SNS topics

☐ Enter SNS topic ARN

SNS topic

sns-email-topic

Cancel Save changes

Upload any file into the S3 bucket

aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) Account ID: 0342-1838-3545 vodlabs/user4570790@gayathribellamkonda19@gmail...

Amazon S3 > Buckets > sns-bucket-cclab > Upload

Drag and drop files and folders you want to upload here, or choose [Add files](#) or [Add folder](#).

Files and folders (1 total, 68.2 KB)

All files and folders in this table will be uploaded.

☐	Name	Folder	Type	Size
☐	car.jpg	-	image/jpeg	68.2 KB

Destination [info](#)

Destination

[s3://sns-bucket-cclab](#)

► Destination details

Bucket settings that impact new objects stored in the specified destination.

► Permissions

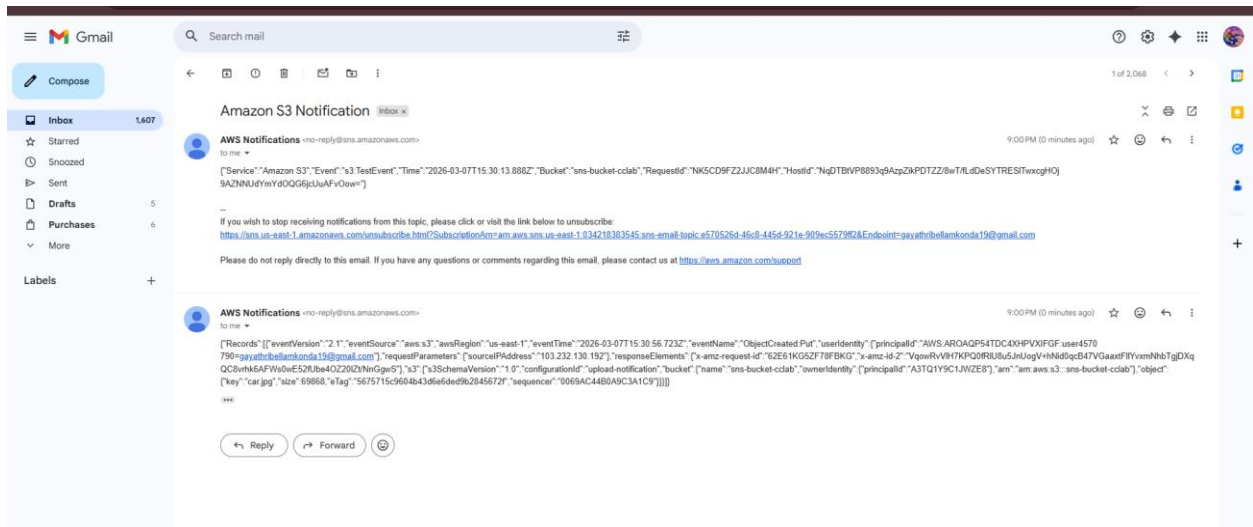
Grant public access and access to other AWS accounts.

► Properties

Specify storage class, encryption settings, tags, and more.

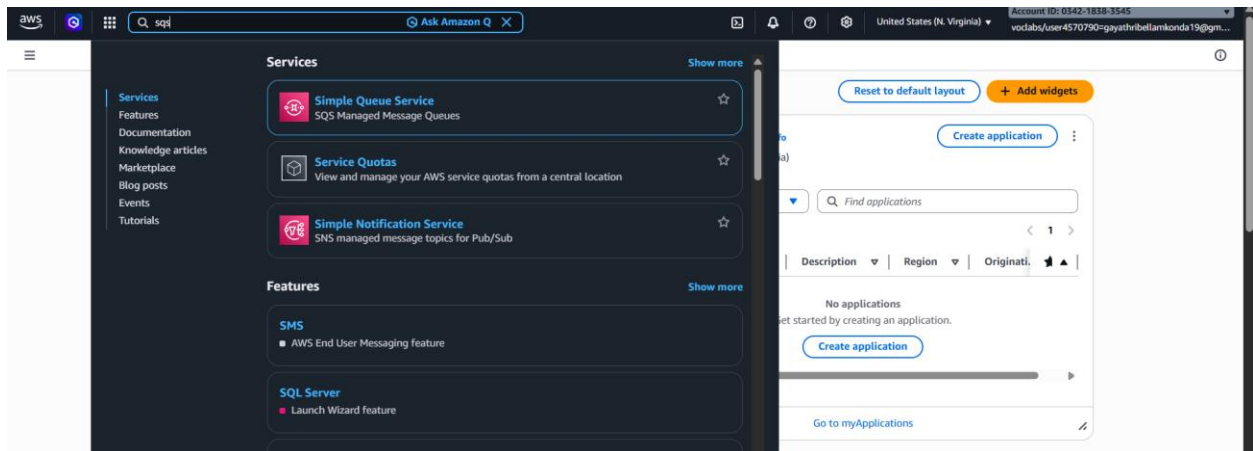
Cancel Upload

Mail about object upload will be received.



Simple Queue Service (SQS)

In the search bar → Type SQS
Open Amazon Simple Queue Service



In the dashboard → Click **Create queue**
Select **Standard** -> **Name : MyQueue**
Leave other settings as **default**
Click **Create queue**

Open the **MyQueue** queue -> Click **Send and receive messages**

In the **Message body**, type : Hello this is SQS test message
Click **Send message**

Send and receive messages

Use this page to send, retrieve and view messages, helping you experiment with various queue features.

Send message [info](#)

Message body
Enter the message to send to the queue.

Hello this is SQS test message

Message group ID - optional, new [info](#)
A group identifier for the message to allow fair processing across message groups in a standard queue.

Message group ID must be 1 to 128 characters. Valid characters are a-z, A-Z, 0-9, and punctuation (!"#%&'()*+,-./:;<=>?@[\]^_`{|}~).

Delivery delay [info](#)
The duration (in seconds) that SQS will postpone the initial delivery of the message. During this delay period, the message is not visible to consumers, allowing you to create a wait time before the message becomes available for processing.

0 Seconds

Should be between 0 seconds and 15 minutes.

► **Message attributes - optional** [info](#)

[Clear content](#) [Send message](#)

Receive messages [info](#)

[Edit poll settings](#) [Stop polling](#) [Poll for messages](#)

Messages available 1 **Polling duration** 30 **Maximum message count** 10 **Polling progress** 0 receives/second 0%

Messages (0)

Q Search messages

◀ 1 ▶

ID	Sent	Size	Receive count
----	------	------	---------------

No messages. To view messages in the queue, poll for messages.

1. In the same page (**Send and receive messages**)
 2. Click **Poll for messages**
- You will see the message displayed.

Receive messages [info](#)

[Edit poll settings](#) [Stop polling](#) [Poll for messages](#)

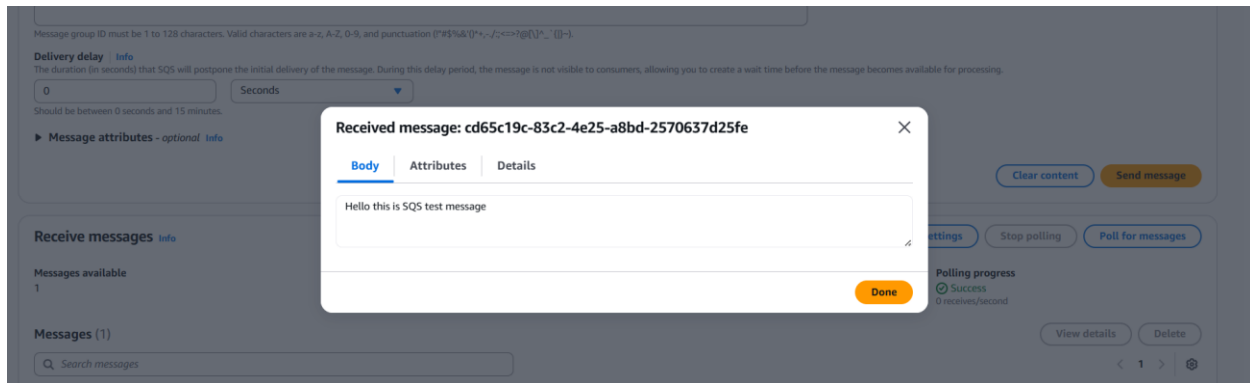
Messages available 1 **Polling duration** 30 **Maximum message count** 10 **Polling progress** 0 receives/second 20%

Messages (1)

Q Search messages

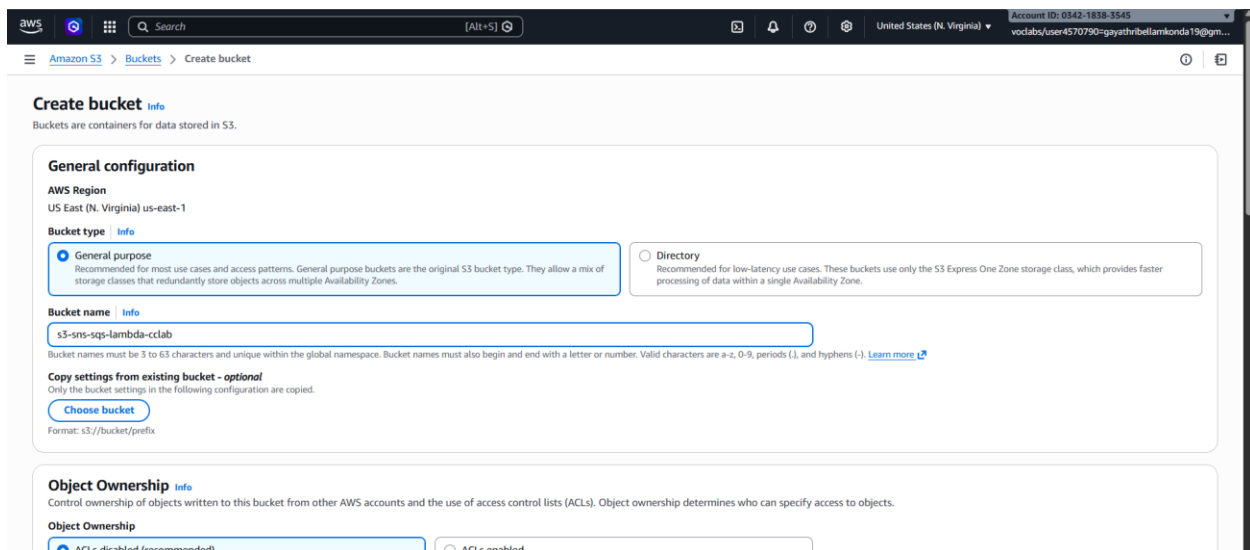
◀ 1 ▶

ID	Sent	Size	Receive count
cd65c19c-83c2-4e25-a8bd-2570637d25fe	2026-03-07T22:02:05:30	30 bytes	3



3) S3 Bucket Event → SNS → SQS → Lambda Processing

- Search S3 → Open Amazon S3 → Click Create bucket
- Bucket name → **s3-sns-sqs-lambda**
- Leave other settings default.



Search SNS in AWS Console → Open Amazon SNS

Click **Topics** → **Create topic**

Configure:

- Type → **Standard**
- Name → **MyS3SNSTopic**

Click **Create topic** -> Copy the **SNS Topic ARN**.

The screenshot shows the 'Create topic' page in the AWS console. The 'Details' section is active, showing the 'Type' as 'Standard' (selected) and 'FIFO (first-in, first-out)' (unselected). The 'Name' field is 's3-sns-sql-lambda'. The 'Display name - optional' field is 'My Topic'. Below the details, there are sections for 'Encryption - optional' and 'Access policy - optional'.

Details

Type [Info](#)
Topic type cannot be modified after topic is created.

☐ FIFO (first-in, first-out)

- Strictly-preserved message ordering
- Exactly-once message delivery
- Subscription protocols: SQS

☒ Standard

- Best-effort message ordering
- At-least once message delivery
- Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints

Name

s3-sns-sql-lambda

Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (_).

Display name - optional [Info](#)
To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.

My Topic

Maximum 100 characters.

► **Encryption - optional** [Info](#)
Amazon SNS provides in-transit encryption by default. Enabling server-side encryption adds at-rest encryption to your topic.

► **Access policy - optional** [Info](#)
This policy defines who can access your topic. By default, only the topic owner can publish or subscribe to the topic.

Search **SQS** in AWS Console -> Open **Amazon SQS** -> Click **Create queue**

Configure:

- Type → **Standard**
- Name → **MyS3Queue**

Click **Create queue** .

The screenshot shows the 'Create queue' page in the AWS console. The 'Details' section is active, showing the 'Type' as 'Standard' (selected) and 'FIFO' (unselected). The 'Name' field is 's3-sns-sql-lambda'. Below the details, there is a 'Configuration' section with 'Visibility timeout' set to 30 seconds and 'Message retention period' set to 4 days.

Create queue

Details

Type [Info](#)
Choose the queue type for your application or cloud infrastructure.

☒ Standard [Info](#)
At-least-once delivery, message ordering isn't preserved

- At-least once delivery
- Best-effort ordering

☐ FIFO [Info](#)
First-in-first-out delivery, message ordering is preserved

- First-in-first-out delivery
- Exactly-once processing

ⓘ You can't change the queue type after you create a queue.

Name

s3-sns-sql-lambda

A queue name is case-sensitive and can have up to 80 characters. You can use alphanumeric characters, hyphens (-), and underscores (_).

Configuration [Info](#)
Set the maximum message size, visibility to other consumers, and message retention.

Visibility timeout [Info](#)

30 Seconds

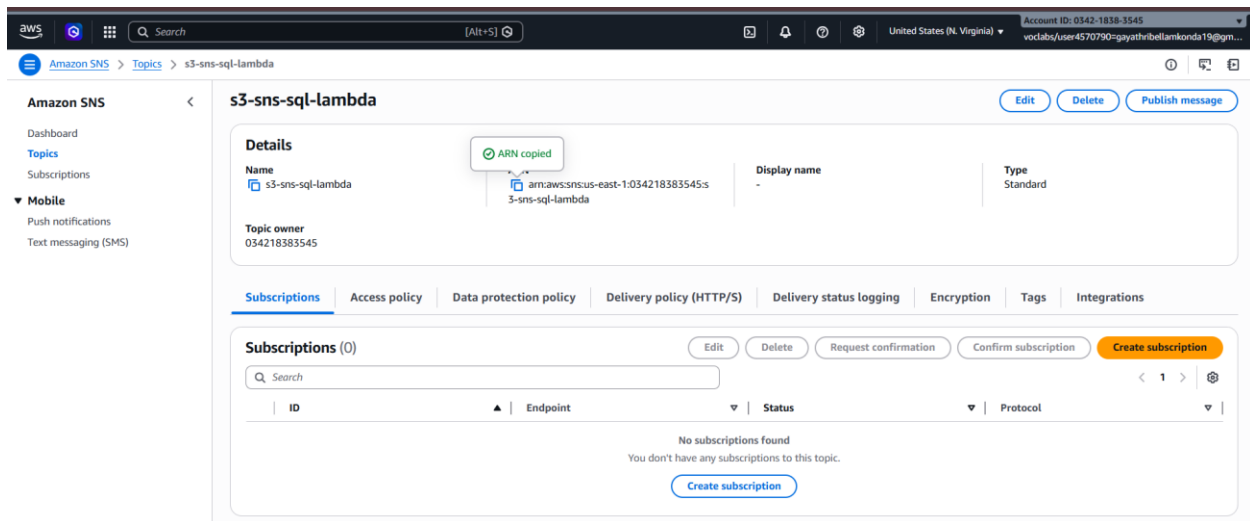
Should be between 0 seconds and 12 hours.

Message retention period [Info](#)

4 Days

Should be between 1 minute and 14 days.

Copy the Queue ARN

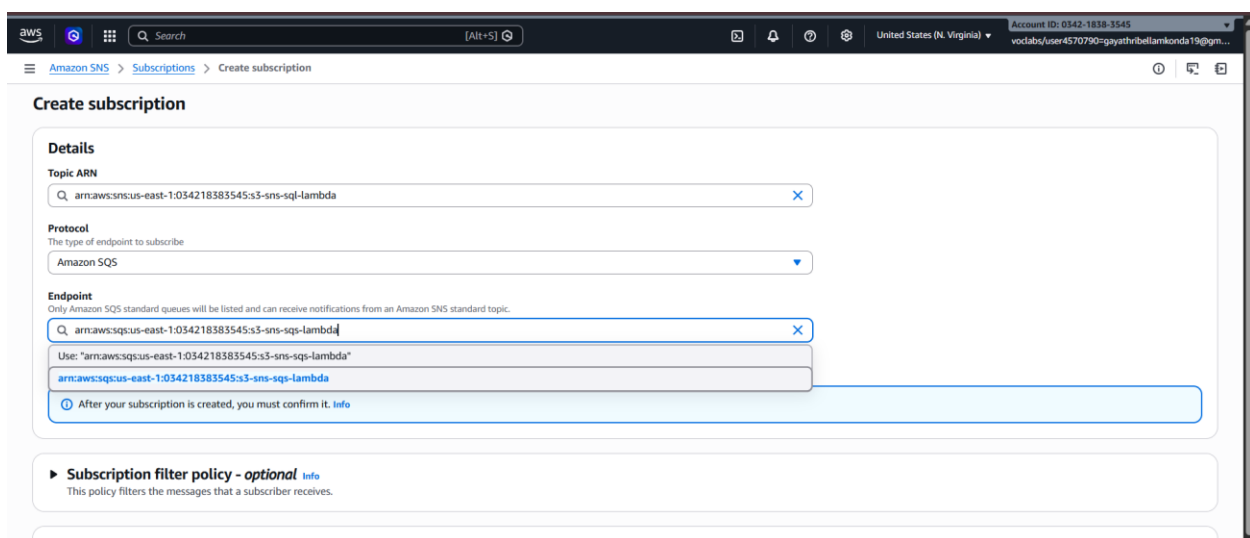


Open **MyS3NSTopic** -> Click **Create subscription**

Configure:

- Protocol -> **Amazon SQS**
- Endpoint -> Select **MyS3Queue**

Click **Create subscription**



Open MyS3Queue → Access Policy → Edit

The screenshot shows the AWS console for the 's3-sns-sqs-lambda' queue. A green notification bar at the top states: 'Queue s3-sns-sqs-lambda created successfully. You can now send and receive messages.' Below this, the queue name 's3-sns-sqs-lambda' is displayed with buttons for 'Edit', 'Delete', 'Purge', 'Send and receive messages', and 'Start DLQ redrive'.

The 'Details' section shows the following information:

Name	Type	ARN
s3-sns-sqs-lambda	Standard	arn:aws:sqs:us-east-1:034218383545:s3-sns-sqs-lambda

Additional details include:

- Encryption: Amazon SQS key (SSE-SQS)
- URL: <https://sqs.us-east-1.amazonaws.com/034218383545/s3-sns-sqs-lambda>
- Dead-letter queue: -

The 'Access policy' section is selected, showing a JSON policy that defines who can access the queue. The policy is:

```
{
  "Version": "2012-10-17",
  "Id": "_default_policy_ID",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "Service": "sns.amazonaws.com"
      },
      "Action": "sqs:SendMessage",
      "Resource": "arn:aws:sqs:us-east-1:034218383545:s3-sns-sqs-lambda",
      "Condition": {
        "ArnEquals": {
          "aws:SourceArn": "arn:aws:sns:us-east-1:034218383545:s3-sns-sqs-lambda"
        }
      }
    }
  ]
}
```

Replace:

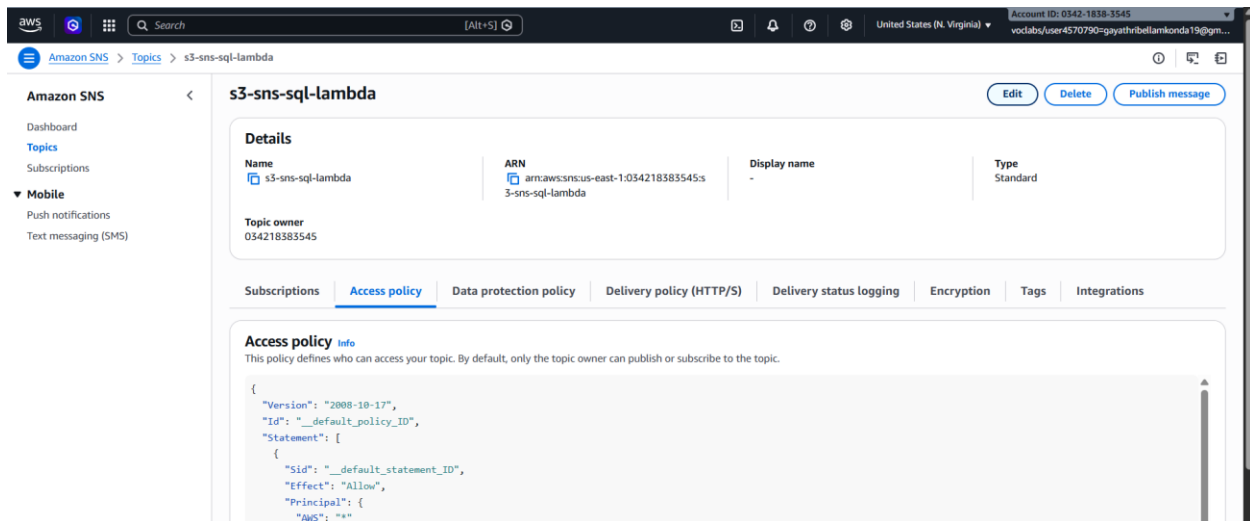
- SQS ARN
- SNS ARN

The screenshot shows the 'Access policy' editor for the 's3-sns-sqs-lambda' queue. The 'Encryption key type' is set to 'Amazon SQS key (SSE-SQS)'. The 'Access policy' section is active, showing a JSON policy that defines who can access the queue. The policy is:

```
{
  "Version": "2012-10-17",
  "Id": "_default_policy_ID",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "Service": "sns.amazonaws.com"
      },
      "Action": "sqs:SendMessage",
      "Resource": "arn:aws:sqs:us-east-1:034218383545:s3-sns-sqs-lambda",
      "Condition": {
        "ArnEquals": {
          "aws:SourceArn": "arn:aws:sns:us-east-1:034218383545:s3-sns-sqs-lambda"
        }
      }
    }
  ]
}
```

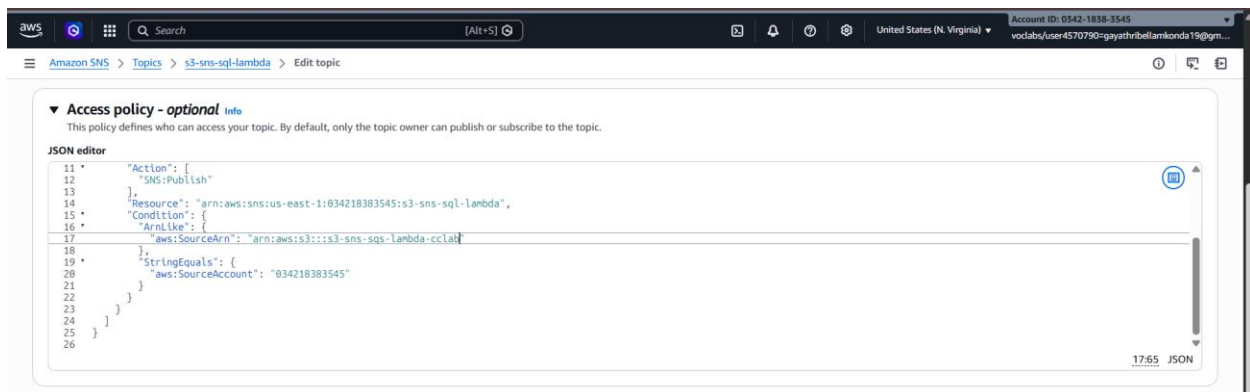
The policy is displayed in a code editor with line numbers 3 through 18. A 'Policy generator' link is visible at the bottom left.

Open SNS Topic → Edit access policy



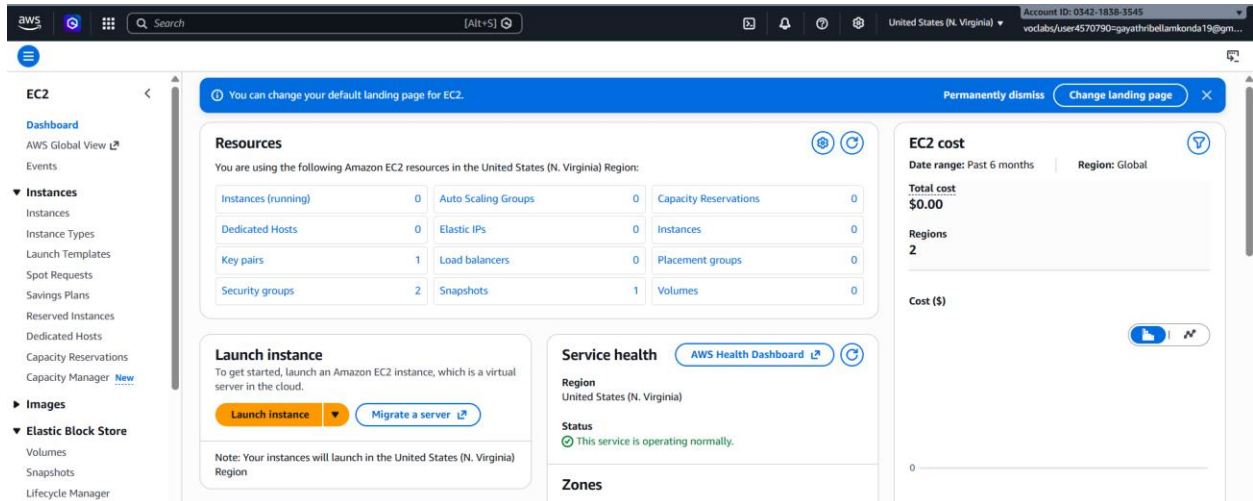
Replace:

- SNS topic ARN
- S3 bucket ARN
- Account ID

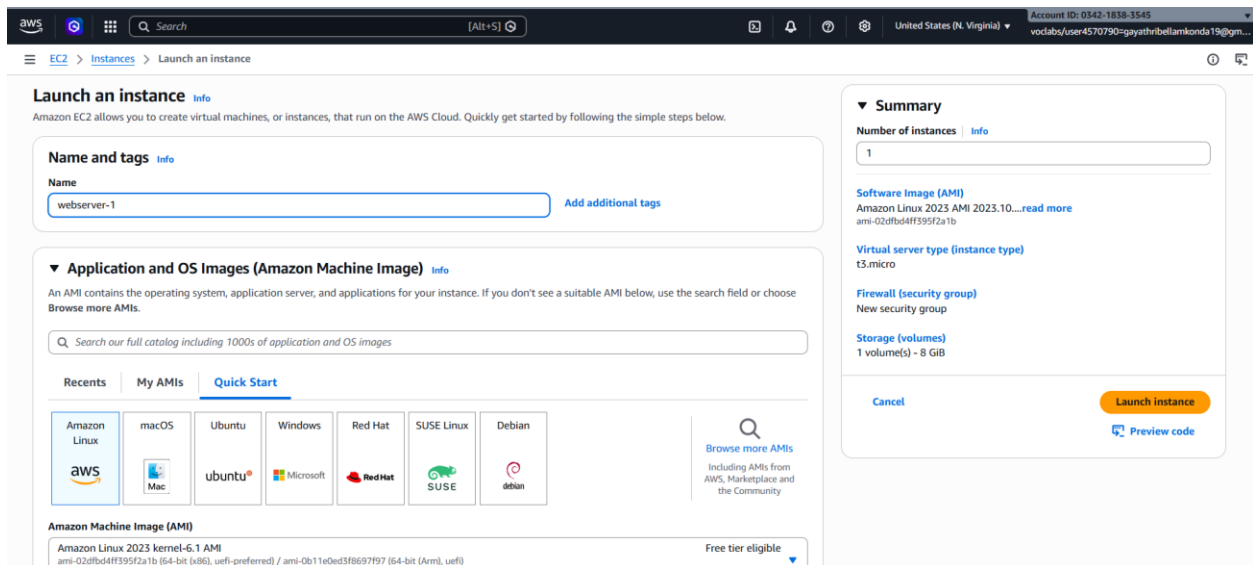


WEEK-9- Implementation of Elastic Load Balancer with Auto Scaling using AWS EC2 Instance

Go to AWS EC2 Console → click on Launch Instance.



Name: webserver-1 -> Choose an AMI (Amazon Linux 2)



Select **Instance Type** (e.g., t2.micro) -> Create Key Pair -> click on edit button of network settings

Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0116 USD per Hour On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour

[Additional costs apply for AMIs with pre-installed software](#)

☒ All generations [Compare instance types](#)

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

webservers [Create new key pair](#)

Network settings [Info](#) [Edit](#)

Network [Info](#)

vpc-063a0b2812e2ff5ff

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.10...[read more](#)

ami-02dfb54ff395f2a1b

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Configure **Security Group**:

- Allow **SSH (22)** from your IP
- Allow **HTTP (80)** from anywhere (0.0.0.0/0)

Click on launch instance

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 103.232.130.149/32) [Remove](#)

Type [Info](#)

ssh

Protocol [Info](#)

TCP

Port range [Info](#)

22

Source type [Info](#)

My IP

Name [Info](#)

[Add CIDR, prefix list or security group](#)

103.232.130.149/32

Description - optional [Info](#)

e.g. SSH for admin desktop

▼ Security group rule 2 (TCP, 80, 0.0.0.0/0) [Remove](#)

Type [Info](#)

HTTP

Protocol [Info](#)

TCP

Port range [Info](#)

80

Source type [Info](#)

Anywhere

Source [Info](#)

[Add CIDR, prefix list or security group](#)

0.0.0.0/0

Description - optional [Info](#)

e.g. SSH for admin desktop

[Add security group rule](#)

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.10...[read more](#)

ami-02dfb54ff395f2a1b

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

[Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.](#)

Run `sudo yum update -y` command in the powershell

Run `sudo yum install httpd -y` command in the powershell

Run `systemctl start httpd`

```
echo "This is Server 1" > /var/www/html/index.html
```

```
ec2-user@ip-172-31-27-38 ~]$ sudo systemctl start httpd
ec2-user@ip-172-31-27-38 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
ec2-user@ip-172-31-27-38 ~]$ echo "This is Server 1" | sudo tee /var/www/html/index.html
This is Server 1
ec2-user@ip-172-31-27-38 ~]$
```

Select the instance and copy the public ip

The screenshot shows the AWS Management Console interface. On the left, the navigation menu includes 'EC2' and 'Instances'. The main panel displays a table of instances with columns: Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, Public IPv4 DNS, and Public. The instance 'webserver-1' with ID 'i-0afbe1269bd5f6f26' is in the 'Running' state. Below the table, the 'Instance summary' section is expanded, showing details like Instance ID, IPv6 address, Hostname type, and Private IP DNS name. A tooltip indicates that the 'Public IPv4 address' has been copied.

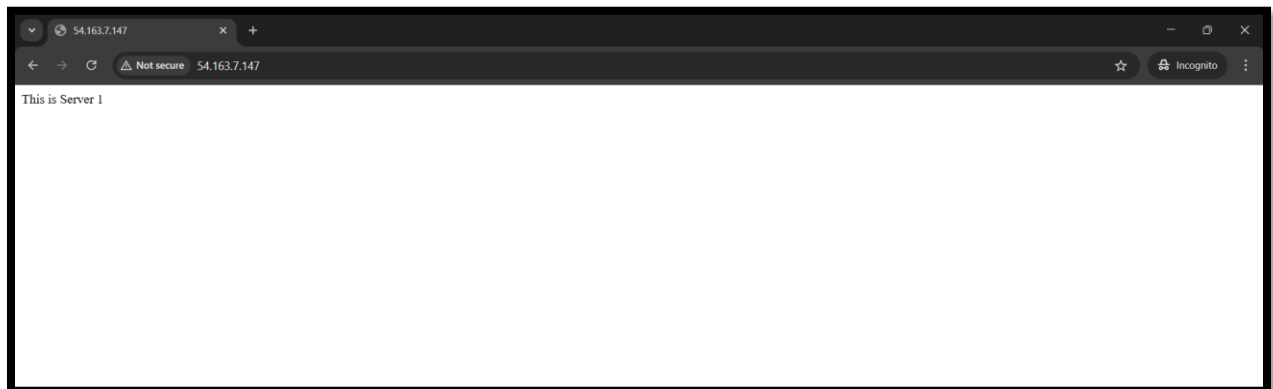
Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
webserver-1	i-0afbe1269bd5f6f26	Running	t2.micro	2/2 checks passed	View alarms	us-east-1c	ec2-54-163-7-147.com...	54.163

i-0afbe1269bd5f6f26 (webserver-1)

Instance summary

- Instance ID: i-0afbe1269bd5f6f26
- IPv6 address: -
- Hostname type: IP name: ip-172-31-27-38.ec2.internal
- Instance state: Running
- Private IP DNS name (IPv4 only): ip-172-31-27-38.ec2.internal
- Private IPv4 addresses: 172.31.27.38
- Public DNS: ec2-54-163-7-147.compute-1.amazonaws.com

Paste the ip address in the incognito window -> the output will be (This is server 1)



Create another instance -> Name : webserver-2 -> select AMI as Amazon Linux

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name
webserver-2 [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents My AMIs **Quick Start**

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10...[read more](#)
ami-02dfb4ff395f2a1b

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Select t2.micro as instance type -> create a key pair -> edit the network settings

Instance type

t2.micro
Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0116 USD per Hour
On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour

[Additional costs apply for AMIs with pre-installed software](#)

☒ All generations [Compare instance types](#)

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required
webserver-1 [Create new key pair](#)

Network settings [Info](#) [Edit](#)

Network [Info](#)
vpc-063a0b2812e2ff5ff

Subnet [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)
Enable

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10...[read more](#)
ami-02dfb4ff395f2a1b

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Configure Security Group:

- Allow **SSH (22)** from anywhere
- Allow **HTTP (80)** from anywhere (0.0.0.0/0)

Click on launch instance

The screenshot shows the 'Launch an instance' page in the AWS Management Console. The 'Inbound Security Group Rules' section is active, showing two rules being configured:

- Security group rule 1 (TCP, 22, 0.0.0.0/0):**
 - Type: ssh
 - Protocol: TCP
 - Port range: 22
 - Source type: Anywhere
 - Source: 0.0.0.0/0
 - Description - optional: e.g. SSH for admin desktop
- Security group rule 2 (TCP, 80, 0.0.0.0/0):**
 - Type: HTTP
 - Protocol: TCP
 - Port range: 80
 - Source type: Anywhere
 - Source: 0.0.0.0/0
 - Description - optional: e.g. SSH for admin desktop

A warning message states: "Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only." Buttons for 'Add security group rule', 'Remove', 'Cancel', 'Launch instance', and 'Preview code' are visible.

Select the instance and click on connect

The screenshot shows the 'Instances' page in the AWS Management Console. A table lists two instances:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IP
webserver-1	i-0afbe1269bd5f6f26	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-54-165-7-147.com...	54.16
webserver-2	i-0c5cdadbe011aa1a1	Running	t2.micro	Initializing	View alarms +	us-east-1c	ec2-98-81-144-134.co...	98.81

The details for instance **i-0c5cdadbe011aa1a1 (webserver-2)** are shown below:

- Instance ID:** i-0c5cdadbe011aa1a1
- IPv6 address:** -
- Hostname type:** IP name: ip-172-31-27-132.ec2.internal
- Answer private resource DNS name:** IPv4 (A)
- Public IPv4 address:** 98.81.144.134 | open address
- Instance state:** Running
- Private IP DNS name (IPv4 only):** ip-172-31-27-132.ec2.internal
- Instance type:** t2.micro
- Private IPv4 addresses:** 172.31.27.132
- Public DNS:** ec2-98-81-144-134.compute-1.amazonaws.com | open address
- Elastic IP addresses:** -

Run sudo yum update -y

```
aws
Search [Alt+S]
United States (N. Virginia) Account ID: 0342-1838-3545
vocabs/user4570790@gayathribellamkonda19@gn...

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-27-132 ~]$ sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
219 kB/s | 30 kB 00:00
```

Run sudo yum install httpd -y

```
aws
Search [Alt+S]
United States (N. Virginia) Account ID: 0342-1838-3545
vocabs/user4570790@gayathribellamkonda19@gn...

[ec2-user@ip-172-31-27-132 ~]$ sudo yum install httpd -y
Last metadata expiration check: 0:00:57 ago on Fri Mar 13 16:03:08 2026.
Dependencies resolved.

Package Architecture Version Repository Size
Installing:
httpd x86_64 2.4.66-1.amzn2023.0.1 amazonlinux 47 k
Installing dependencies:
apr x86_64 1.7.5-1.amzn2023.0.4 amazonlinux 129 k
apr-util x86_64 1.6.3-1.amzn2023.0.2 amazonlinux 97 k
apr-util-ldap x86_64 1.6.3-1.amzn2023.0.2 amazonlinux 13 k
generic-logos-httpd noarch 18.0.0-12.amzn2023.0.3 amazonlinux 19 k
httpd-core x86_64 2.4.66-1.amzn2023.0.1 amazonlinux 1.4 M
httpd-filesystem noarch 2.4.66-1.amzn2023.0.1 amazonlinux 13 k
httpd-tools x86_64 2.4.66-1.amzn2023.0.1 amazonlinux 81 k
libbrotli x86_64 1.0.9-4.amzn2023.0.2 amazonlinux 315 k
mailcap noarch 2.1.49-3.amzn2023.0.3 amazonlinux 33 k
Installing weak dependencies:
apr-util-openssl x86_64 1.6.3-1.amzn2023.0.2 amazonlinux 15 k
mod_http2 x86_64 2.0.27-1.amzn2023.0.3 amazonlinux 166 k
mod_lua x86_64 2.4.66-1.amzn2023.0.1 amazonlinux 166 k

Transaction Summary
Install 13 Packages
Total download size: 2.4 M
```

Run systemctl start httpd

systemctl enable httpd

echo "This is Server 1" > /var/www/html/index.html

```
[ec2-user@ip-172-31-27-132 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-27-132 ~]$ sudo systemctl enable httpd
[ec2-user@ip-172-31-27-132 ~]$ echo "This is Server 2" | sudo tee /var/www/html/index.html
This is Server 2
[ec2-user@ip-172-31-27-132 ~]$
```

Select the instance and copy the public ip address

The screenshot shows the AWS Management Console for the 'Instances' page. A table lists two instances: 'webserver-1' and 'webserver-2'. 'webserver-2' is selected, and a tooltip indicates its public IPv4 address '98.81.144.134' is being copied. Below the table, the details for 'webserver-2' (ID: i-0c5cdadbe011aa1a1) are displayed, showing it is in a 'Running' state with type 't2.micro'. The 'Networking' tab is active, showing the public IPv4 address '98.81.144.134' and the public DNS 'ec2-98-81-144-134.compute-1.amazonaws.com'.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IP
webserver-1	i-0afbe1269bd5f6f26	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-54-163-7-147.com...	54.163.7...
webserver-2	i-0c5cdadbe011aa1a1	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-98-81-144-134.co...	98.81.144...

i-0c5cdadbe011aa1a1 (webserver-2)

Instance summary

Instance ID: i-0c5cdadbe011aa1a1

IPv6 address: -

Hostname type: IP name: ip-172-31-27-132.ec2.internal

Public IPv4 address copied: 98.81.144.134 | open address

Instance state: Running

Private IP DNS name (IPv4 only): ip-172-31-27-132.ec2.internal

Private IPv4 addresses: 172.31.27.132

Public DNS: ec2-98-81-144-134.compute-1.amazonaws.com | open address

Paste the ip address in the incognito window -> the output will be (This is server 2)

The screenshot shows a web browser window with the address bar displaying '98.81.144.134'. The page content displays 'This is Server 2'.

Go to EC2 → Security Groups → Click on Create Security Group

The screenshot shows the AWS Management Console for the 'Security Groups' page. A green notification bar at the top states 'Security group (sg-071c8c80cfa2fa36 | lb-securitygroup) successfully deleted'. Below the notification, a table lists three security groups: 'default', 'launch-wizard-2', and 'launch-wizard-1'. The 'default' group is highlighted.

Name	Security group ID	Security group name	VPC ID	Description
-	sg-0519e36d5421230bb	default	ypc-063a0b2812e2ff5ff	default VPC security group
-	sg-06cecb9d5dbeb8d9ea	launch-wizard-2	ypc-063a0b2812e2ff5ff	launch-wizard-2 created 2026-03-13T1...
-	sg-09b110ecf50ce5ff5	launch-wizard-1	ypc-063a0b2812e2ff5ff	launch-wizard-1 created 2026-03-13T1...

Select a security group

Name : loadbalancer-sg -> give description

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

loadbalancer-sg

Name cannot be edited after creation.

Description [Info](#)

load balancer security group

VPC [Info](#)

vpc-063a0b2812e2ff5ff

Inbound rules [Info](#)

This security group has no inbound rules.

[Add rule](#)

click on add rule of inbound rule -> Allow **HTTP (80)** from anywhere

Leave the outbound rules as default

Click on create security group

Inbound rules [Info](#)

Type	Protocol	Port range	Source	Description - optional	
HTTP	TCP	80	Anywhere...		Delete

[Add rule](#)

Rules with source of 0.0.0.0/0 or :::0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules [Info](#)

Type	Protocol	Port range	Destination	Description - optional	
All traffic	All	All	Custom		Delete

[Add rule](#)

Rules with destination of 0.0.0.0/0 or :::0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

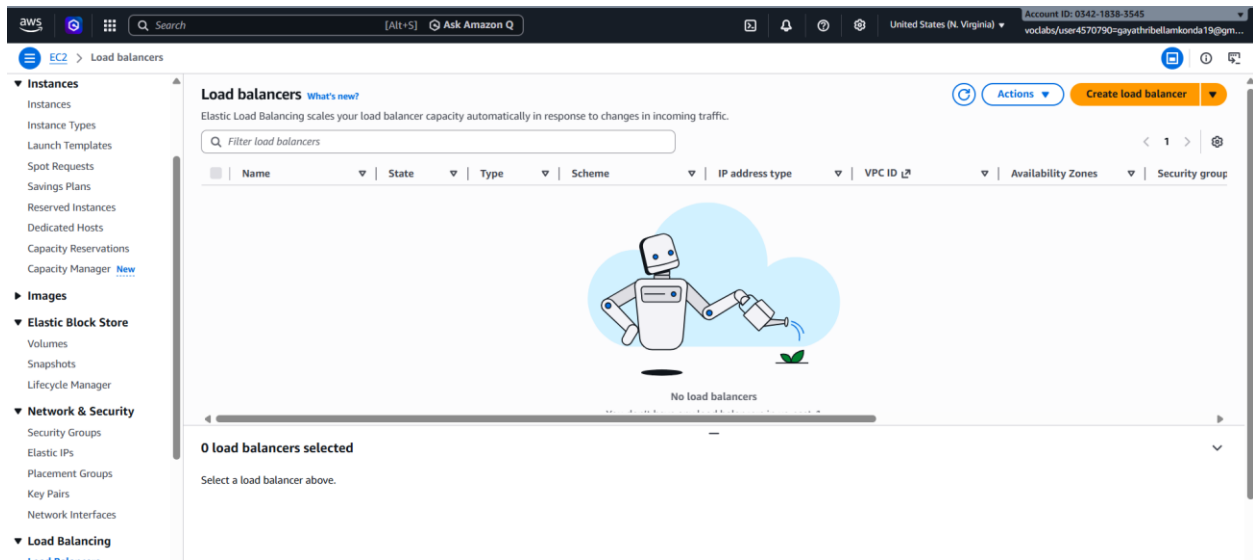
No tags associated with the resource.

[Add new tag](#)

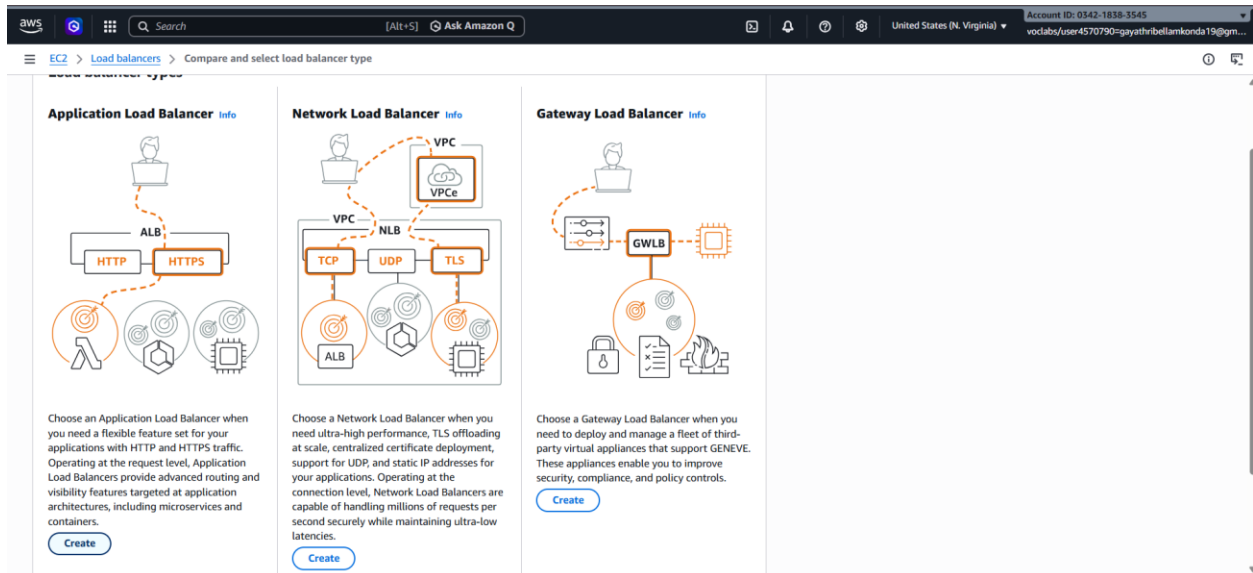
You can add up to 50 more tags

[Cancel](#) [Create security group](#)

Go to EC2 → Load Balancers → Click on Create Load Balancer



Click on create of Application Load Balancer



Name: my-loadbalancer -> Scheme: Internet-facing

IP address type: IPv4

Create Application Load Balancer [Info](#)

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

► **How Application Load Balancers work**

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

my-loadbalancer

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type [Info](#)
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**

Includes only IPv4 addresses.

☐ **Dualstack**

Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.

Select the default VPC where your EC2 instances exist

Select all subnets (one for each instance if possible)

Choose the loadbalancer-sg security group created earlier

Subnets

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

☒ **us-east-1d (use1-az6)**

Subnet

subnet-0c32f1f5a05da3f44
172.31.16.0/20

☒ **us-east-1e (use1-az3)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0285963a158c2194e
172.31.32.0/20

☒ **us-east-1f (use1-az5)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0d0c5ad7ca7d255e6
172.31.48.0/20

Security groups [Info](#)
A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

☐ launch-wizard-1 (sg-09b110ecf50ce5ff5)

☐ launch-wizard-2 (sg-06ceb9d5dbbeb8d9ea)

☒ **loadbalancer-sg (sg-0c4da404f065a5d12)**

☐ default (sg-0519e36d3421230bb)

balancer routes requests to its registered targets.

Click on create target group

The screenshot shows the 'Listeners and routing' section of the 'Create Application Load Balancer' page. A listener named 'HTTP:80' is configured with Protocol 'HTTP' and Port '80'. The default action is 'Forward to target groups'. Under 'Forward to target group', a target group is selected from a dropdown menu. The weight is set to 1 (100%). There is an option to 'Add target group' and a checkbox for 'Turn on target group stickiness' which is currently unchecked. The page also shows 'Listener tags - optional' at the bottom.

Name: my-targetgroup -> Target type: Instance

Protocol: HTTP -> Port: 80

The screenshot shows the 'Create target group' page. On the left, a progress bar indicates the steps: 'Step 1: Create target group' (selected), 'Step 2 - recommended: Register targets', and 'Step 3: Review and create'. The main section is titled 'Create target group' and includes a note: 'A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.' Under 'Settings - immutable', there is a section for 'Target type' with four options: 'Instances' (selected), 'IP addresses', 'Lambda function', and 'Application Load Balancer'. Each option has a description and suitable load balancer types (ALB, NLB, GWLB). Below this, the 'Target group name' is entered as 'my-targetgroup'. At the bottom, the 'Protocol' is set to 'HTTP' and the 'Port' is set to '80'.

Select ipv4 -> select the default vpc -> click on next

aws [Search] [Alt+S] United States (N. Virginia) Account ID: 0342-1838-3545 vocabs/user4570790=gayathribellamkonda19@gm...

EC2 > Target groups > Create target group

IP address type
Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ IPv6
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-063a0b2812e2ff5ff (default) [Create VPC](#)

Protocol version

☒ HTTP1
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ HTTP2
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ gRPC
Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Health checks
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol
HTTP

Health check path

Select all the instances -> Click **Include as pending** → Click on next

aws [Search] [Alt+S] United States (N. Virginia) Account ID: 0342-1838-3545 vocabs/user4570790=gayathribellamkonda19@gm...

EC2 > Target groups > Create target group

Step 1
Create target group

Step 2 - recommended
Register targets

Step 3
Review and create

Register targets - recommended
This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/2)

Filter instances

☒	Instance ID	Name	State	Security groups	Zone	Private IPv4
☒	i-0c5cdadbe011aa1a1	webserver-2	Running	launch-wizard-2	us-east-1c	172.31.27.1
☒	i-0afbe1269bd5f6f26	webserver-1	Running	launch-wizard-1	us-east-1c	172.31.27.3

2 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80
1-65535 (separate multiple ports with commas)

[Include as pending below](#)

Review targets

Review targets

Targets (2)

Filter targets ☐ Show only pending [Remove all pending](#)

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-0c5cdadbe011aa1a1	webserver-2	80	Running	launch-wizard-2	us-east-1c	172.31.27.132	subnet-0c32f1f5a05da3f44	March 13, 2026, 21
i-0afbe1269bd5f6f26	webserver-1	80	Running	launch-wizard-1	us-east-1c	172.31.27.38	subnet-0c32f1f5a05da3f44	March 13, 2026, 21

2 pending

[Cancel](#) [Previous](#) [Next](#)

Review all settings → Click **Create target group**

aws [Search] [Alt+S] United States (N. Virginia) Account ID: 0342-1838-3545

EC2 > Target groups > Create target group

Review and create

Target group details

Name	my-targetgroup	Target type	Instance	Protocol : Port	HTTP: 80	Protocol version	HTTP1
VPC	vpc-063a0b2812e2ff5ff						
		IP address type	IPv4				

Health check details

Health check protocol	HTTP	Health check path	/	Health check port	traffic-port	Interval	30 seconds
Timeout	5 seconds	Healthy threshold	5	Unhealthy threshold	2	Success codes	200

Step 2: Register targets

Targets (2)

Instance ID	Name	Port	Zone
i-0c5cda0be011aa1a1	webserver-2	80	us-east-1c
i-0a0be1269bd5f6f26	webserver-1	80	us-east-1c

Cancel Previous Create target group

Go to **Load Balancers** → **Select the loadbalancer** → **Description** → copy the DNS name

aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) Account ID: 0342-1838-3545

EC2 > Load balancers > my-loadbalancer

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

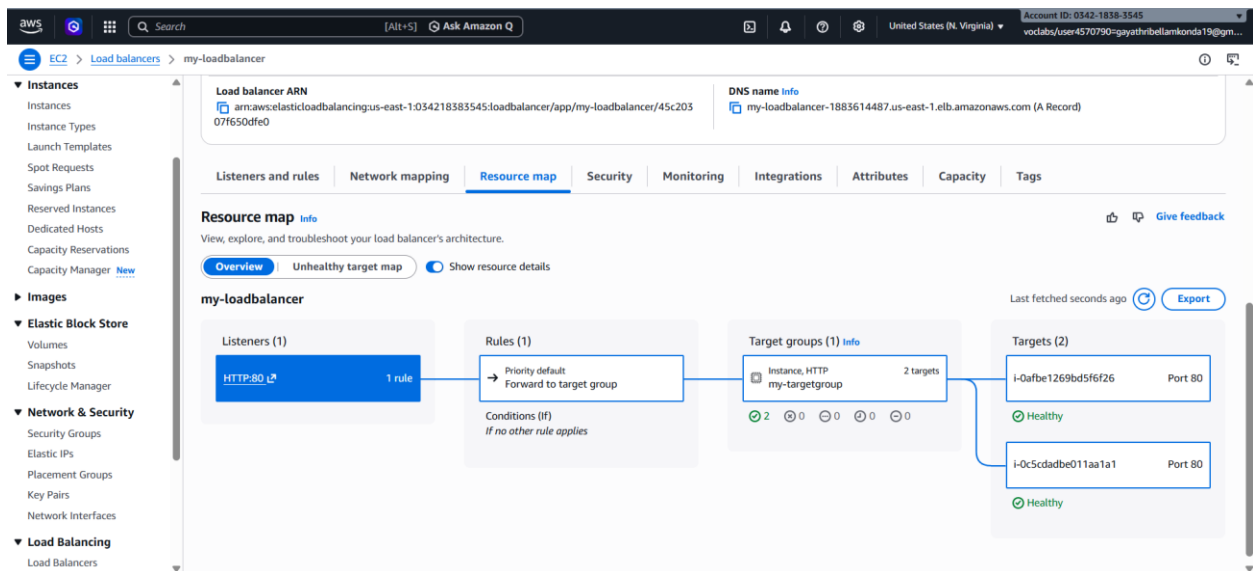
Introducing ALB target optimizer

Target optimizer lets you enforce a maximum number of requests per target using an ALB-provided agent, improving success rates, latency, and efficiency. [Learn more](#)

my-loadbalancer

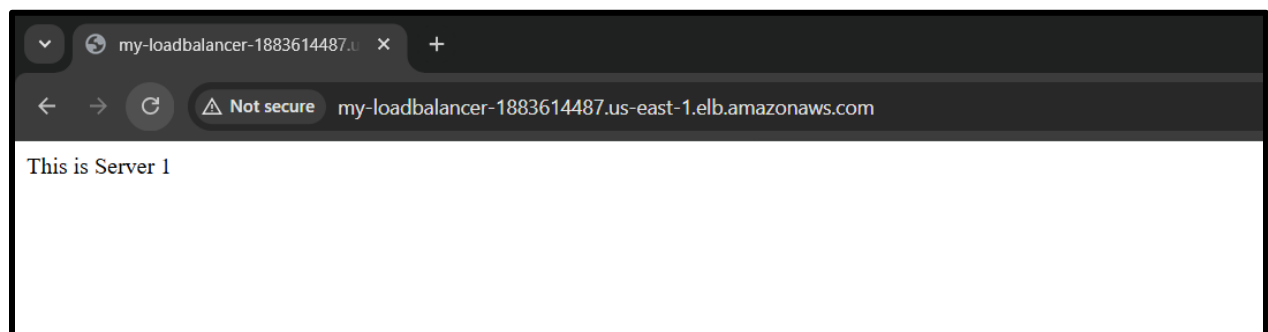
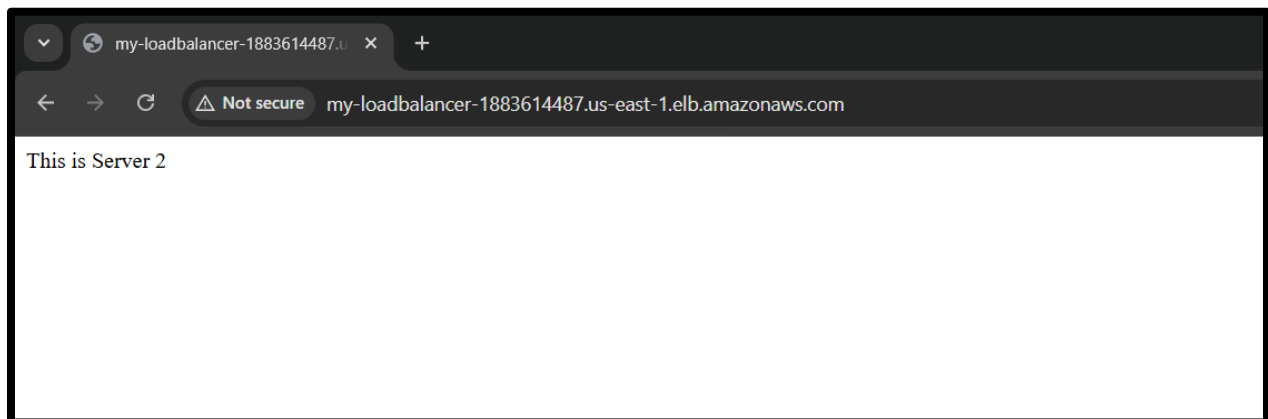
Details

Load balancer type	Application	Status	Provisioning	VPC	vpc-063a0b2812e2ff5ff	Load balancer IP address type	IPv4
Scheme	Internet-facing	Hosted zone	Z35XDOTRQ7X7K	Availability Zones	subnet-0d0c5ad7ca7d255e6 us-east-1e (use1-az3) subnet-0285963a158c2194e us-east-1d (use1-az6) subnet-090a2755c7a41fc2c us-east-1a (use1-az1) subnet-07a991dc9d333c9f4 us-east-1b (use1-az2) subnet-0c32f1f5a05da3f44 us-east-1c (use1-az4) subnet-005e9d9a41c352eab us-east-1f (use1-az5)	Date created	March 13, 2026, 22:25 (UTC+05:30)
Load balancer ARN	arn:aws:elasticloadbalancing:us-east-1:034218383545:loadbalancer/app/my-loadbalancer/45c20307f650df60			DNS name	my-loadbalancer-1883614487.us-east-1.elb.amazonaws.com (A Record)		



Paste it in a browser → You should see either **WebServer-1** or **WebServer-2** page.

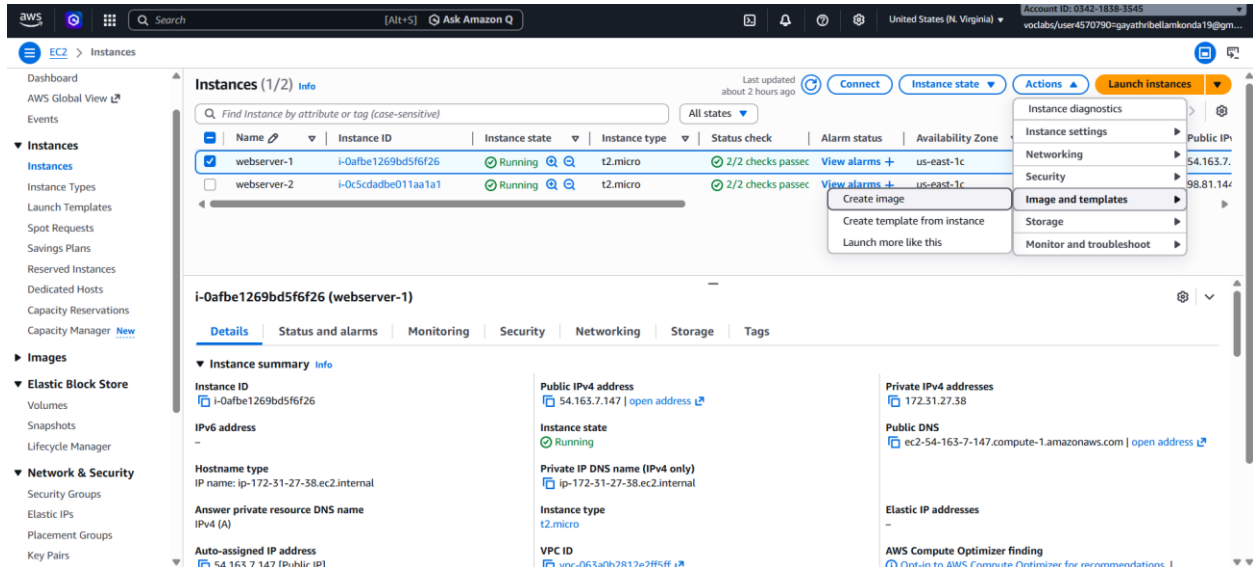
Refresh multiple times → Load balancer distributes traffic between the two instances



Auto Scaling Group with Load Balancer

Go to **EC2 Console** → **Instances** → Select the EC2 instance

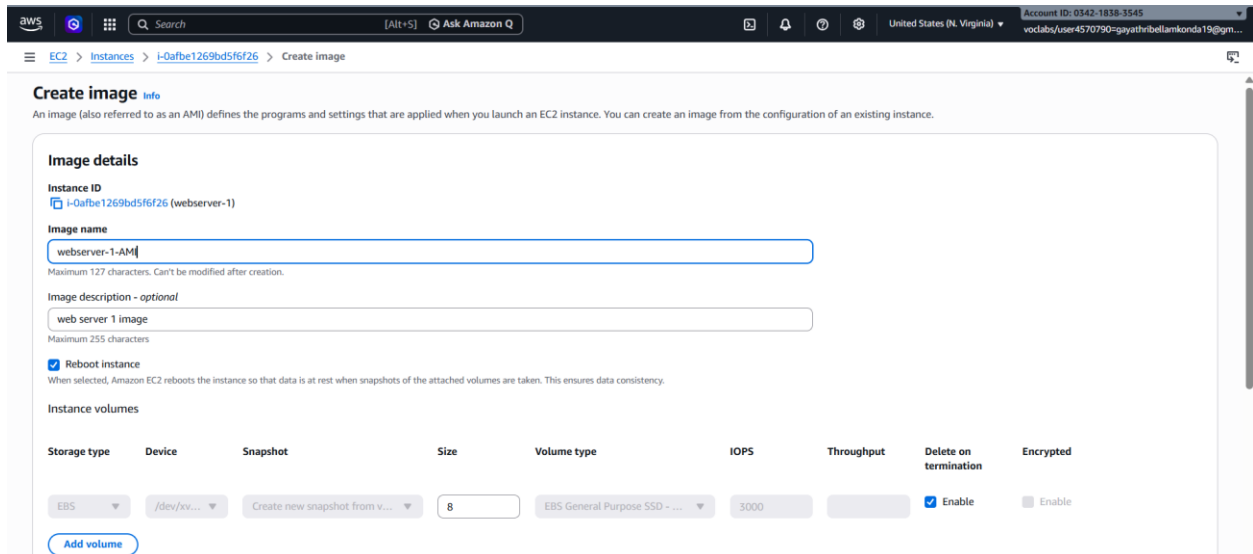
Click **Actions** → **Image and Templates** → **Create Image**



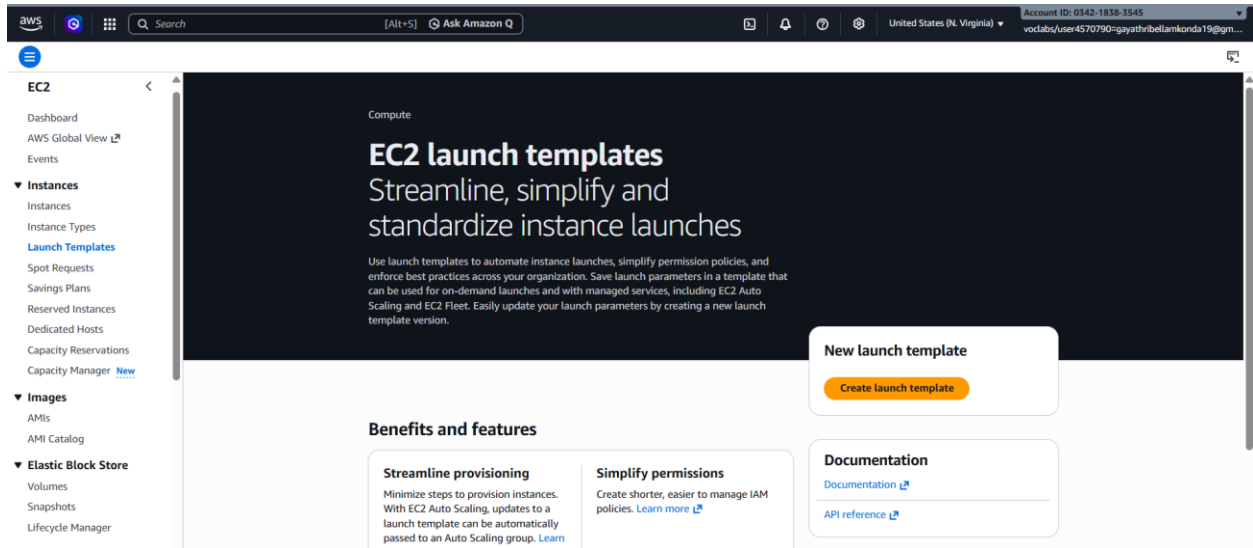
Provide Image Name: e.g., my-webserver-AMI

Optional: Add description

Click **Create Image**

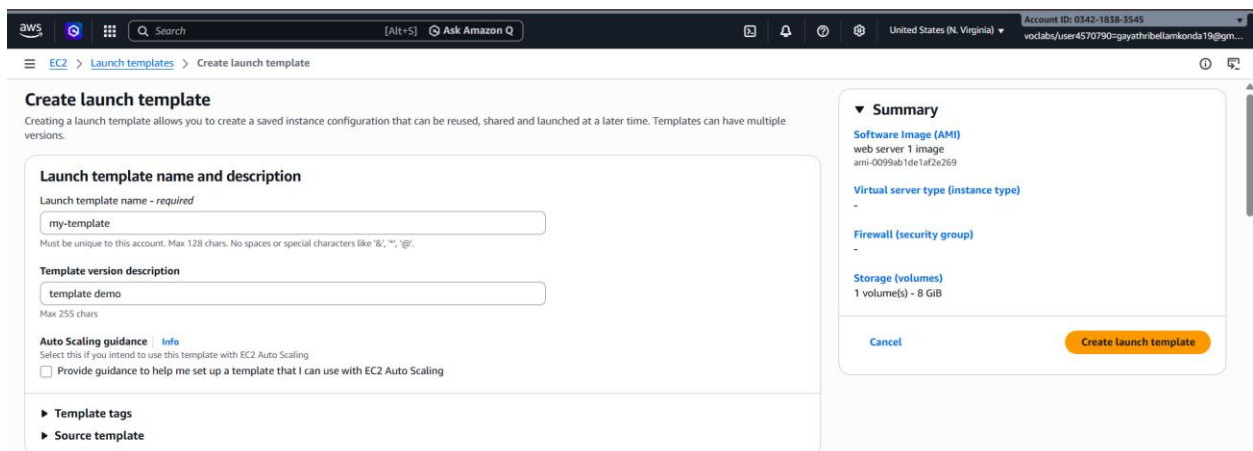


Go to EC2 → instances->Launch Templates → Create Launch Template



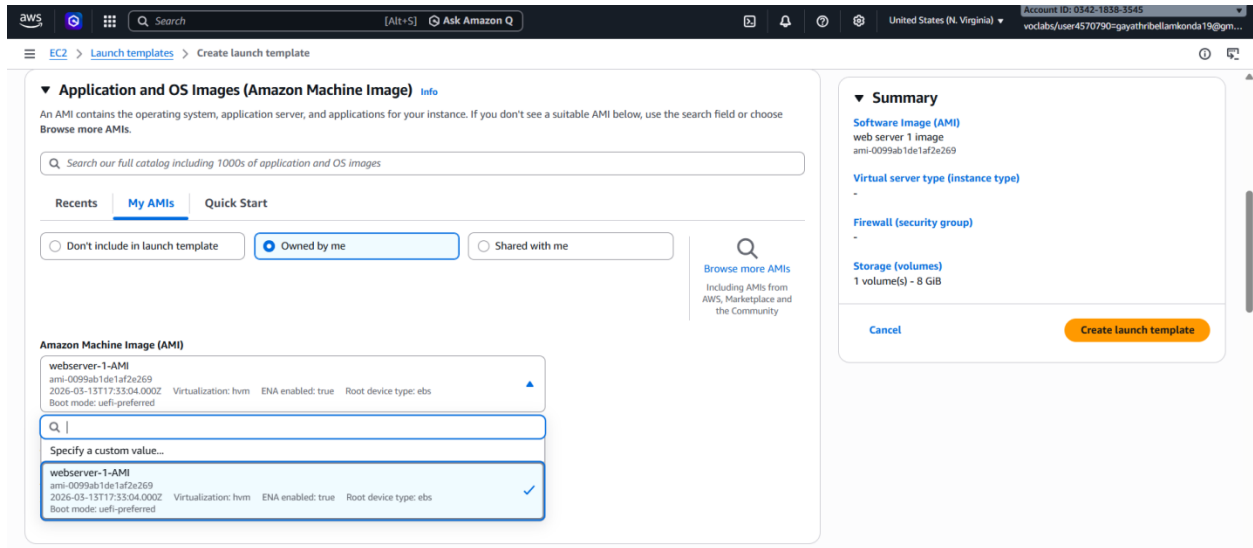
Launch Template Name: my-template

Add description



Select My AMIs -> select owned by me

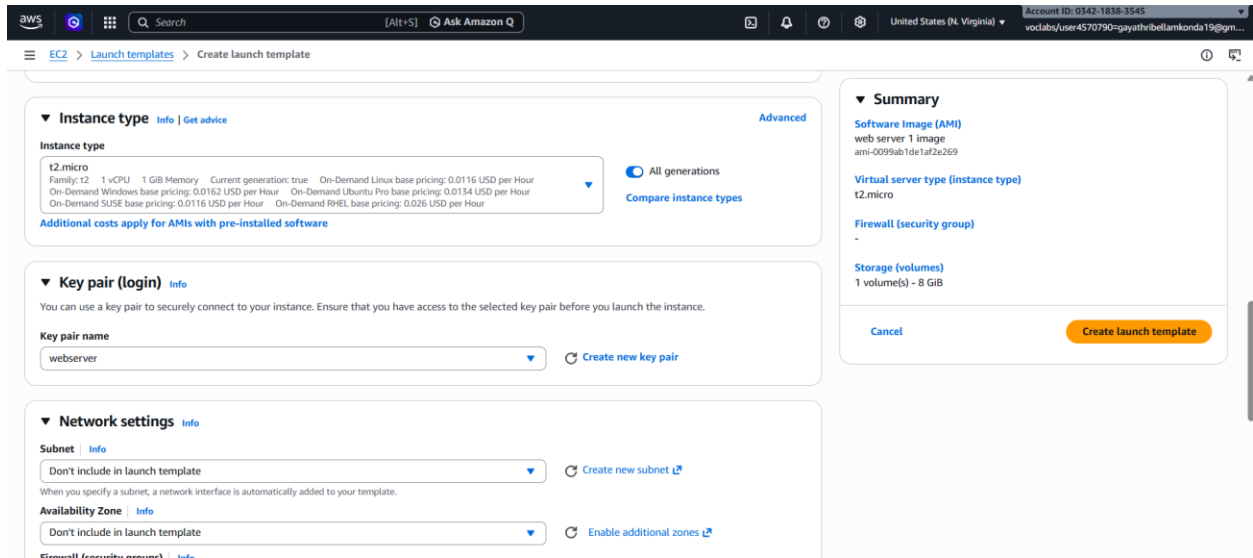
AMI: Select the AMI created in Step 1 (webserver-1-AMI)



The screenshot shows the 'Create launch template' page in the AWS Management Console. The 'Application and OS Images (Amazon Machine Image)' section is active. It features a search bar with the text 'Search our full catalog including 1000s of application and OS images'. Below the search bar are tabs for 'Recents', 'My AMIs', and 'Quick Start'. Under 'My AMIs', there are radio buttons for 'Don't include in launch template', 'Owned by me' (selected), and 'Shared with me'. A search icon and a link 'Browse more AMIs' are also present. The 'Amazon Machine Image (AMI)' list shows 'webserver-1-AMI' with details: ami-0099ab1de1af2e269, 2026-03-13T17:33:04.000Z, Virtualization: hvm, ENA enabled: true, Root device type: ebs, and Boot mode: uefi-preferred. A search bar below the list has 'webserver-1-AMI' entered and selected. The 'Summary' section on the right shows 'Software Image (AMI)' as 'web server 1 image', 'Virtual server type (instance type)' as '-', 'Firewall (security group)' as '-', and 'Storage (volumes)' as '1 volume(s) - 8 GiB'. A 'Create launch template' button is at the bottom right.

Instance Type: t2.micro

Key Pair: Select your existing key pair



The screenshot shows the 'Create launch template' page in the AWS Management Console, Step 2. The 'Instance type' section shows 't2.micro' selected. The 'Key pair (login)' section shows 'webserver' selected. The 'Network settings' section shows 'Don't include in launch template' selected for both 'Subnet' and 'Availability Zone'. The 'Summary' section on the right shows 'Software Image (AMI)' as 'web server 1 image', 'Virtual server type (instance type)' as 't2.micro', 'Firewall (security group)' as '-', and 'Storage (volumes)' as '1 volume(s) - 8 GiB'. A 'Create launch template' button is at the bottom right.

Set **Health Check Grace Period**: 3 seconds

Click **Next**

The screenshot shows the 'Create Auto Scaling group' page in the AWS Management Console. The 'Health checks' section is expanded, showing options for 'EC2 health checks' and 'Additional health check types'. The 'Health check grace period' is set to 3 seconds. The 'Next' button is highlighted.

Application Recovery Controller (ARC) zonal shift - new [info](#)
During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.

☐ **Enable zonal shift**
New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.

Health checks
Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks
[Always enabled](#)

Additional health check types - optional [info](#)

☐ **Turn on Elastic Load Balancing health checks** [Recommended](#)
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

☐ **Turn on VPC Lattice health checks**
VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

☐ **Turn on Amazon EBS health checks**
EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

Health check grace period [info](#)
This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

3 seconds

[Cancel](#) [Skip to review](#) [Previous](#) [Next](#)

Desired Capacity: 2 (initial instances)

Minimum Capacity: 1 -> **Maximum Capacity**: 4

Click **Next**

The screenshot shows the 'Configure group size and scaling' section of the 'Create Auto Scaling group' page. The 'Desired capacity' is set to 2. The 'Min desired capacity' is 1 and the 'Max desired capacity' is 4. The 'Automatic scaling' section is also visible.

Configure group size and scaling - optional [info](#)
Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [info](#)
Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity
Specify your group size.

2

Scaling [info](#)
You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits
Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity
1
Equal or less than desired capacity

Max desired capacity
4
Equal or greater than desired capacity

Automatic scaling - optional
Choose whether to use a target tracking policy [info](#)
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

Leave notifications and tags as default -> click next



Review all settings

Click **Create Auto Scaling Group**

Disabled

Auto Scaling group deletion protection

None (default)

Disabled

Capacity Reservation preference

Preference

Default

Capacity Reservation IDs

-

Resource Groups

-

Step 5: Add notifications

Notifications

No notifications

Step 6: Add tags

Tags (0)

Key	Value	Tag new instances
No tags		

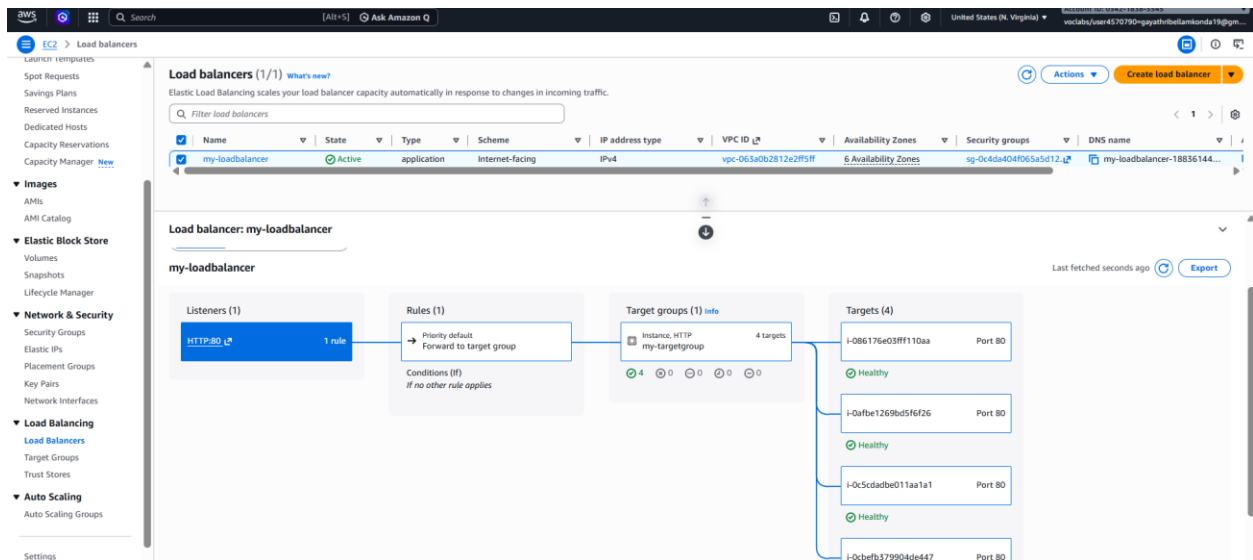
Previous code

Cancel Previous Create Auto Scaling group

Go to **Load Balancers** → **Select your Load Balancer**

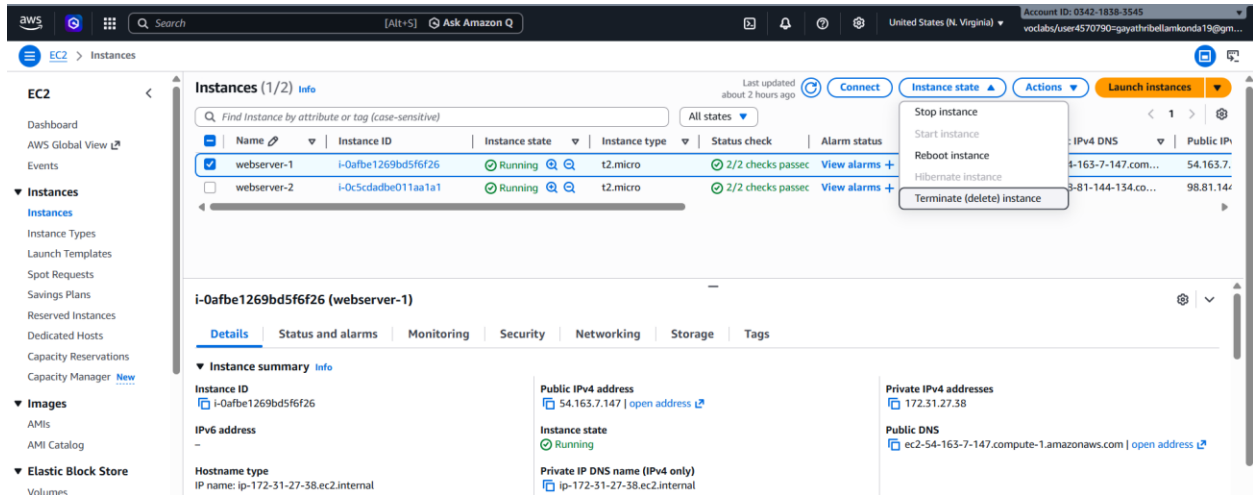
Check **Auto Scaling Group** → **Instances**

- You may see up to **4 instances** if scaling triggers



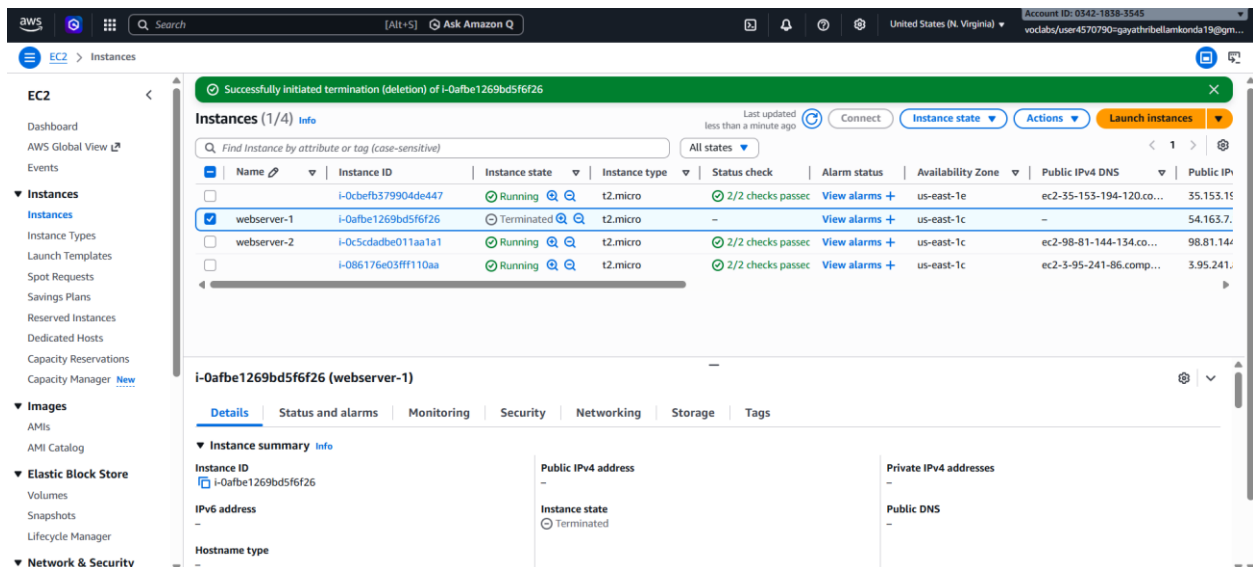
Test auto recovery:

Select the instance -> Terminate the instance



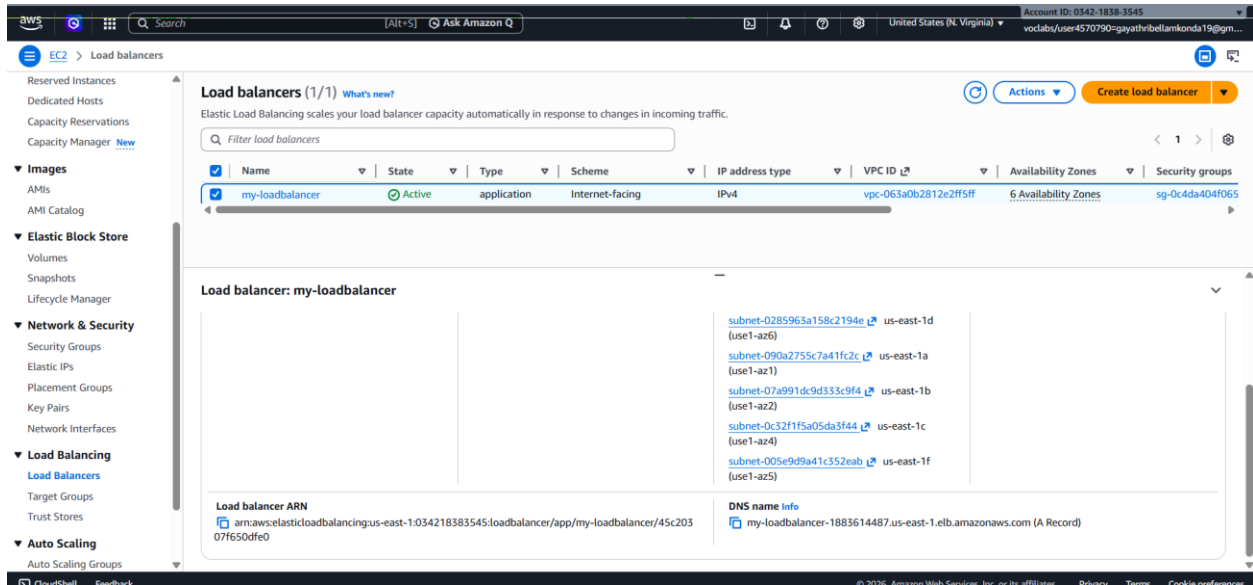
The screenshot shows the AWS Management Console for EC2 Instances. The left sidebar contains navigation links for EC2, Dashboard, AWS Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager, Images, AMIs, AMI Catalog, Elastic Block Store, Volumes, Snapshots, Lifecycle Manager, and Network & Security. The main content area displays the 'Instances (1/2)' table. The table has columns for Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, Public IPv4 DNS, and Public IP. Two instances are listed: 'webserver-1' (i-0afbe1269bd5f6f26) and 'webserver-2' (i-0c5cdadbe011aa1a1), both in a 'Running' state. The 'Actions' dropdown menu is open, showing options like 'Stop instance', 'Start instance', 'Reboot instance', 'Hibernate instance', and 'Terminate (delete) instance'. The details for 'webserver-1' are shown below the table, including its Instance ID, Public IPv4 address (54.163.7.147), and Instance state (Running).

2 instances will be created automatically by the auto scaling



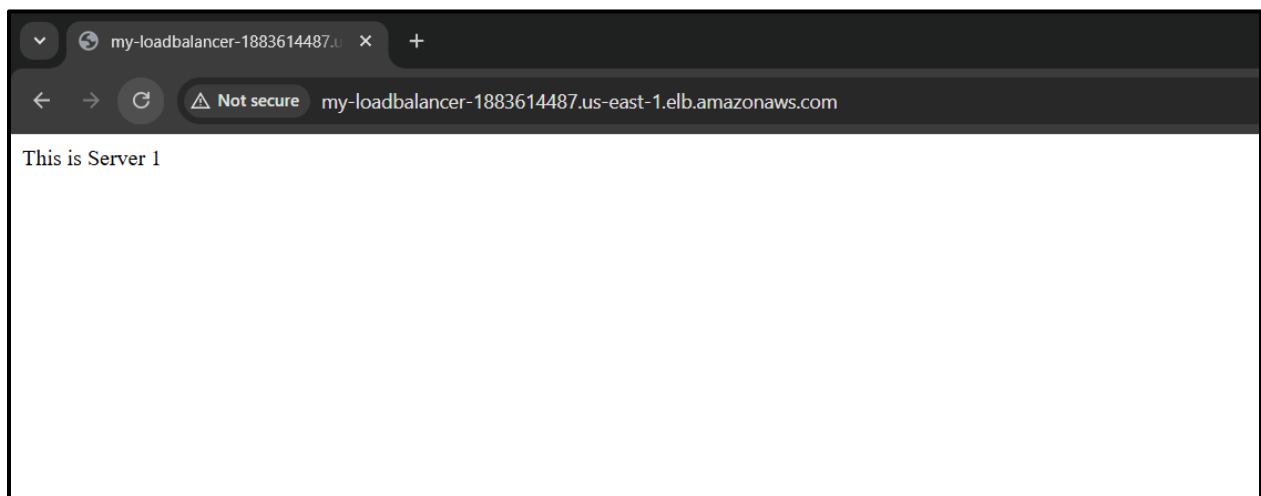
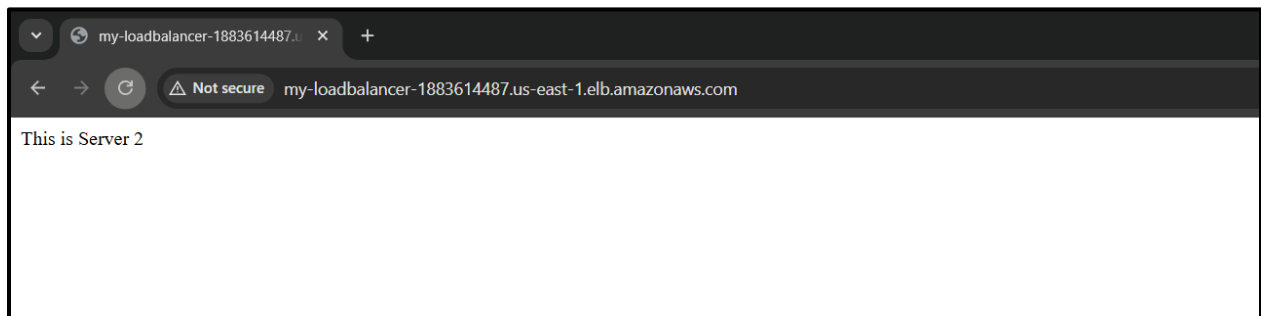
The screenshot shows the AWS Management Console for EC2 Instances after terminating 'webserver-1'. A green banner at the top states 'Successfully initiated termination (deletion) of i-0afbe1269bd5f6f26'. The 'Instances (1/4)' table now shows four instances: 'webserver-1' (i-0afbe1269bd5f6f26) is 'Terminated', while 'webserver-2' (i-0c5cdadbe011aa1a1) and two new instances (i-0cbe1b379904de447 and i-086176e03fff110aa) are in a 'Running' state. The details for 'webserver-1' are shown below the table, indicating its Instance state is 'Terminated'.

Select the load balancer -> copy the DNS name



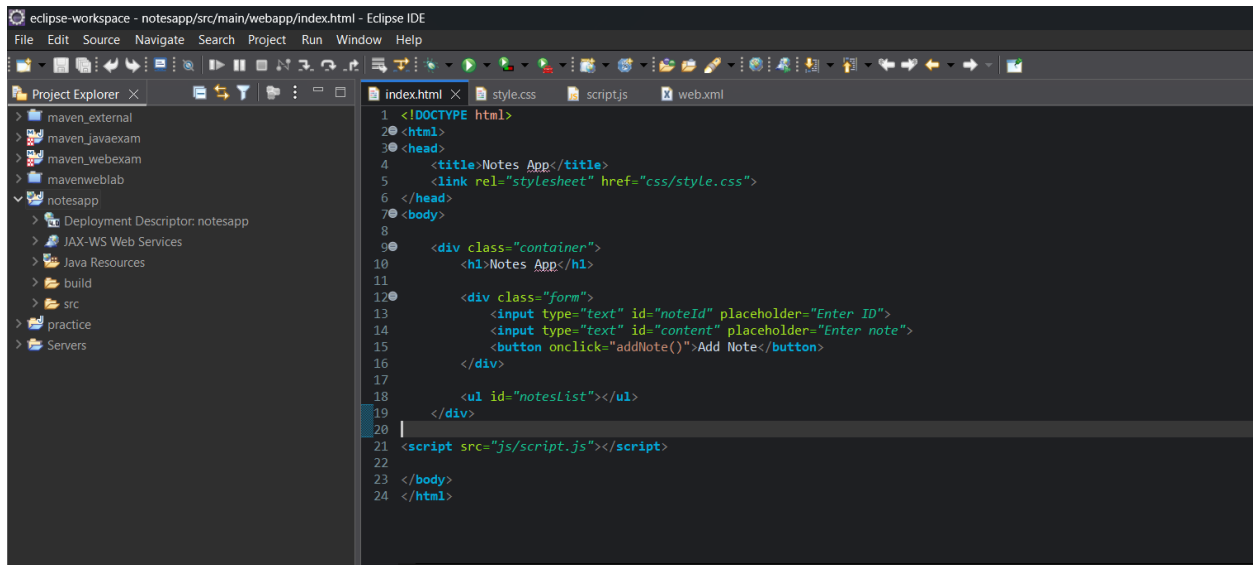
Paste the DNS in the incognito mode

You should see your web page from one of the instances -> Refresh multiple times → traffic alternates between instances

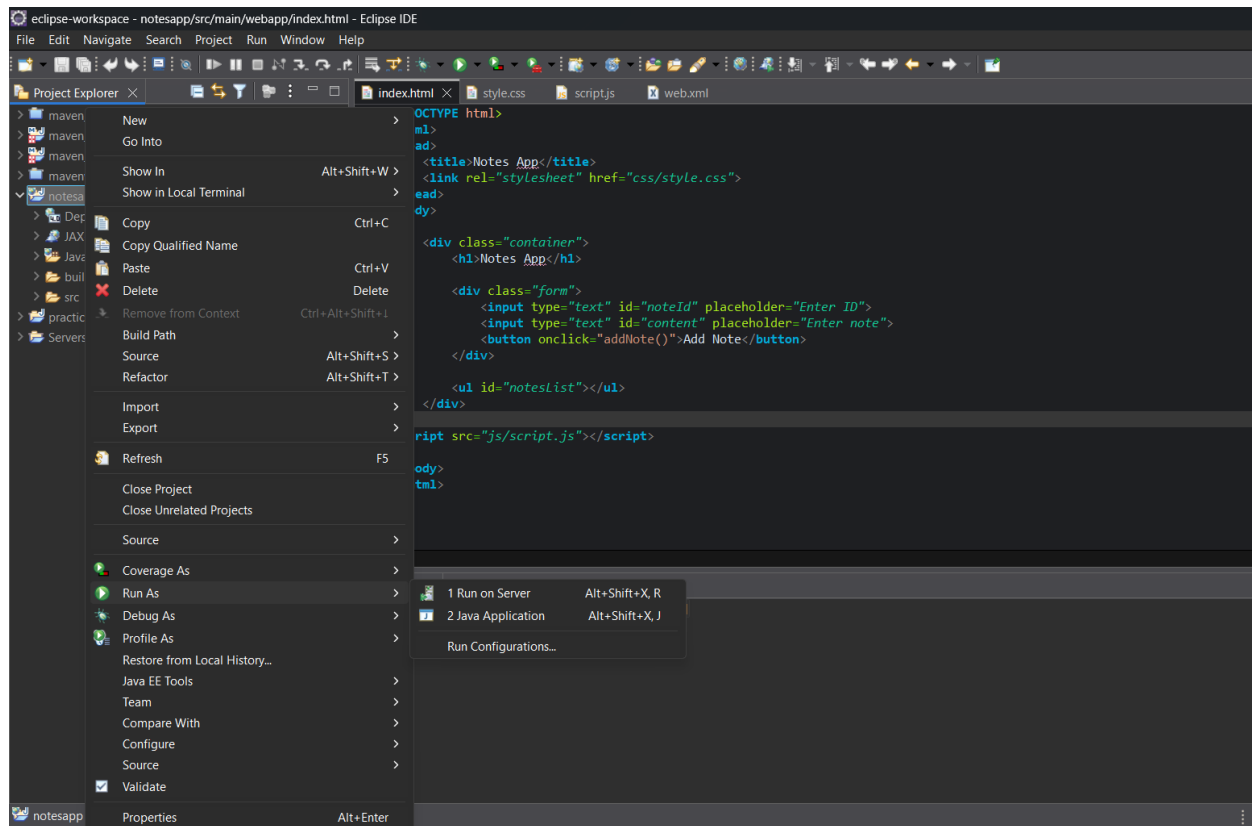


WEEK -10 : ELASTIC BEANSTALK

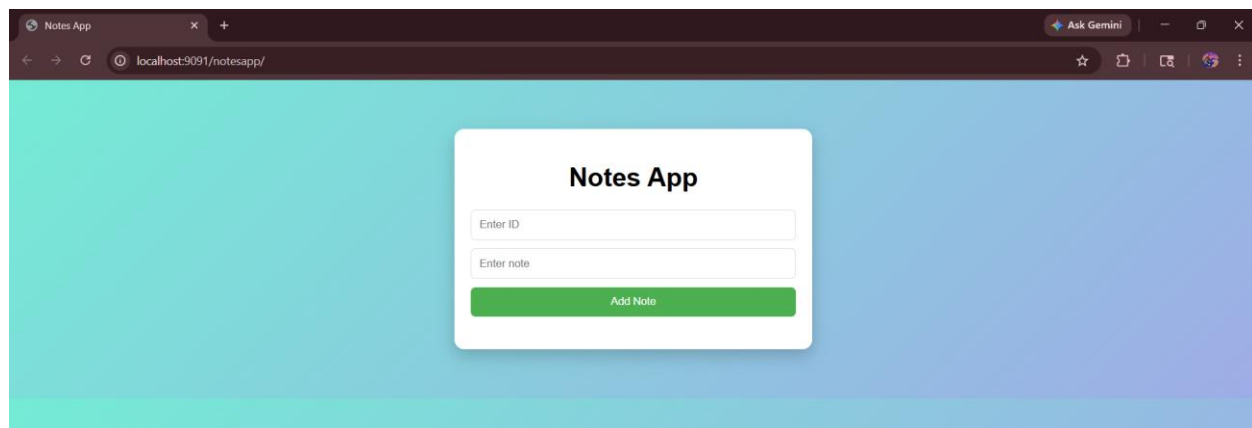
- Open Eclipse
- Go to: **File** → **New** → **Dynamic Web Project**
 - Project Name: NotesApp
 - Target Runtime: (Select Apache Tomcat)
- Click **Finish**



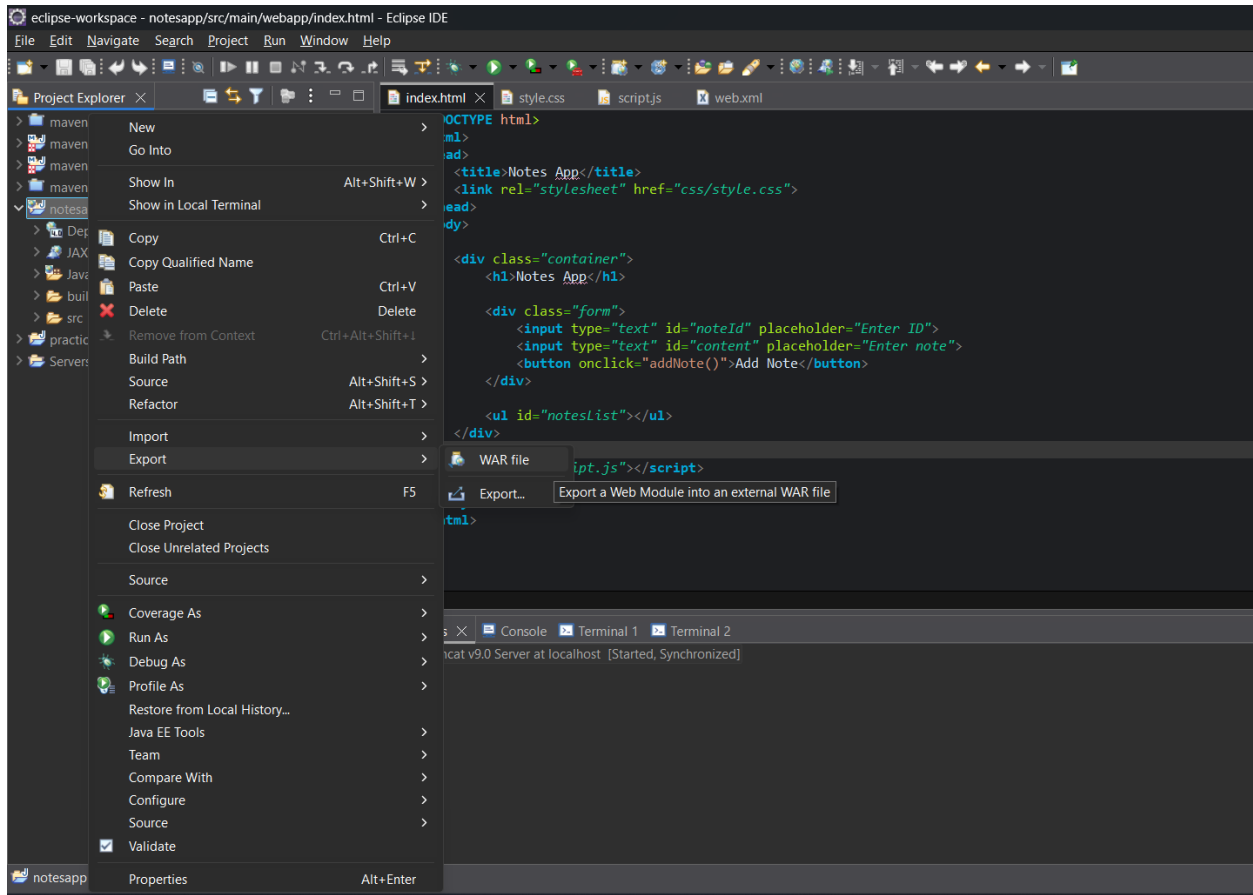
- Right-click project
- **Run As** → **Run on Server**
- Choose **Apache Tomcat**



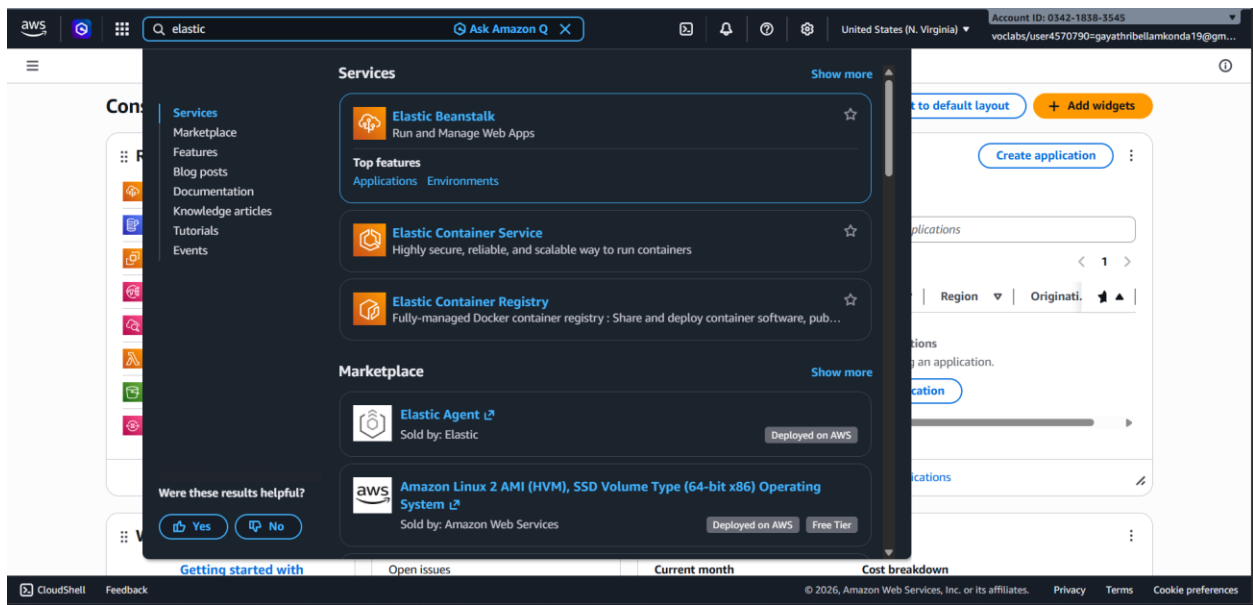
You should see your Notes App running locally.



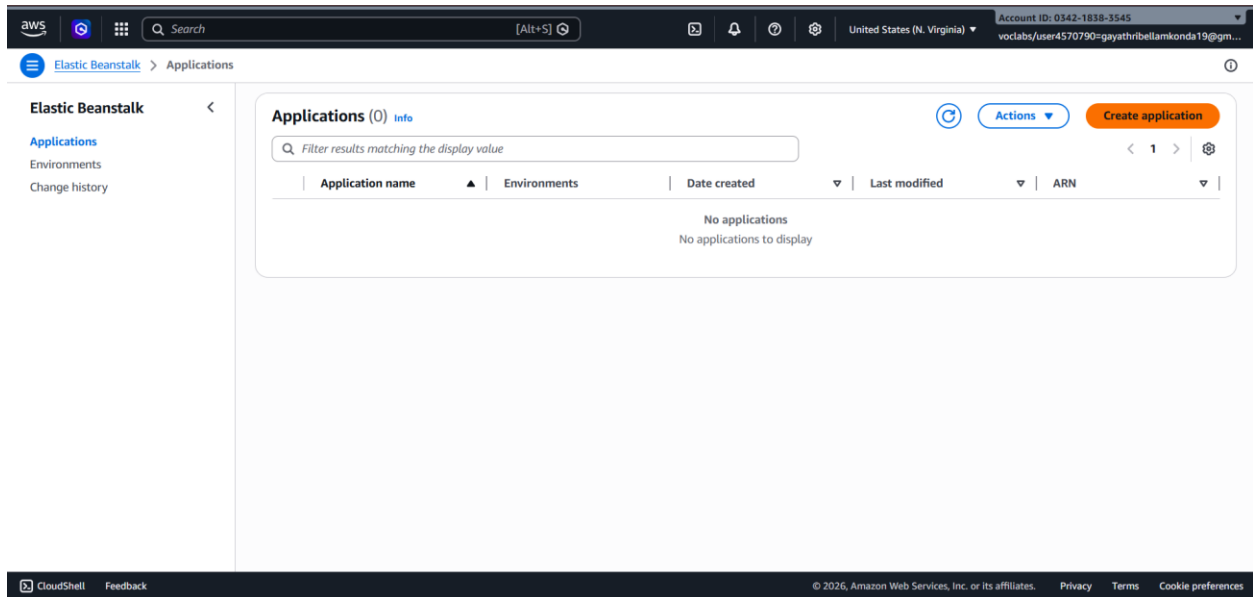
- Right-click project
- **Export** → **WAR file**
- Choose destination:



Open AWS console and search for Elastic Beanstalk



Click on create application



- Go to Elastic Beanstalk , Create a new application name it “Notes app”
- Add Description
- Click on create

Search

[Alt+S]

United States (N. Virginia)

Account ID: 0342-1838-3545
voclabs/user4570790=gayathribellamkonda19@gm...

Elastic Beanstalk

Create application

Elastic Beanstalk

Applications

Environments

Change history

Create new application

Application information

Application name

notes app

Maximum length of 100 characters.

Description

This is a simple notes app

Tags

Apply up to 50 tags. You can use tags to group and filter your resources. A tag is a key-value pair. The key must be unique within the resource and is case-sensitive. [Learn more](#)

No tags associated with the resource.

Add new tag

You can add up to 50 tags.

Cancel

Create

Application will be created

Search

[Alt+S]

United States (N. Virginia)

Account ID: 0342-1838-3545
voclabs/user4570790=gayathribellamkonda19@gm...

Elastic Beanstalk

Applications

Elastic Beanstalk

Applications

Environments

Change history

Applications (1)

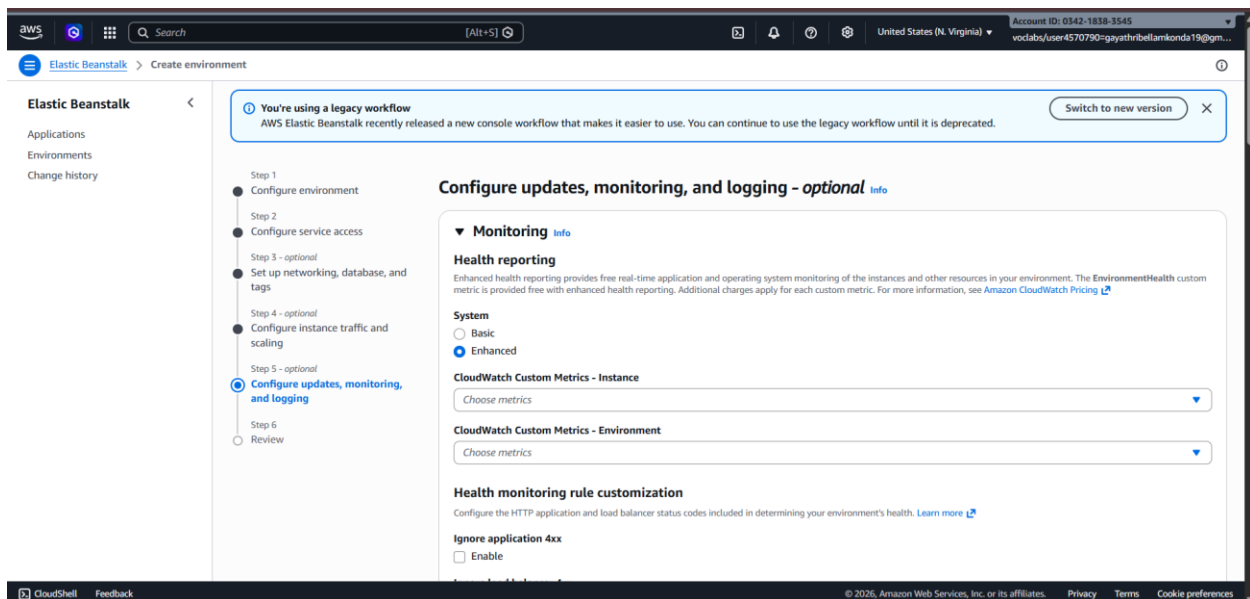
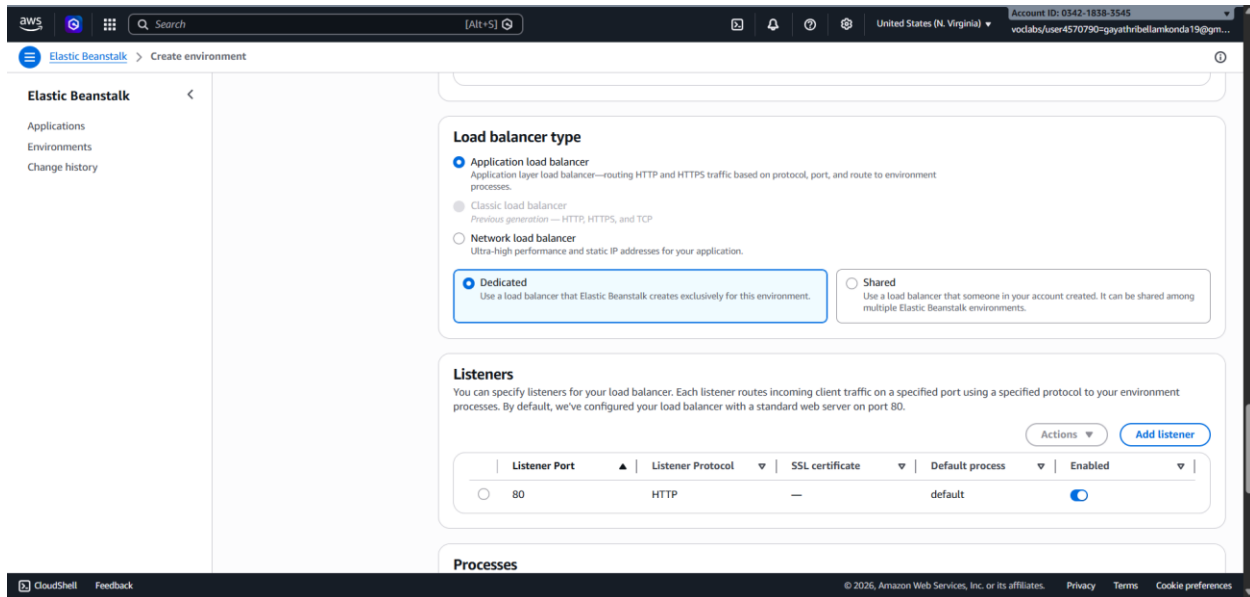
Filter results matching the display value

Application name	Environments	Date created	Last modified	ARN
notes app	-	March 23, 2026 10:54:49 (U...	March 23, 2026 10:54:49 (U...	arn:aws:elasticb...

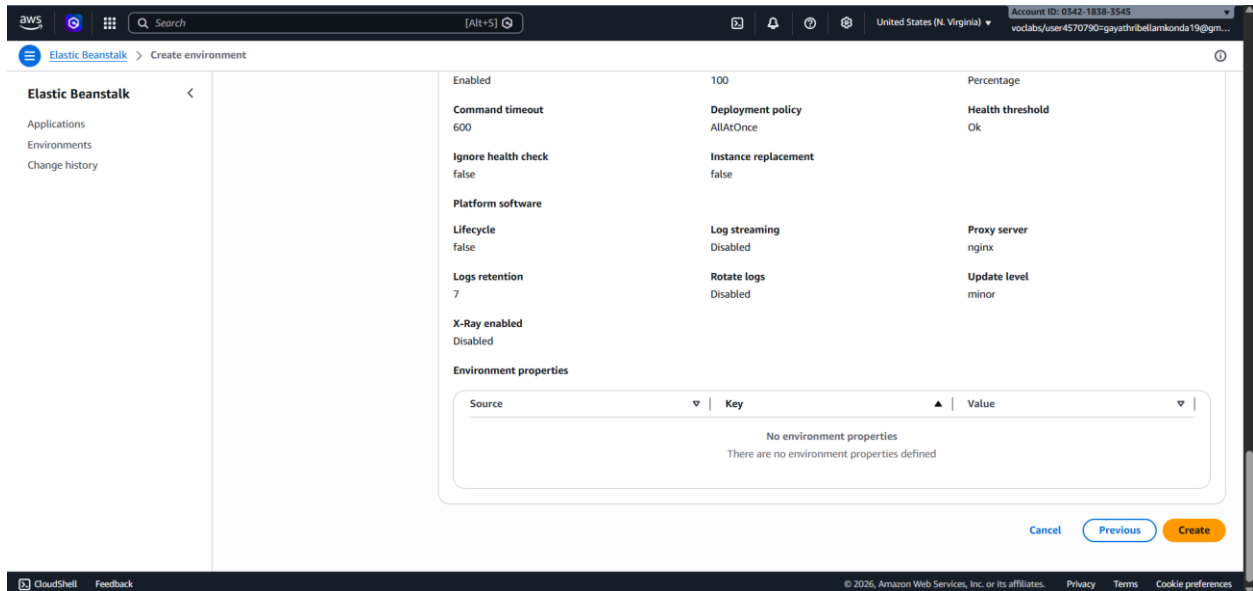
Create application

Click on the application -> click on create new environment

Keep every thing as default and click on next

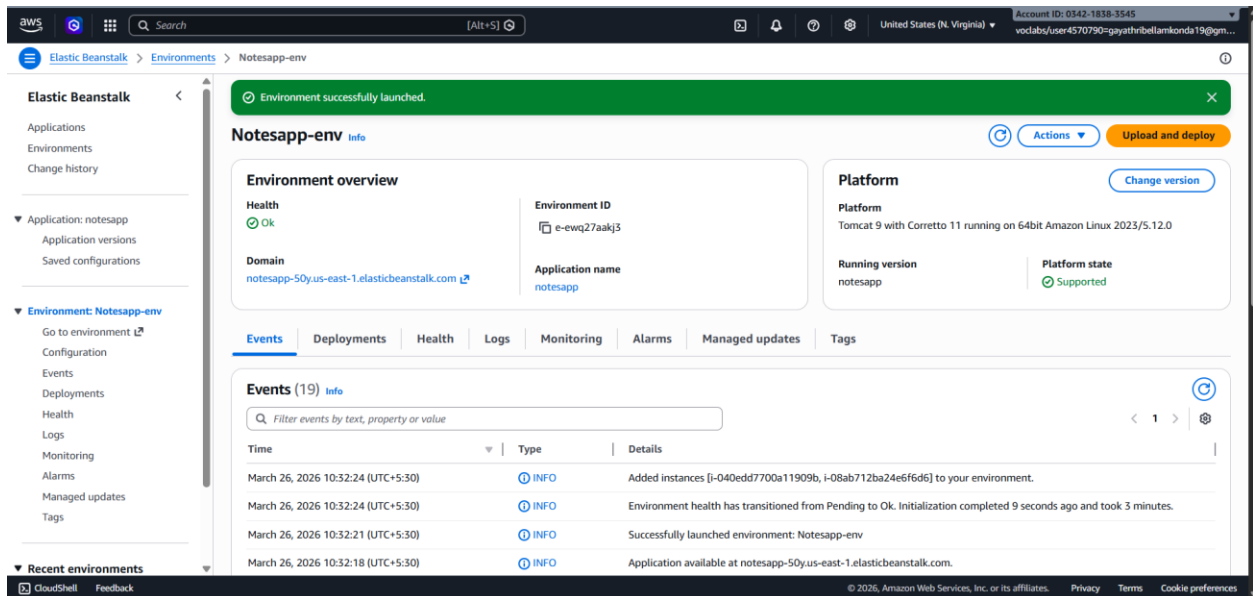


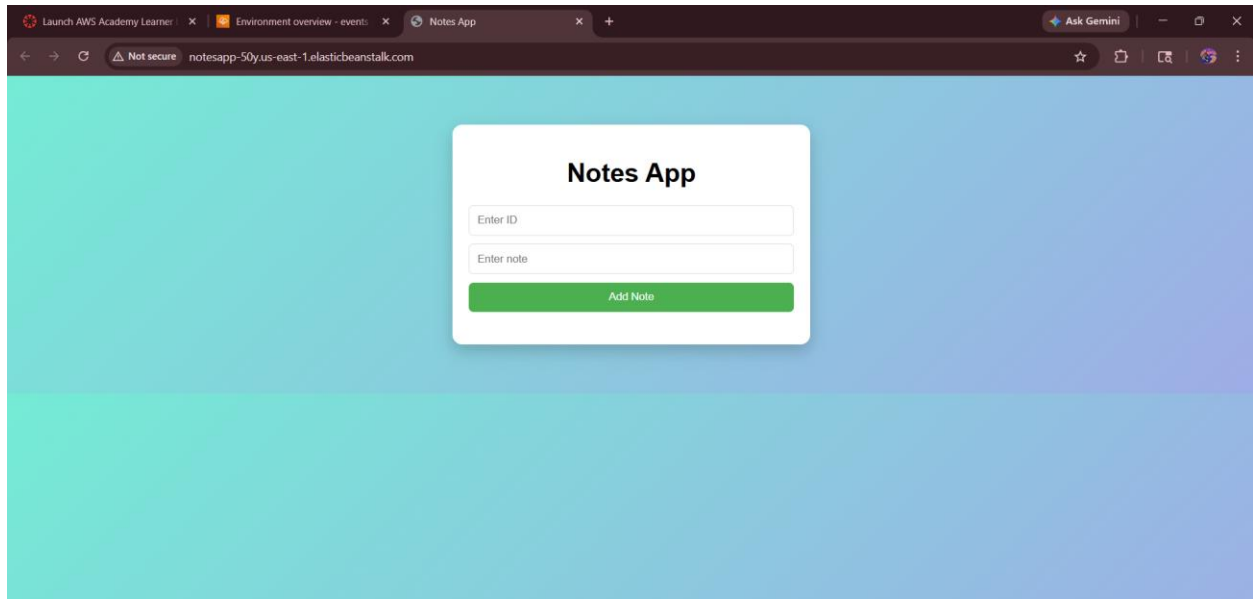
Review the environment and click on create



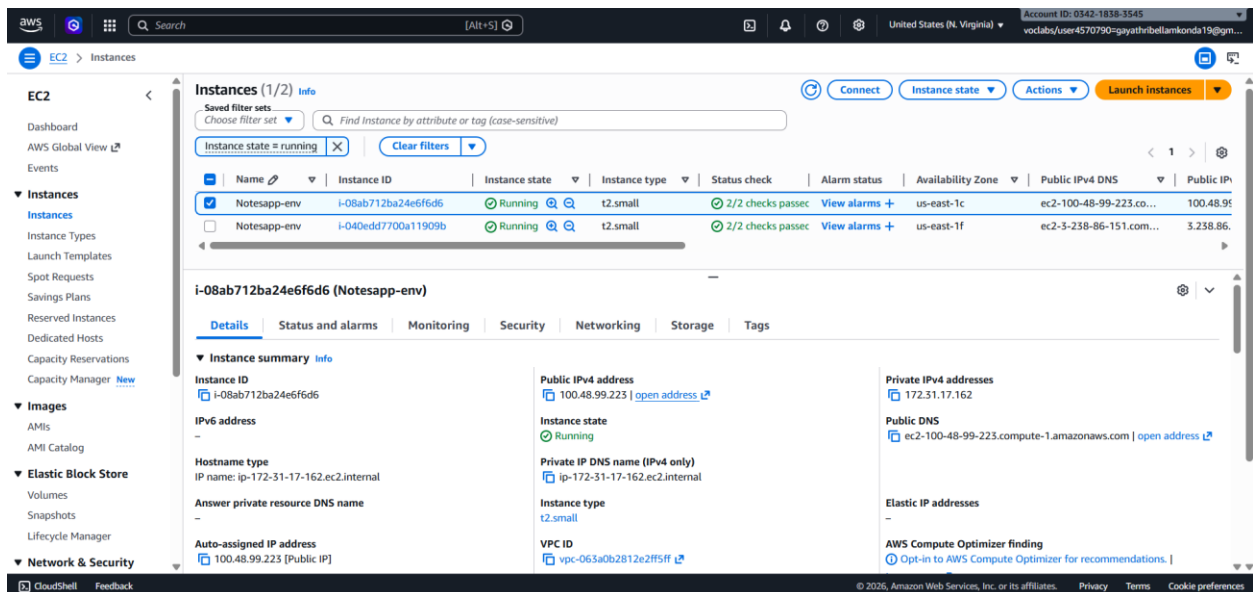
Environment will be successfully created after some time

Click on domain link it will navigate to the application output



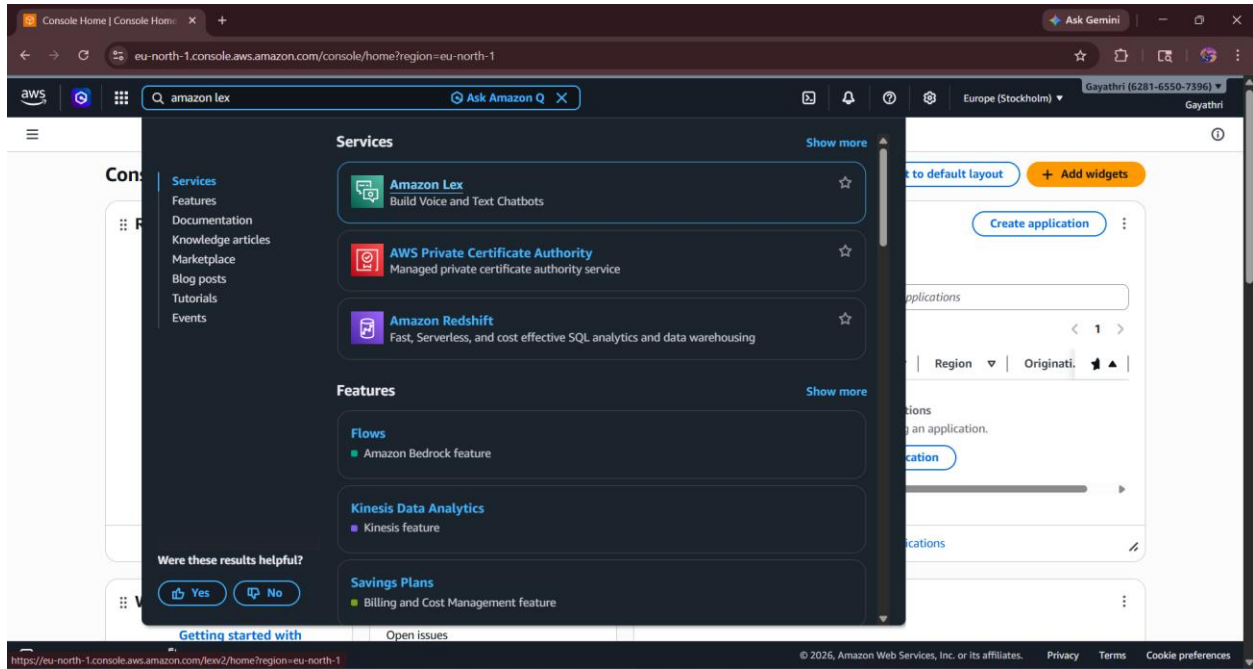


Instances will be automatically gets created

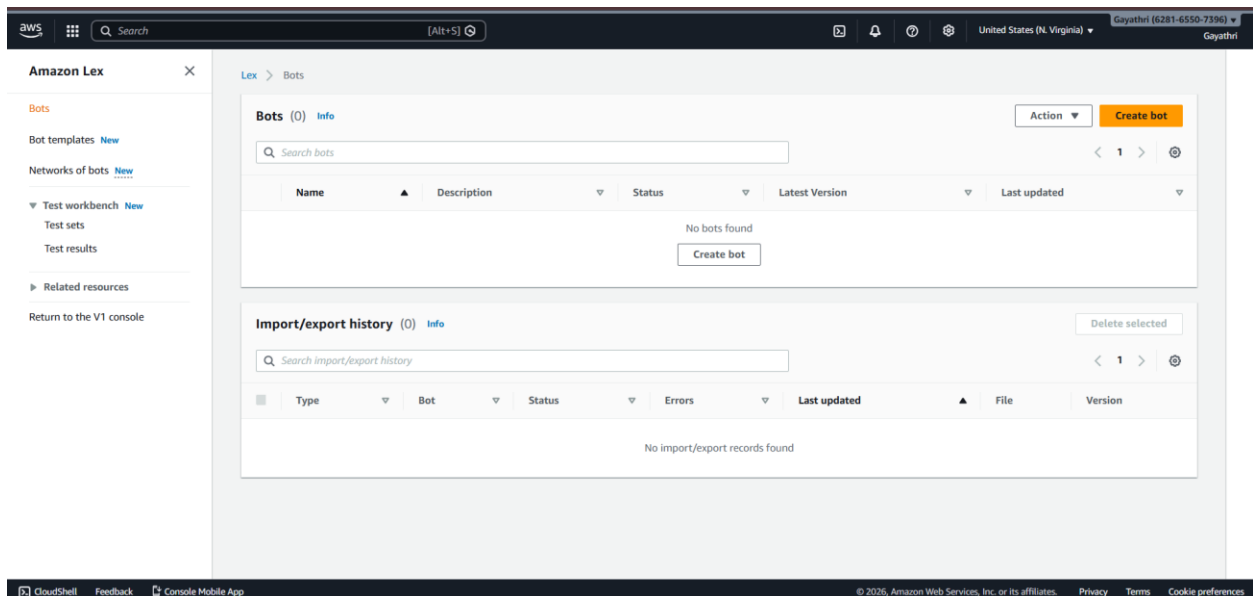


WEEK-11—Amazon LEX

Login to AWS Console -> Search **Amazon Lex**



Choose: **Create a blank bot**



Enter: Bot name: HotelBookingBot -> Description is optional

Step 1
Configure bot settings

Step 2
Add languages

Configure bot settings

Creation method

☒ **Create a blank bot**
Create a basic bot with no preconfigured languages, intents, and slot types.

☐ **Start with an example**
An example bot has preconfigured languages, intents, and slot types. You can change these settings.

☐ **Start with transcripts**
Automatically generate intents from conversation transcripts that you upload. Only English (US) language is available when starting with a transcript.

Bot configuration

Bot name

Maximum 100 characters. Valid characters: A-Z, a-z, 0-9, -, _

Description - optional
This description appears on bot list page. It can help you identify the purpose of your bot.

Maximum 2000 characters.

IAM role → Create new role

IAM permissions

IAM roles are used to access other services on your behalf.

Runtime role
Choose a role that defines permissions for your bot. To create a custom role, use the IAM console.

☒ **Create a role with basic Amazon Lex permissions.**

☐ Use an existing role.

Creating a role takes a few minutes. Don't delete the role or edit the trust or permissions policies in this role until we've finished creating it.

New role
Amazon Lex creates a runtime role with permission to upload to Amazon CloudWatch Logs.
AWSRoleForLexV2Bots_APAYVB06ZKD

Bot error logging
Debug unexpected issues on Lex bots.

Error logs
☐ Enabled
☒ Disabled
[Learn more about error logs](#)
[Go to error logs](#)

Remaining default → Click next

This screenshot shows the configuration page for an Amazon Lex bot named 'Gayathri' in the 'United States (N. Virginia)' region. The page includes several settings sections:

- Error logs:** A radio button interface with 'Disabled' selected. Links for 'Learn more about error logs' and 'Go to error logs' are provided.
- Children's Online Privacy Protection Act (COPPA):** A section with an 'info' link and a radio button interface where 'No' is selected, indicating the bot is not subject to COPPA.
- Idle session timeout:** A section explaining that the session duration can be configured. It features a text input with '5' and a dropdown menu set to 'minute(s)'. A note states: 'By default, session duration is 5 minutes, but you can specify any duration between 1 and 1440 minutes (24 hours).' Below this is a link for 'Advanced settings - optional'.

At the bottom of the configuration area are 'Cancel' and 'Next' buttons. The footer of the console shows '© 2026, Amazon Web Services, Inc. or its affiliates.' along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

Language: English -> Voice (optional)

Click **Done**

This screenshot shows the 'Add language to bot' configuration page for the 'Gayathri' bot. The left sidebar indicates 'Step 2: Add languages' is the current step. The main configuration area includes:

- Language: English (US):** A section header for the current language configuration.
- Select language:** A dropdown menu currently showing 'English (US)'.
- Description - optional:** A text input field with a note: 'Maximum 2000 characters.'
- Enable speech-to-speech inference:** An unchecked checkbox.
- Voice interaction:** A section with a note 'The text-to-speech voice that your bot uses to interact with users.' It includes a dropdown menu set to 'Danielle'.
- Voice sample:** A text input containing 'Hello, my name is Danielle. Let me know how I can assist you.' with a 'Play' button next to it.
- Speech model preference:** A section with a note 'The preferred speech model to use for speech-to-text' and a dropdown menu set to 'None'.
- Intent classification confidence score threshold:** A text input set to '0.40', with a note 'Min: 0.00, max: 1.00.'

At the bottom of the configuration area are 'Cancel', 'Add another language', and 'Done' buttons. The footer of the console shows '© 2026, Amazon Web Services, Inc. or its affiliates.' along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

Go to **Intents** -> Name: BookHotel

Click **Create**

Amazon Lex

Back to intents list (2)

Search

Sort by last updated

NewIntent (Unsaved)

FallbackIntent

Intent: NewIntent

An intent represents an action that fulfills a user's request. Intents can have arguments called slots that represent variable information.

Conversation flow

Intent details

Intent name

BookHotel

Maximum 100 characters. Valid characters: A-Z, a-z, 0-9, -, _

Display name

BookHotel

Recommended for Intent Disambiguation.

Maximum 100 characters. Valid characters: A-Z, a-z, 0-9, -, _

Intent and utterance generation description

Describe the purpose of your intent. This will also be used when generating utterances for your intent.

Maximum 2000 characters.

ID: CWERQ62HYO

Editor Visual builder New

Save intent

Add user sentences (how user talks):

I want to book a hotel ,Book a room ,Reserve hotel , I need a room

These are **Utterances**

Amazon Lex

Back to intents list (2)

Search

Sort by last updated

NewIntent (Unsaved)

FallbackIntent

Sample utterances (2)

What's this? Generate utterances

Representative phrases that you expect a user to speak or type to invoke this intent. Amazon Lex extrapolates based on the sample utterances to interpret any user input that may vary from the samples. The priority order of the sample utterances is not used to determine intent classification output.

Filter

Sort by added (ascending)

Preview Plain text

I want to book a hotel

Book a room

Reserve hotel

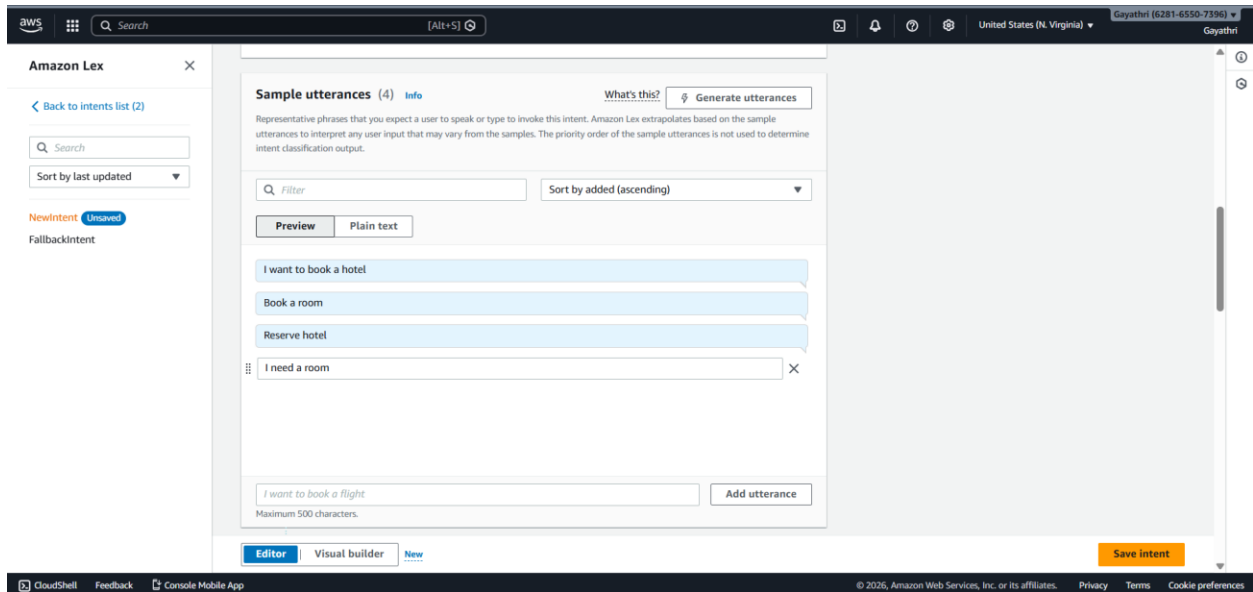
Add utterance

Maximum 500 characters.

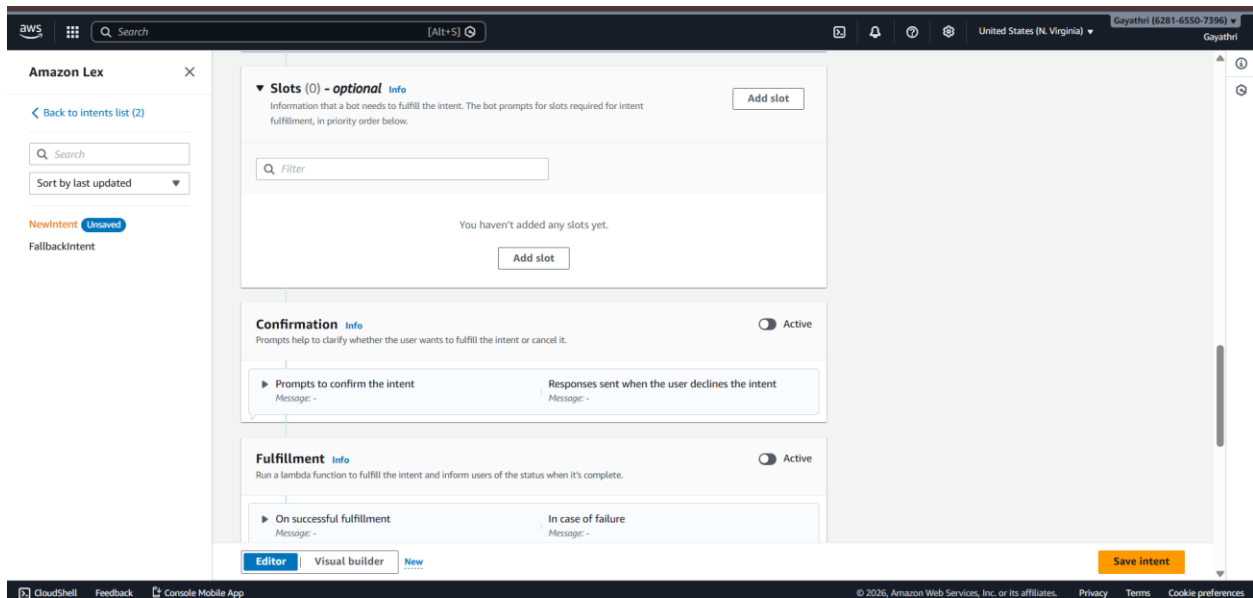
Editor Visual builder New

Save intent

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Click **Add slot**



Fill: Slot name: age , Slot type: AMAZON.Number , Prompt: *Which city do you want?*

Mark as **Required** -> click on **Add**

×

☒ Required for this intent

Name

Slot type

1

Which city do you want?

Add

Click on Age slot

below.

or the use

e intent

pt the us

r request

er says M

ot be submitted.

Add slot

A slot is used to capture information from the user to fulfill the intent.

☒ Required for this intent
The bot will prompt for this slot during the conversation if a value is not provided by the user.

Name

Slot type

Prompts

aws

Search [Alt+S]

United States (N. Virginia)

Gayathri (6281-6550-7296)

Gayathri

Amazon Lex

[Back to intents list \(2\)](#)

Search

Sort by last updated

BookHotel (Unsaved)

FallbackIntent

▼ Slots (4) - optional info

Information that a bot needs to fulfill the intent. The bot prompts for slots required for intent fulfillment, in priority order below.

Filter

▶ Prompt for slot: Age Message: Which city do you want?	Slot type AMAZON.Number	×
▶ Prompt for slot: Location Message: Which city do you want?	Slot type AMAZON.City	×
▶ Prompt for slot: Checkin Message: What is your check-in date	Slot type AMAZON.Date	×
▶ Prompt for slot: Nights Message: How many nights will you stay?	Slot type AMAZON.Number	×

Click on back to instance list

The screenshot shows the Amazon Lex console interface. The left sidebar contains navigation options: Bots, Bot templates, Networks of bots, HotelBookingBot, Bot versions, Draft version, All languages, English (US), Intents, Slot types, Deployment, Aliases, Channel integrations, Analytics, Overview, Conversation dashboard, Conversation flows, Conversations, and Performance dashboard. The main content area is titled 'Intents (2)' and includes a search bar, a table of intents, and buttons for 'Delete', 'Add intent', 'Analyze', 'Build', and 'Test'. The table lists two intents: 'BookHotel' and 'FallbackIntent'.

Name	Description	Last edited
BookHotel	-	2 minutes ago
FallbackIntent	Default intent when no other intent matches	22 minutes ago

Click on slot types -> click on Add slot type -> click on Add blank slot type

The screenshot shows the Amazon Lex console interface for the 'Slot types' page. The left sidebar is the same as the previous screenshot. The main content area is titled 'Slot types (0)' and includes a search bar, a table of slot types, and buttons for 'Delete', 'Add slot type', 'Analyze', 'Build', and 'Test'. The 'Add slot type' button is expanded, showing options: 'Add blank slot type', 'Use built-in type', 'Add grammar slot type', and 'Add composite slot type'. The table is empty, with a message stating 'No slot types found'.

Name	Description	Type	Last edited
------	-------------	------	-------------

Click on Add values and add “single” ,”double” , “suite” values

The screenshot shows the Amazon Lex console interface. On the left, the 'Amazon Lex' sidebar is visible with a search bar and a 'Slot types (1)' section. The main content area is titled 'Slot type values' and includes a search bar for slot type values. Below the search bar, there are two input fields: 'Single Double Suite' and 'Double', each with a close button (X). A third input field labeled 'Suite' is shown with a placeholder text: 'Maximum 140 characters. Valid characters: A-Z, a-z, 0-9, @, #, \$'. To the right of this field is an 'Add value' button. At the bottom, there is a checkbox labeled 'Use slot values as custom vocabulary' with an 'Info' link.

Click on bookhotel

The screenshot shows the Amazon Lex console interface for the 'HotelBookingBot'. The left sidebar shows the 'Intents' section under 'Bot versions'. The main content area is titled 'Intents (2)' and includes a search bar for intents. Below the search bar, there is a table with columns 'Name', 'Description', and 'Last edited'. The table contains two rows: 'BookHotel' and 'Fallbackintent'. The 'BookHotel' row is selected, and its description is '-'. The 'Fallbackintent' row has a description of 'Default intent when no other intent matches'. To the right of the table, there are buttons for 'Delete' and 'Add intent'. The top of the console shows the 'Draft version' and 'English (US)' language settings, along with a 'Not built' status.

Name	Description	Last edited
BookHotel	-	4 minutes ago
Fallbackintent	Default intent when no other intent matches	24 minutes ago

- Go to **Slot** -> **click on add slot**
- Slot name: RoomType
- Slot type: RoomType

Add slot

A slot is used to capture information from the user to fulfill the intent.

☒ **Required for this intent**
The bot will prompt for this slot during the conversation if a value is not provided by the user.

Name: RoomType Slot type: RoomType

Prompts: What type of room would you like?

Cancel Add

Click on advance options

Amazon Lex

Back to intents list (2)

Search

Sort by last updated

BookHotel

FallbackIntent

Slots (5) - optional

Information that a bot needs to fulfill the intent. The bot prompts for slots required for intent fulfillment, in priority order below.

Filter

Prompt for slot: RoomType

Message: What type of room would you like?

☒ **Required for this intent**
The bot will prompt for this slot during the conversation if a value is not provided by the user.

Name: RoomType Slot type: RoomType

Prompts: What type of room would you like?

You can use the advanced options setting to configure rich messages such as a custom payload, card groups, and SSML.

Advanced options

Confirmation

Prompts help to clarify whether the user wants to fulfill the intent or cancel it.

Active

Editor Visual builder New

Save intent

Go to slot prompts -> more prompt options

The screenshot displays the Amazon Lex console interface. On the left, the 'Amazon Lex' sidebar is visible with a search bar and a list of intents: 'BookHotel' and 'FallbackIntent'. The main content area is divided into two panels. The left panel, titled 'Slots (5) - optional', shows a list of slots. The right panel, titled 'Slot: RoomType', provides detailed configuration options for the 'RoomType' slot. In this panel, the 'Required for this intent' checkbox is checked. The 'Name' field is set to 'RoomType'. The 'Prompts' section contains the text 'What type of room would you like?'. Below the prompts, there is a section for 'Variations - optional' with a button labeled 'More prompt options'.

Amazon Lex

Back to intents list (2)

Search

Sort by last updated

BookHotel

FallbackIntent

Slots (5) - optional

Filter

Prompt for slot: RoomType

Message: What type of room would you like?

Slot type: RoomType

☒ Required for this intent

The bot will prompt for this slot during the conversation if a value is not provided.

Name: RoomType

Prompts: What type of room would you like?

You can use the advanced options setting to configure rich messages such as cards and images.

Advanced options

Slot: RoomType

Slot prompts

Prompts to elicit the slot.

Bot elicits information

Message: What type of room would you like?

☐ Play the messages in order

Messages will be used in the predefined order as slot prompts by your bot.

Message group

You can define a text message group to respond using plain text.

Message: What type of room would you like?

Variations - optional

More prompt options

Add custom payloads, SSML, and card groups.

Click **Add** -> add card groups

The screenshot displays the Amazon Lex console interface. On the left, the 'Amazon Lex' sidebar shows a 'Back to intents list (2)' link and a search bar. Below the search bar, there are two intents listed: 'BookHotel' and 'FallbackIntent'. The main content area is divided into two panels. The left panel, titled 'Slots (5) - optional', shows a list of slots. The 'RoomType' slot is selected, and its details are shown. The 'RoomType' slot has a prompt 'What type of room would you like?' and is marked as 'Required for this intent'. The right panel, titled 'Slot: RoomType > Slot prompts editor', shows the 'Prompts settings' section. It includes checkboxes for 'Users can interrupt the prompt when it is being read' (checked), 'Play the messages in order' (unchecked), and 'Invoke Lambda code hook after each elicitation' (checked). Below these settings, there is a text input field for 'Invocation label - optional' and a numeric input field for 'Maximum number of retries' set to 4. At the bottom of the right panel, there is a 'Slot prompts' section with a toggle for 'Preview' and an 'Add' button. A dropdown menu is open from the 'Add' button, showing options: 'Add text message group', 'Add card group', 'Add custom payload', and 'Add SSML language'. The footer of the console shows the AWS logo, search bar, and various utility links like 'CloudShell', 'Feedback', and 'Console Mobile App'.

Amazon Lex

Back to intents list (2)

Search

Sort by last updated

BookHotel

FallbackIntent

Slots (5) - optional

Filter

Prompt for slot: RoomType

Message: What type of room would you like?

Slot type: RoomType

Required for this intent

Name: RoomType

Prompts

What type of room would you like?

Advanced options

Confirmation

Editor Visual builder New

Slot: RoomType > Slot prompts editor

Prompts settings

Additional settings for how your bot uses the prompts.

☒ Users can interrupt the prompt when it is being read

☐ Play the messages in order

☒ Invoke Lambda code hook after each elicitation

Invocation label - optional

Maximum number of retries

4

Slot prompts

Prompts to elicit the slot.

☐ Preview

Add

Add text message group

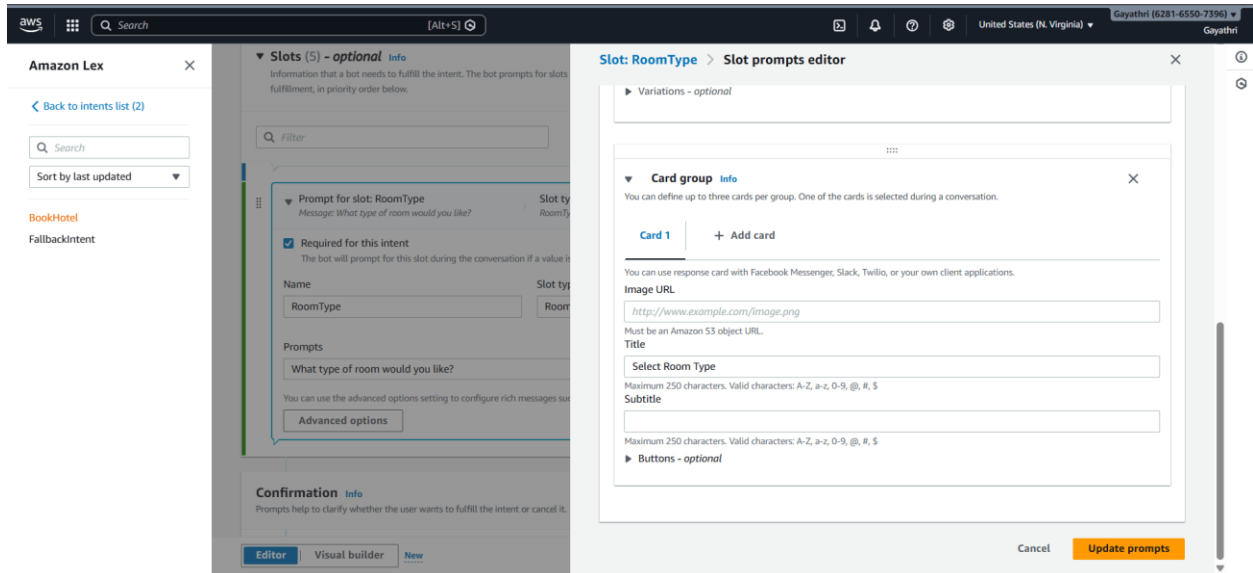
Add card group

Add custom payload

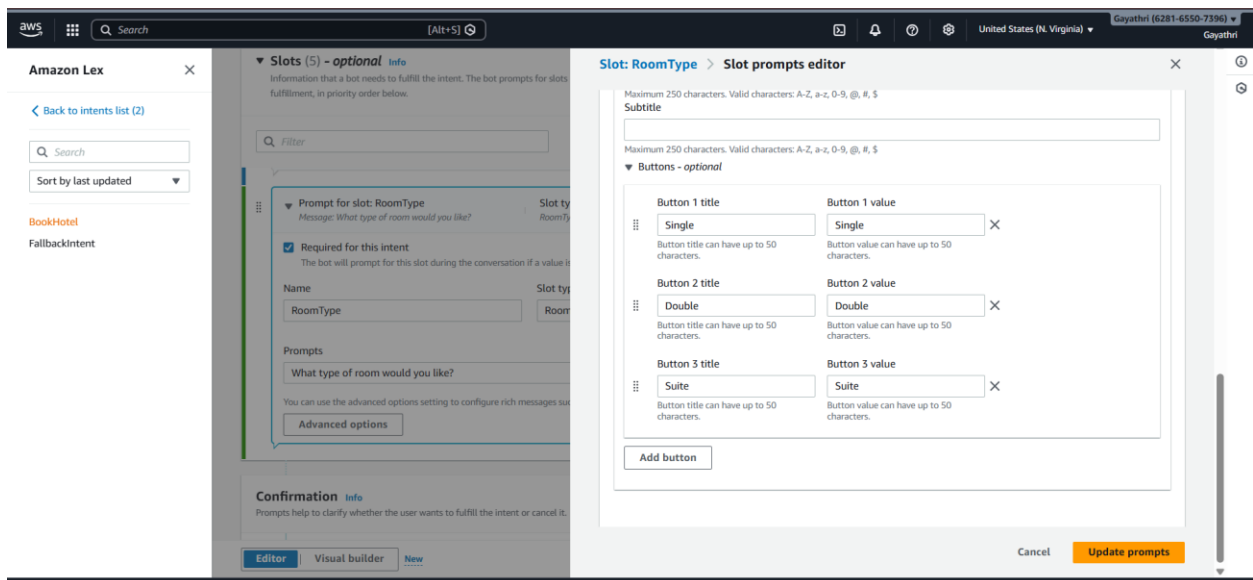
Add SSML language

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Add title -> click on buttons



Add titles and values -> update prompts



Initial Response:

Welcome to Hotel Booking! What is your name?

The screenshot shows the Amazon Lex console interface for configuring the 'Initial response' for the 'BookHotel' intent. The left sidebar contains the 'Amazon Lex' header, a search bar, a 'Back to intents list (2)' link, and a list of intents including 'BookHotel' (marked as 'Unsaved') and 'FallbackIntent'. The main content area is titled 'Initial response' and includes an 'Info' icon. It contains a section for 'Response to acknowledge the user's request' with a message field containing 'Welcome to Hotel Booking! What is your name?'. Below this is a 'Message group' section with a 'Message - optional' field containing the same text. There are also 'Variations - optional' and 'Advanced options' sections. The top of the console shows the AWS logo, a search bar, and the account name 'Gayathri (6281-6550-7396)'.

Confirmation:

Do you want to confirm booking in {location} for {nights} nights?

The screenshot shows the Amazon Lex console interface for configuring the 'Confirmation' and 'Fulfillment' for the 'BookHotel' intent. The left sidebar is the same as the previous screenshot. The main content area is titled 'Confirmation' and includes an 'Info' icon and an 'Active' toggle. It contains a section for 'Prompts to confirm the intent' with a message field containing 'Do you want to confirm booking in {Location} for {Nights} nights?'. Below this is a 'Decline response' section with a message field containing 'Okay. Your request will not be submitted.' and an 'Advanced options' button. The 'Fulfillment' section is also visible, with an 'Active' toggle and a message field for 'On successful fulfillment'. The bottom of the console shows the 'Editor' tab selected, and a 'Save intent' button. The footer of the console shows the AWS logo, a search bar, and the account name 'Gayathri (6281-6550-7396)'.

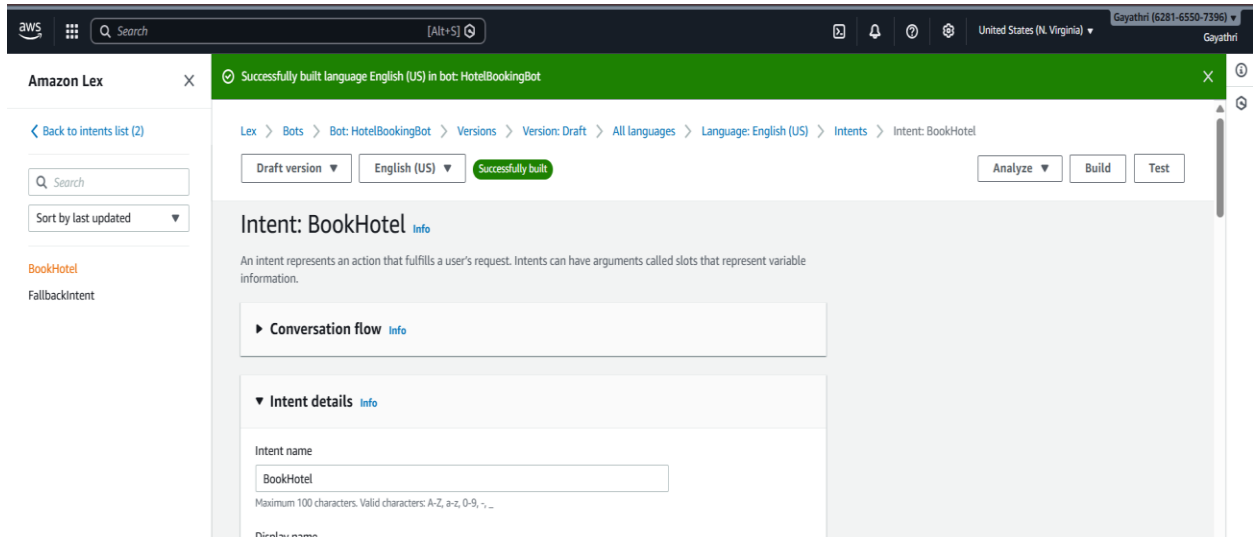
Add closing response (optional) -> click on save intent

The screenshot shows the Amazon Lex console interface. On the left, there's a sidebar with 'Amazon Lex' and a search bar. The main area is titled 'Closing response' and includes an 'Active' toggle. It contains sections for 'Response sent to the user after the intent is fulfilled' (with a message 'Your hotel has been booked successfully!'), 'Message group' (with a text input), 'Variations - optional' (with a 'More response options' button), 'Set values' (with a 'Next step in conversation' dropdown), and 'Code hooks - optional' (with a checkbox for 'Use a Lambda function for initialization and validation'). A 'Save intent' button is at the bottom right.

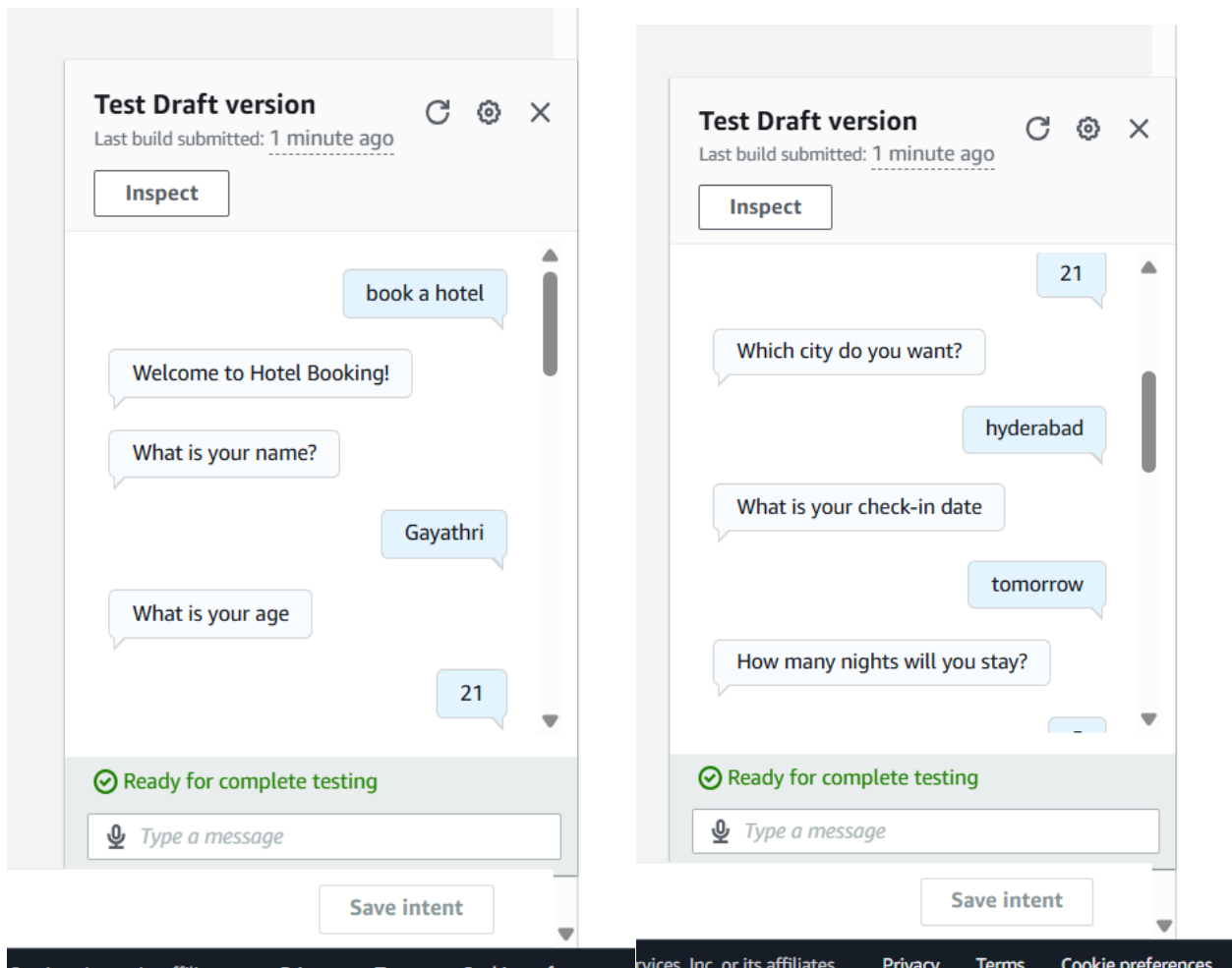
Click on build

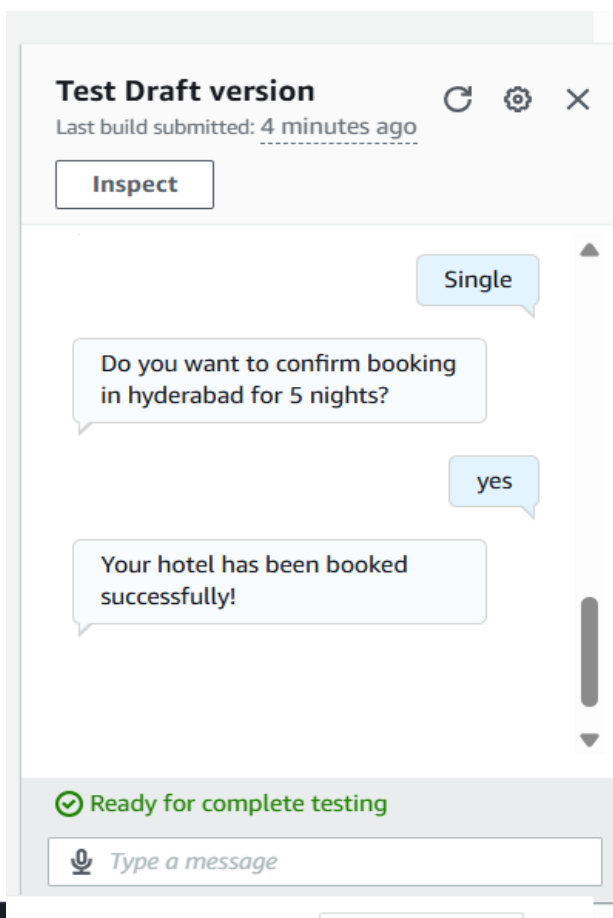
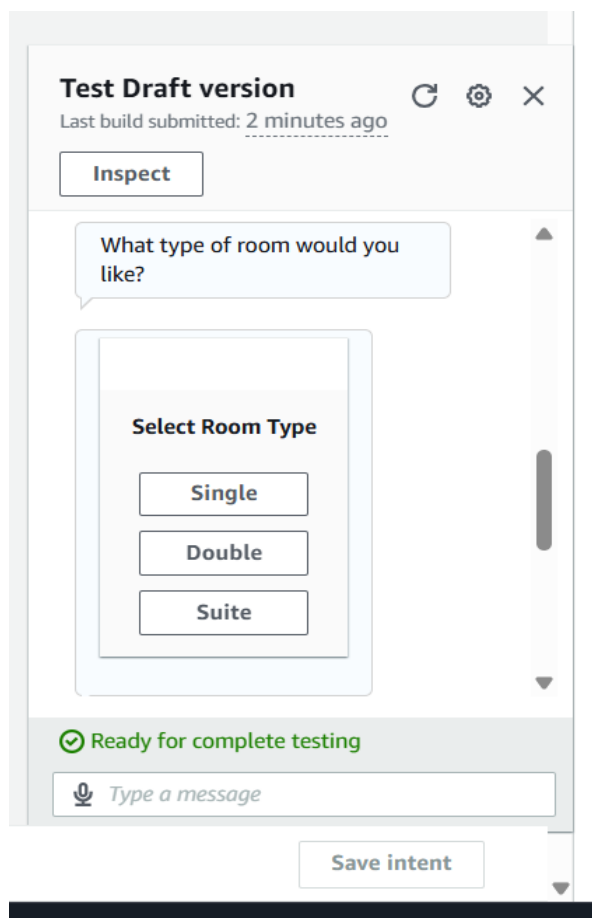
The screenshot shows the Amazon Lex console interface for the 'Intent: BookHotel'. The breadcrumb trail is 'Lex > Bots > Bot: HotelBook... > Versions > Version: DRAFT > All languages > Language: English (US) > Intents > Intent: BookHotel'. The page has tabs for 'Draft version', 'English (US)', and 'Not built'. There are buttons for 'Analyze', 'Build', and 'Test'. The main content area is titled 'Intent: BookHotel' and includes a description: 'An intent represents an action that fulfills a user's request. Intents can have arguments called slots that represent variable information.' It has sections for 'Conversation flow', 'Intent details' (with fields for 'Intent name' and 'Display name'), and 'Intent and utterance generation description'. A 'Save intent' button is at the bottom right.

After the build is successful -> click on test



Test the chatbot





Test Draft version

Last build submitted: 9 minutes ago



Inspect

Gayathri

What is your age

12

Your not eligible for hotel
booking.

✓ Ready for complete testing



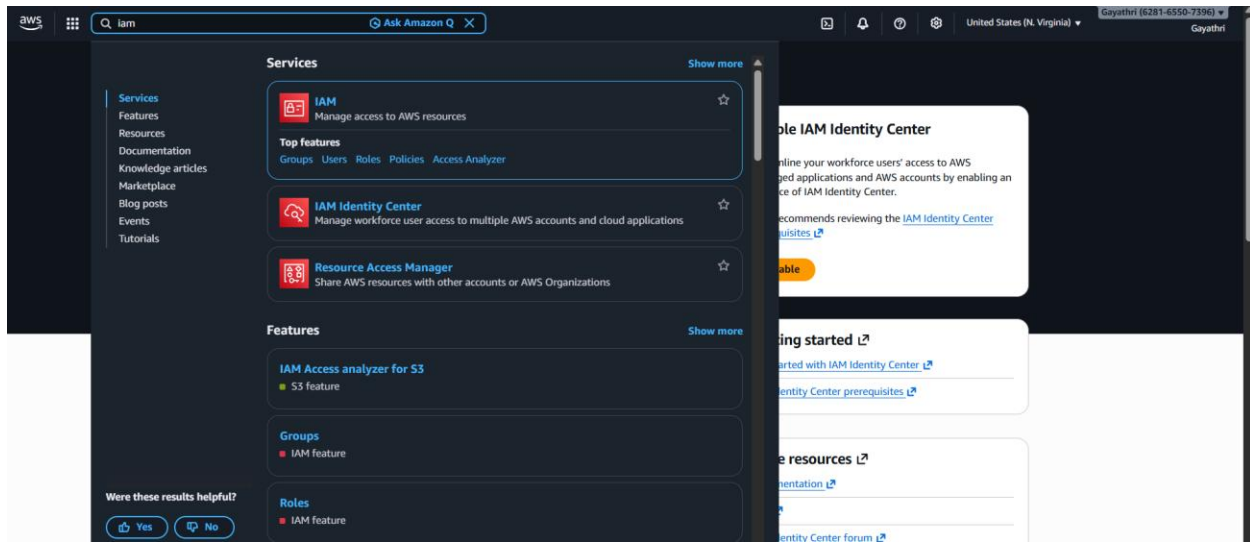
Type a message

Save intent

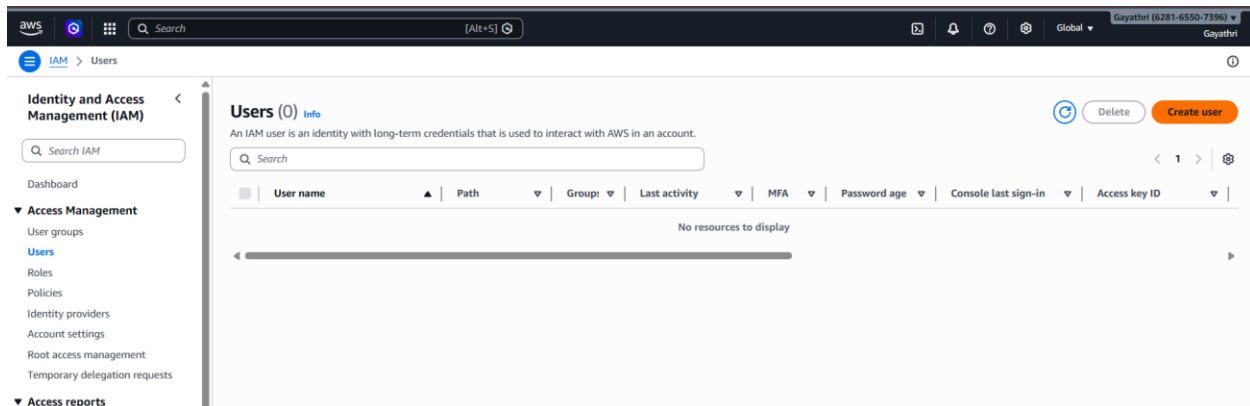
WEEK-12 –Creating IAM USER with permission Access it with GUI and CLI

Log in to the AWS Management Console as the **Root User**.

Search for **IAM** in search bar.



Navigate to Users and click Create user.



- User Details: Enter a name .Check the box for "Provide user access to the AWS Management Console." – custom password.
- Click next

Specify user details

User details

User name

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and +, =, ., @, _ (hyphen)

☒ **Provide user access to the AWS Management Console - optional**
In addition to console access, users with `SigInLocalDevelopmentAccess` permissions can use the same console credentials for programmatic access without the need for access keys.

Console password

☐ **Autogenerated password**
You can view the password after you create the user.

☒ **Custom password**
Enter a custom password for the user.

☒ **Show password**

☒ **Users must create a new password at next sign-in - Recommended**
Users automatically get the `IAMUserChangePassword` policy to allow them to change their own password.

❗ If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user.
[Learn more](#)

[Cancel](#) [Next](#)

Set Permissions: Select "Attach policies directly." Search for and select AmazonS3FullAccess.

Click Next

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☐ **Add user to group**
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ **Copy permissions**
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1484)
Choose one or more policies to attach to your new user.

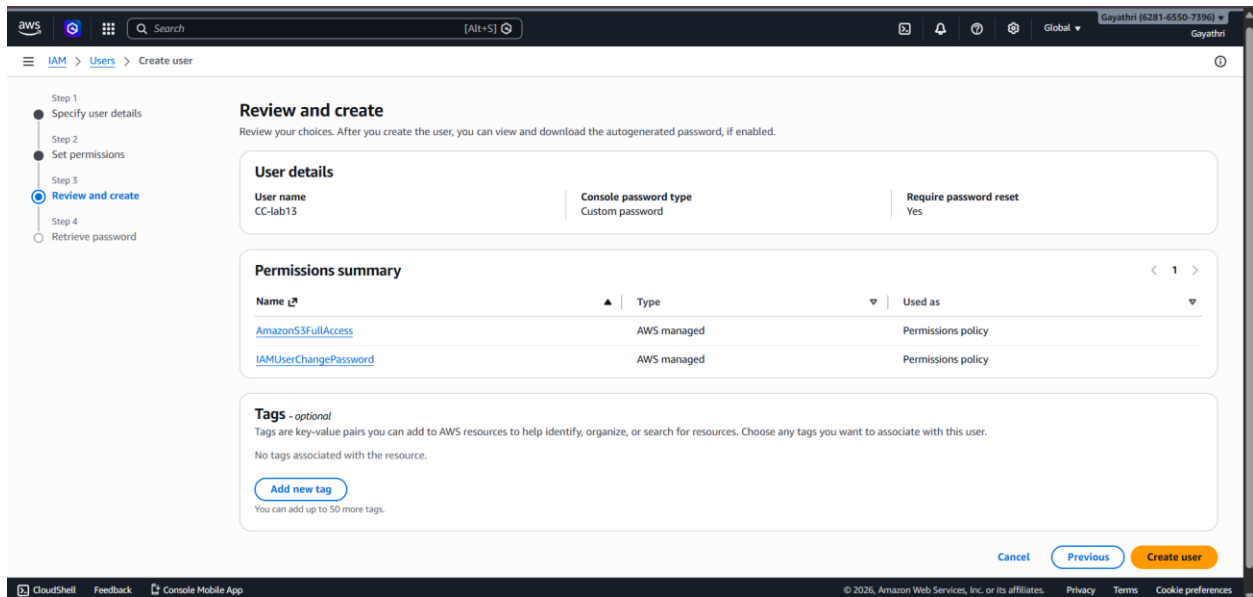
Filter by Type
 1 match

☒	Policy name	Type	Attached entities
☒	AmazonS3FullAccess	AWS managed	0

▶ Set permissions boundary - optional

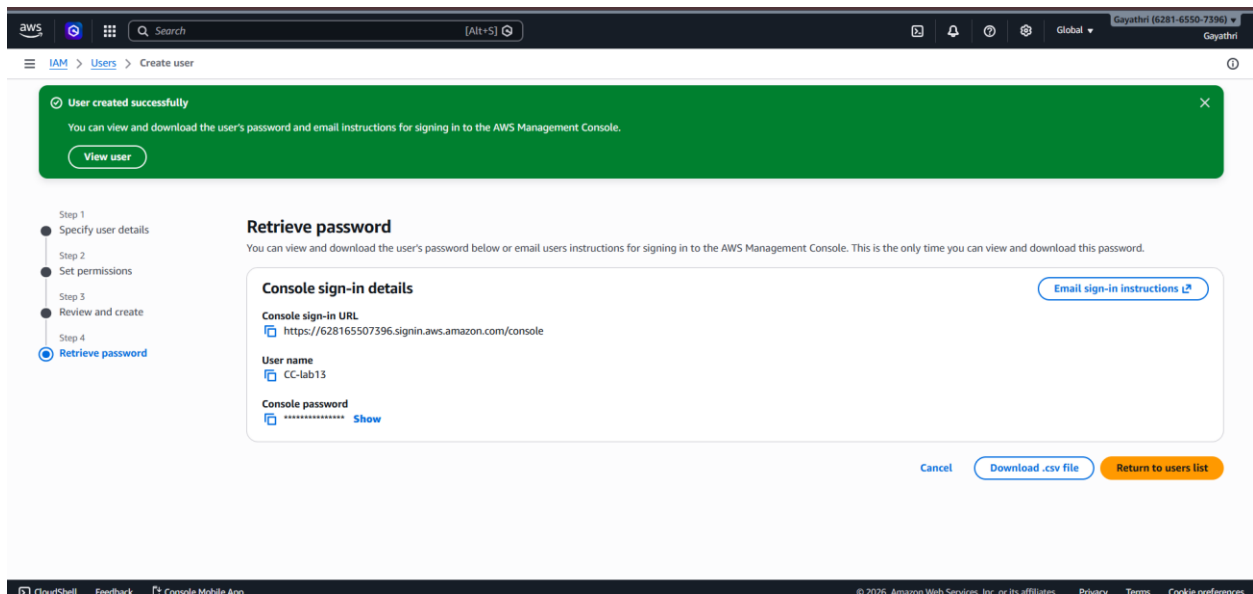
[Cancel](#) [Previous](#) [Next](#)

- **Review & Create:** Complete the wizard. On the final page, click Download .csv file.



Click on download .csv file

Copy your 12-digit Account ID (found in the top-right dropdown or the .csv) .





File Home Insert Page Layout Formulas Data Review View

Clipboard: Cut, Copy, Paste, Format Painter

Font: Calibri, 11, Bold, Italic, Underline, Text Color, Background Color

Alignment: Wrap Text, Merge & Center

Number: General, Currency, Percentage, Decimals, Fractions

Conditional Formatting, Format as Table

	A	B	C	D	E	F	G	H
1	User name	Password	Console sign-in URL					
2	CC-lab13	Gayathri@08	https://628165507396.signin.aws.amazon.com/console					
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								

Sign out the root account

Global ▾

Gayathri (6281-6550-7396) ▲
Gayathri

AWS Management Console.

/ or email users instructions for signing in to the AWS Management Console. This is the only time

m/console

Free plan status

Credits remaining
\$99.95 USD

Days remaining
160 days

Your free access to AWS services will end on Sep 26, 2026 or when you have depleted all credits. To ensure uninterrupted AWS access, see [upgrading your plan](#) for details.

Account ID
📄 6281-6550-7396

Account name
📄 Gayathri

Account color
● Unset

Account

Organization

Service Quotas

Billing and Cost Management

Security credentials

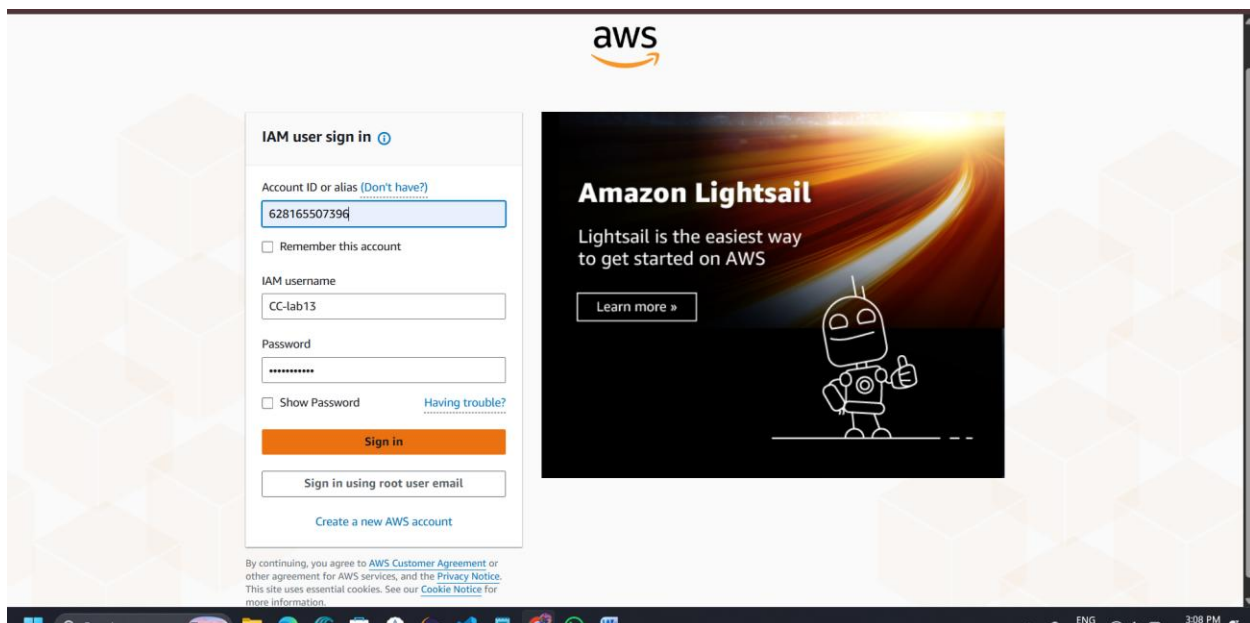
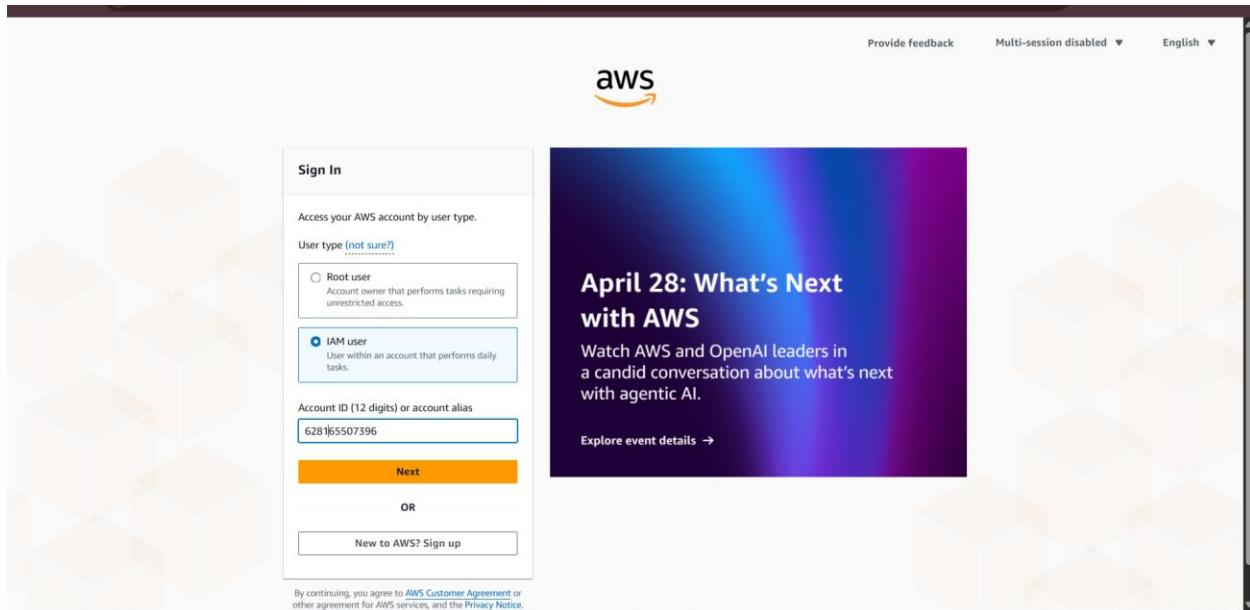
Console Mobile App

Cancel

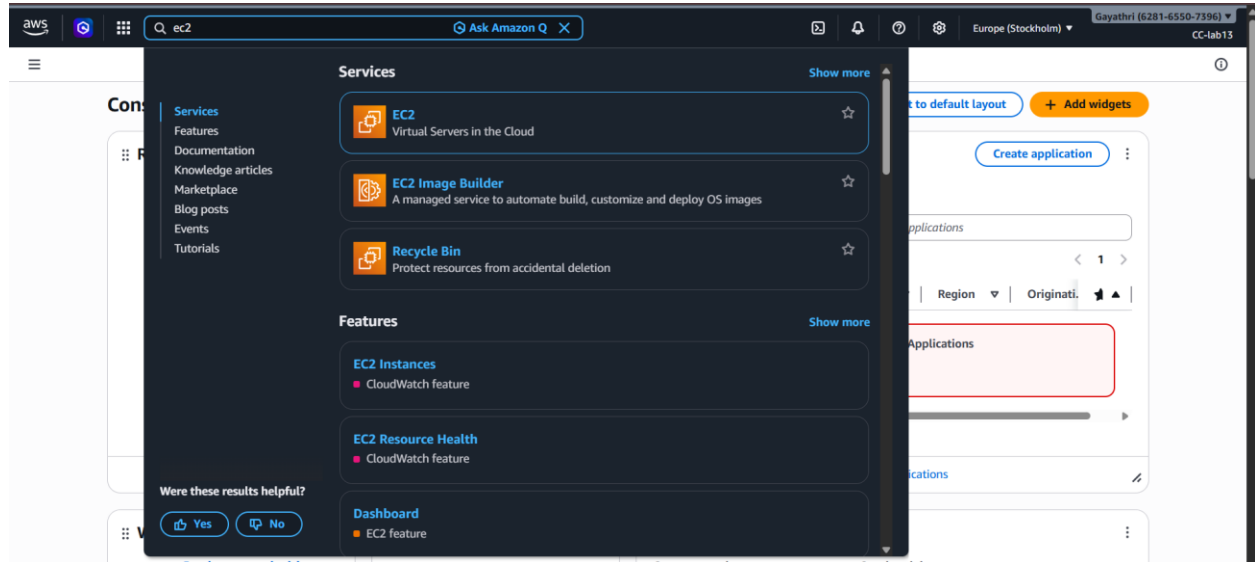
Turn on multi-session support

Sign out

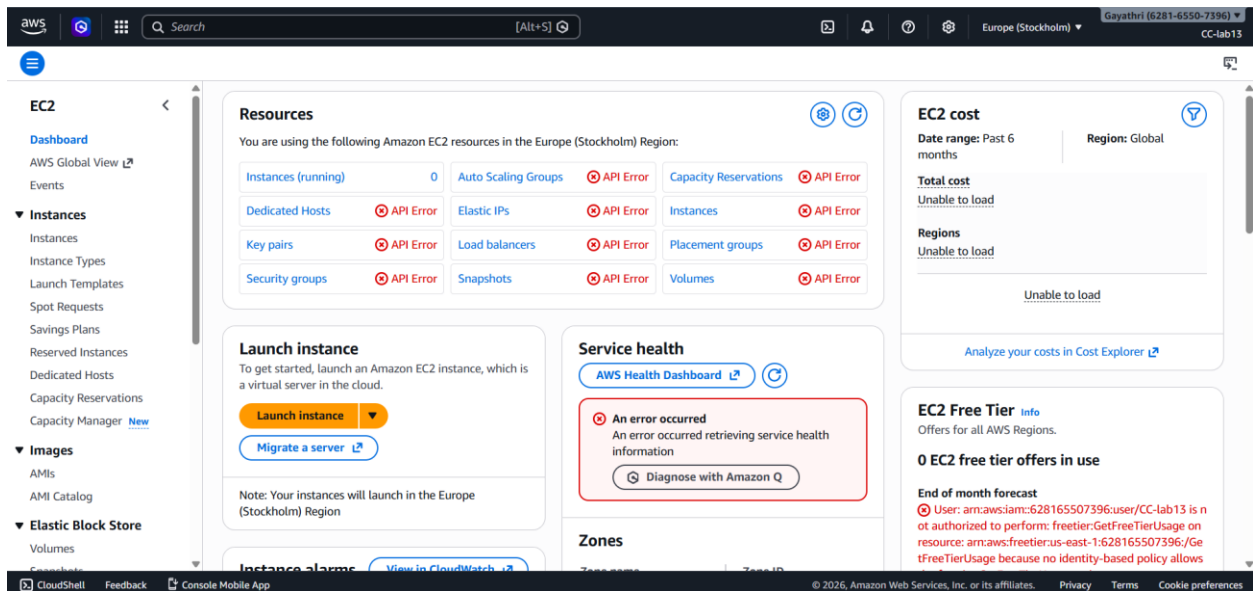
- Use the Sign-in URL from the .csv (or go to the AWS sign-in page and select **IAM User**).
- Enter the **Account ID**, then the **IAM Username** and **Password** from the .csv.



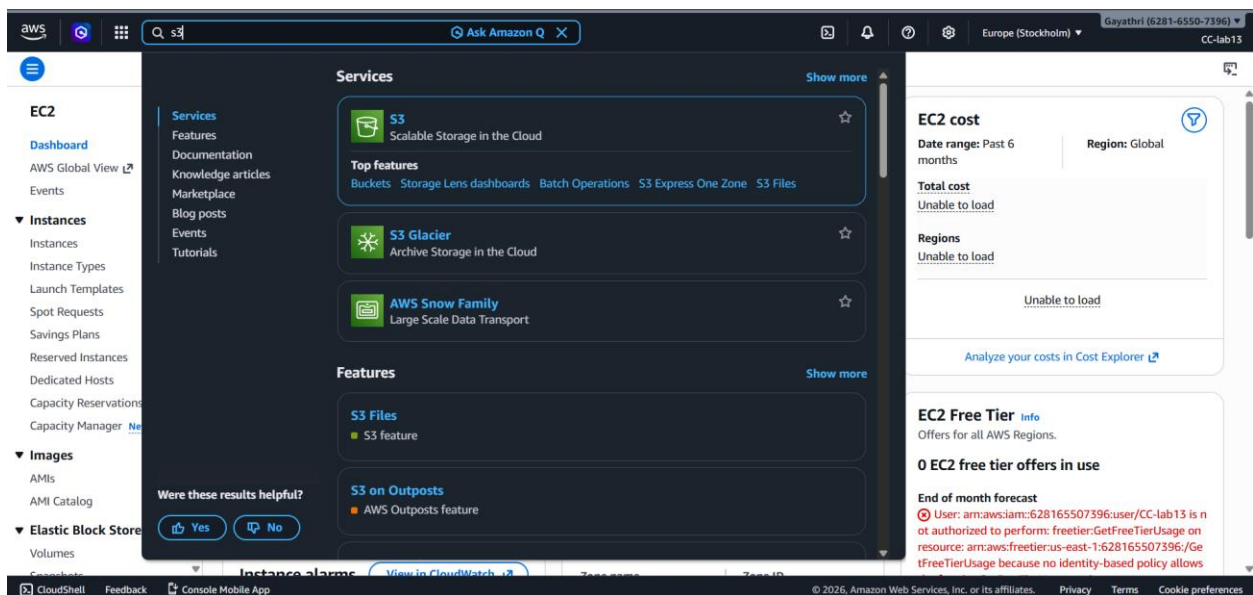
Test 1 (Failure): Navigate to EC2.



- Try to view instances. You will see "API Error" or "Access Denied." This proves the user is restricted



Test 2 (Success): Navigate to S3.



- Create a new bucket. It will work perfectly because of the attached policy.

aws    Search [Alt+S]     Europe (Stockholm) Gayathri (6281-6550-7396) CC-lab13

Amazon S3 > Buckets > Create bucket

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
Europe (Stockholm) eu-north-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [Info](#)

cclab-12

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

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aws    Search [Alt+S]     Europe (Stockholm) Gayathri (6281-6550-7396) CC-lab13

Amazon S3 > Buckets > Create bucket

Object Ownership [Info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership
Bucket owner enforced

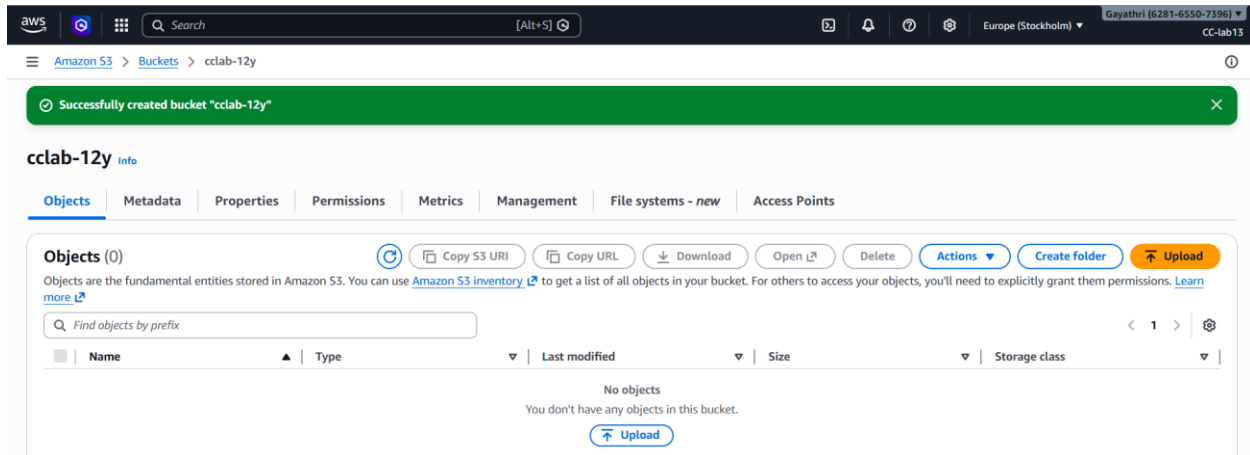
Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

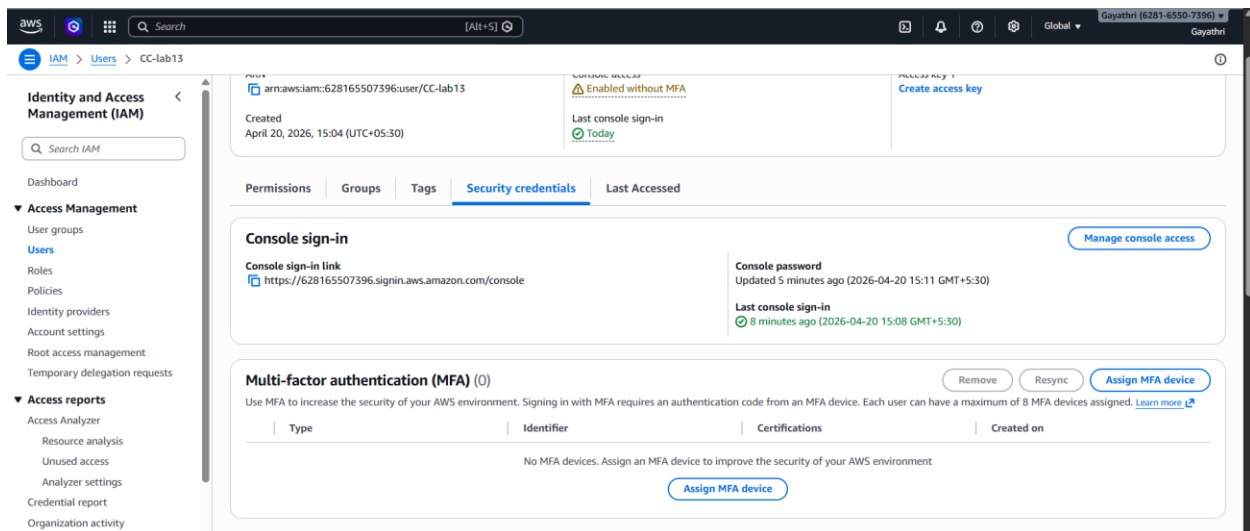
☒ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

- ☒ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
- ☒ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
- ☒ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
- ☒ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects.

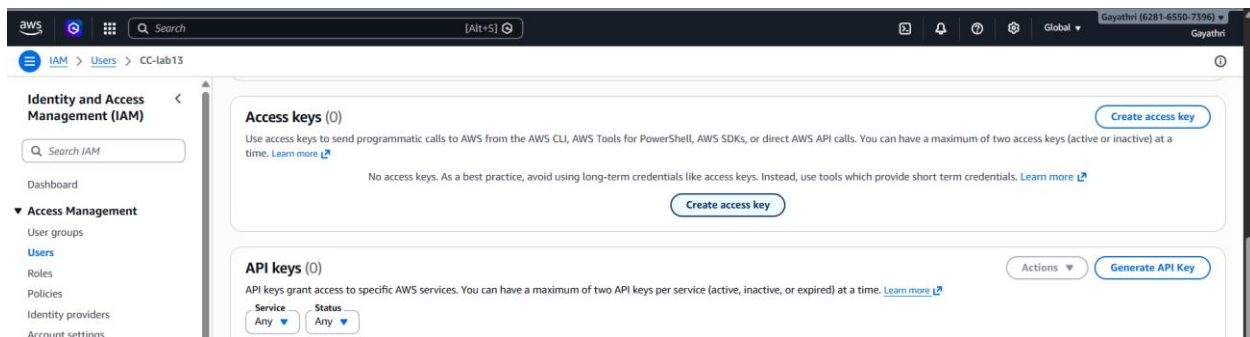
CloudShell Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences



Login to the root user -> click on users -> click on security credentials



Navigate to Access keys -> click on create access key



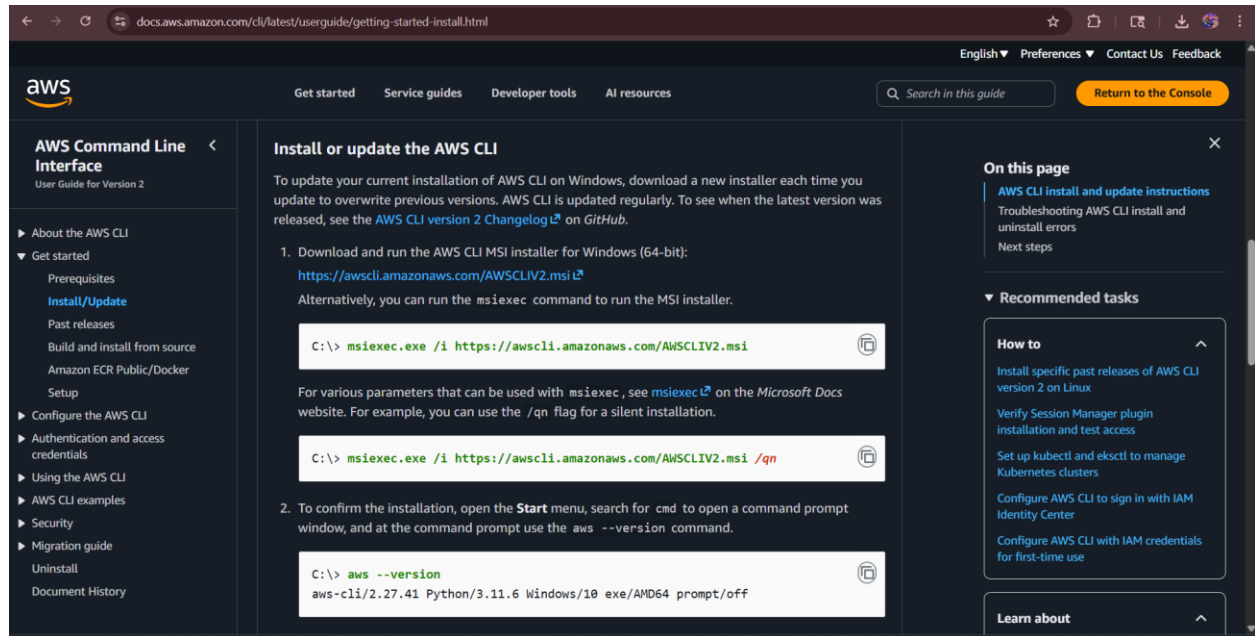
Select command line interface -> Acknowledge the warning and click **Next**.

The screenshot shows the AWS IAM console 'Create access key' page. The breadcrumb trail is IAM > Users > CC-lab13 > Create access key. The left sidebar shows a progress indicator with three steps: 'Set description tag', 'Step 3', and 'Retrieve access keys'. The main content area is titled 'Use case' and contains several radio button options: 'Command Line Interface (CLI)' (selected), 'Local code', 'Application running on an AWS compute service', 'Third-party service', 'Application running outside AWS', and 'Other'. Below these is a yellow box titled 'Alternatives recommended' with two bullet points: 'Use AWS CLI V2 and the `aws` `login` command to use your existing console credentials in the CLI. [Learn more](#)' and 'Use AWS CloudShell, a browser-based CLI, to run commands. [Learn more](#)'. At the bottom, a 'Confirmation' section has a checked checkbox with the text 'I understand the above recommendation and want to proceed to create an access key.'

Download the .csv file. > **Critical:** This contains the **Access Key ID** and **Secret Access Key**.

The screenshot shows the AWS IAM console 'Retrieve access keys' page. A green banner at the top states: 'This is the only time that the secret access key can be viewed or downloaded. You cannot recover it later. However, you can create a new access key any time.' The left sidebar shows a progress indicator with three steps: 'Access key best practices & alternatives', 'Step 2 - optional: Set description tag', and 'Step 3: Retrieve access keys' (selected). The main content area is titled 'Retrieve access keys' and contains an 'Access key' section with the text 'If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.' Below this, the 'Access key' is displayed as AKIAZEQMU7FCHCZOKIQ8 and the 'Secret access key' is masked with asterisks and a 'Show' button. An 'Access key best practices' section lists four bullet points: 'Never store your access key in plain text, in a code repository, or in code.', 'Disable or delete access key when no longer needed.', 'Enable least-privilege permissions.', and 'Rotate access keys regularly.' At the bottom right, there are two buttons: 'Download .csv file' and 'Done'.

Install the **AWS CLI Tool** from google on your computer.



Open your Command Prompt (CMD) or Powershell.

Type `aws configure` and press Enter.

Provide the following from your .csv:

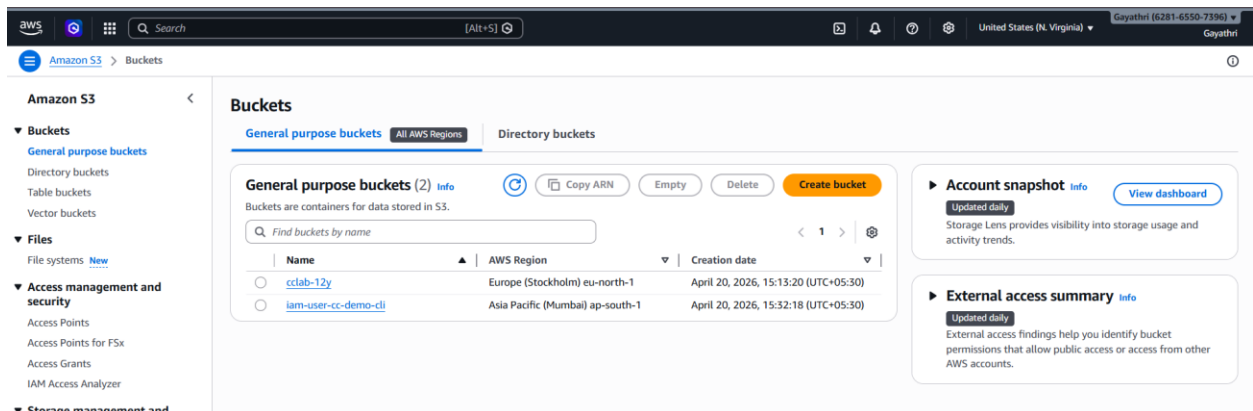
- **Access Key ID**
- **Secret Access Key**
- **Default region: (e.g., ap-south-1)**
- **Default output: json**

```
PS C:\Users\Admin> aws configure
AWS Access Key ID [*****KIQB]: AKIAZEQMU7FCHCZOKIQB
AWS Secret Access Key [*****woEL]: 9+4/dgjL5nezgLf4irx8VLgcJmFbKcFIszIOwoEL
Default region name [None]: ap-south-1
Default output format [None]: json
PS C:\Users\Admin> S|
```

- Run `aws s3 ls`. This should list your buckets.
- Run `aws s3 mb s3://your-unique-bucket-name`. (Success)

```
PS C:\Users\Admin> aws s3 ls
2026-04-20 15:13:21 cclab-12y
PS C:\Users\Admin> aws s3 mb s3://iam-user-cc-demo-cli
make_bucket: iam-user-cc-demo-cli
PS C:\Users\Admin> |
```

Login to the root user the bucket will be created .



- Run `aws iam list-users`.
 - **Result:** You will get an **(AccessDenied)** error.

This confirms that the CLI is respecting the same IAM policy as the GUI.

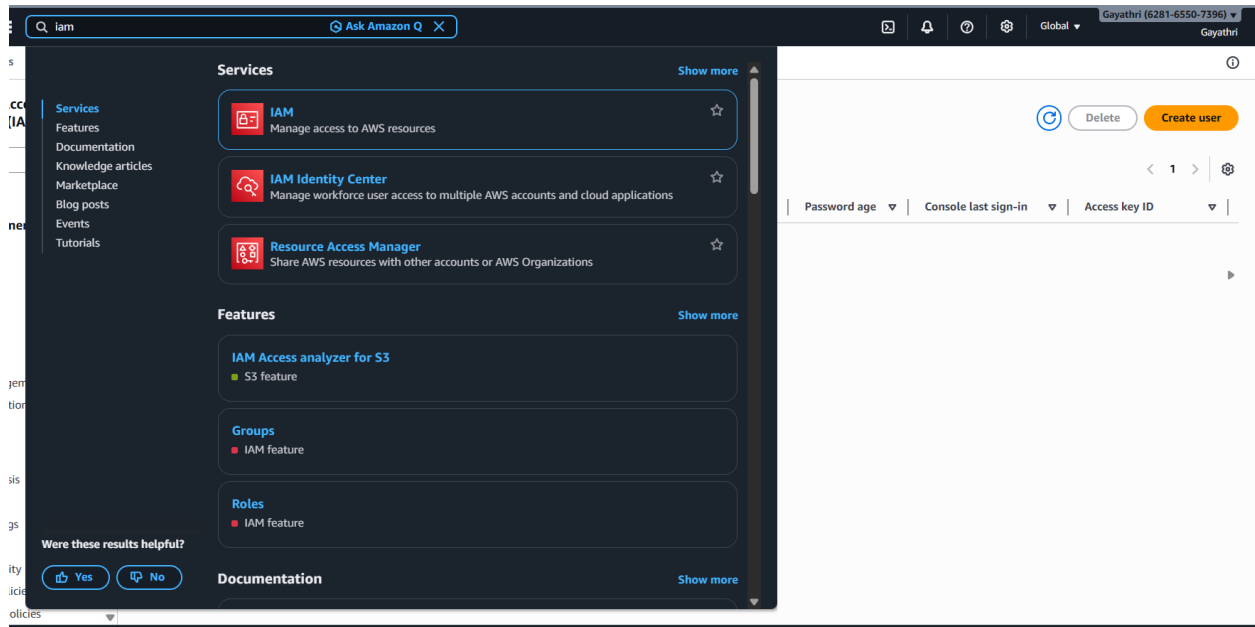

```
PS C:\Users\Admin> aws iam list-users
```

```
aws: [ERROR]: An error occurred (AccessDenied) when calling the ListUsers operation: User: arn:aws:iam::628165507396:user/CC-lab13 is not authorized to perform: iam:ListUsers on resource: arn:aws:iam::628165507396:user/ because no identity-based policy allows the iam:ListUsers action
```

```
PS C:\Users\Admin> |
```

WEEK-13 : Attach an IAM Role to an EC2 Instance (S3 access without keys)

Login in to the aws account -> search for iam service in console



Open IAM Console → Roles → Create role

Roles (5) Info

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Role name	Trusted entities	Last activity
AWSServiceRoleForLexV2Bots_APAYVB06ZKD	AWS Service: lexv2 (Service-Linked R	-
AWSServiceRoleForLexV2Bots_INXY110FDOO	AWS Service: lexv2 (Service-Linked R	-
AWSServiceRoleForResourceExplorer	AWS Service: resource-explorer-2 (S	11 minutes ago
AWSServiceRoleForSupport	AWS Service: support (Service-Link	-
AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service	-

Roles Anywhere Info

Authenticate your non AWS workloads and securely provide access to AWS services.

Access AWS from your non AWS workloads

Operate your non AWS workloads using the same authentication and authorization strategy that you use within AWS.

X.509 Standard

Use your own existing PKI infrastructure or use [AWS Certificate Manager Private Certificate Authority](#) to authenticate identities.

Temporary credentials

Use temporary credentials with ease and benefit from the enhanced security they provide.

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- Select AWS Service
- Use case: Choose EC2

Step 1: Select trusted entity

Step 2: Add permissions

Step 3: Name, review, and create

Select trusted entity Info

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

EC2

Filter service or use case

Commonly used services

- EC2
- Lambda

Click Next

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
EC2

Choose a use case for the specified service.

Use case

- ☒ **EC2**
Allows EC2 instances to call AWS services on your behalf.
- ☐ **EC2 Role for AWS Systems Manager**
Allows EC2 instances to call AWS services like CloudWatch and Systems Manager on your behalf.
- ☐ **EC2 Spot Fleet Role**
Allows EC2 Spot Fleet to request and terminate Spot Instances on your behalf.
- ☐ **EC2 - Spot Fleet Auto Scaling**
Allows Auto Scaling to access and update EC2 spot fleets on your behalf.
- ☐ **EC2 - Spot Fleet Tagging**
Allows EC2 to launch spot instances and attach tags to the launched instances on your behalf.
- ☐ **EC2 - Spot Instances**
Allows EC2 Spot Instances to launch and manage spot instances on your behalf.
- ☐ **EC2 - Spot Fleet**
Allows EC2 Spot Fleet to launch and manage spot fleet instances on your behalf.
- ☐ **EC2 - Scheduled Instances**
Allows EC2 Scheduled Instances to manage instances on your behalf.

Cancel Next

- Search policy: AmazonS3FullAccess
- Select it → Next

Add permissions [Info](#)

Permissions policies (1/1146) [Info](#)

Choose one or more policies to attach to your new role.

Filter by Type: All types 1 match

☒	Policy name 🔗	Type	Description
☒	AmazonS3FullAccess	AWS managed	Provides full access to all buckets via the ...

► Set permissions boundary - optional

Cancel Previous Next

- Name: EC2-S3-Role

- Click Create role

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.
ec2-s3-role
Maximum 64 characters. Use alphanumeric and '+', '@', '_' characters.

Description
Add a short explanation for this role.
Allows EC2 instances to call AWS services on your behalf.
Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=, @-/\[\]#\$%^&*~`''

Step 1: Select trusted entities [Edit](#)

Trust policy

```

1 - {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
8       ],
9       "Principal": {
10        "Service": [
11          "ec2.amazonaws.com"

```

Go to EC2 Dashboard

Services [Show more](#)

- EC2**
Virtual Servers in the Cloud
- EC2 Image Builder**
A managed service to automate build, customize and deploy OS images
- Recycle Bin**
Protect resources from accidental deletion

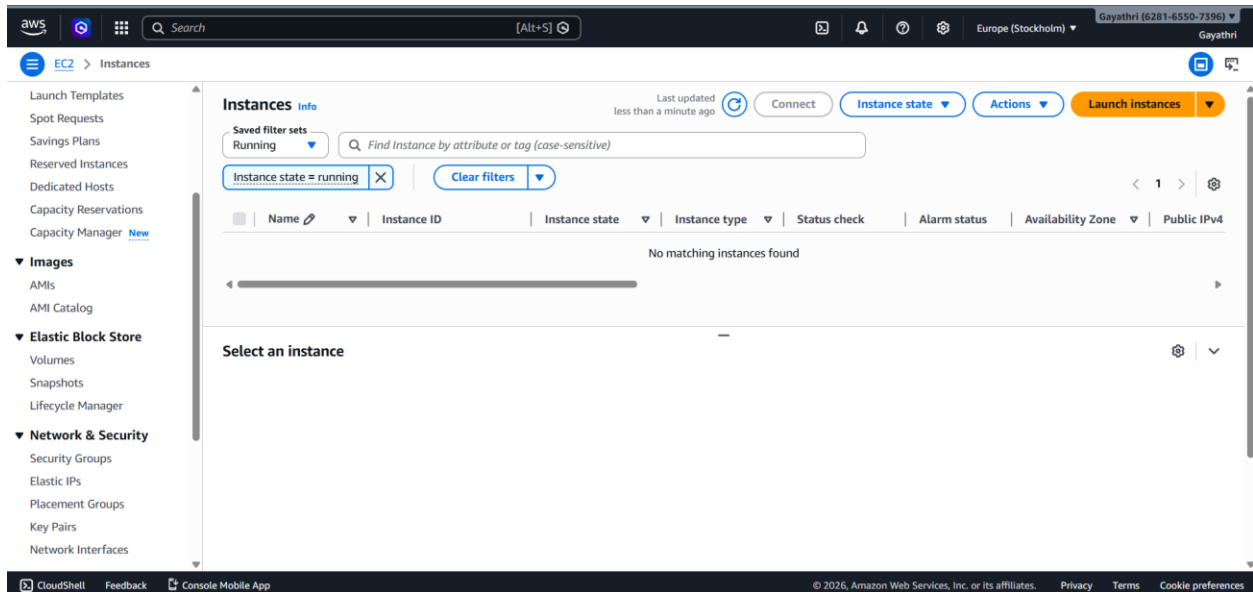
Features [Show more](#)

- EC2 Instances**
CloudWatch feature
- EC2 Resource Health**
CloudWatch feature
- Dashboard**
EC2 feature

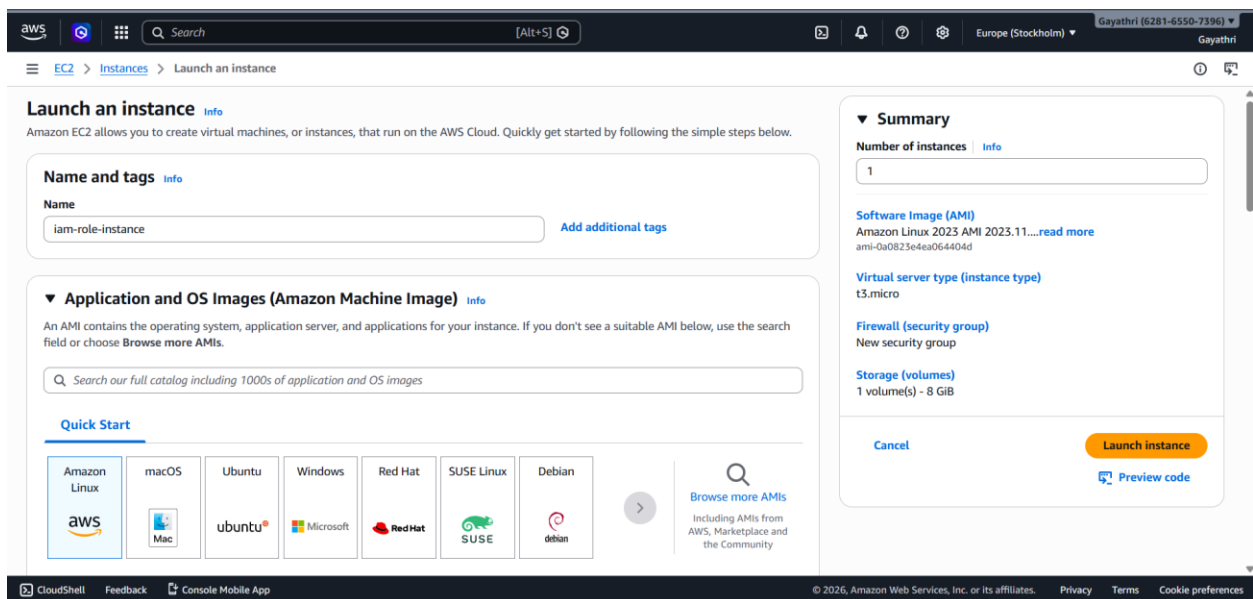
Were these results helpful?
[Yes](#) [No](#)

Resources in eu-north-1 / for a focused search [Show more in Resource Explorer](#)

Click on launch instance



Name the instance and click on launch instance



Select the running instance

Click on Actions → Security → Modify IAM role

The screenshot shows the AWS Management Console 'Instances' page. The left sidebar contains navigation links for Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager, Images, Elastic Block Store, and Network & Security. The main content area shows a table of instances with columns for Name, Instance ID, Instance state, Instance type, Status check, and Alarm state. The instance 'iam-role-insta...' with ID 'i-0ab804983a58c8770' is selected. The 'Actions' menu is open, showing options like Instance diagnostics, Instance settings, Networking, Security, Image and templates, Storage, and Monitor and troubleshoot. The 'Security' option is highlighted, and the 'Modify IAM role' option is selected. Below the table, the details for instance 'i-0ab804983a58c8770 (iam-role-instance)' are shown, including its Instance ID, Public IPv4 address, Private IPv4 addresses, Instance state (Running), and Public DNS.

- Select EC2-S3-Role → Update

The screenshot shows the 'Modify IAM role' page in the AWS Management Console. The breadcrumb trail indicates the path: EC2 > Instances > i-0ab804983a58c8770 > Modify IAM role. The page title is 'Modify IAM role' with an 'Info' link. Below the title, it says 'Attach an IAM role to your instance.' The 'Instance ID' is 'i-0ab804983a58c8770 (iam-role-instance)'. The 'IAM role' dropdown is set to 'ec2-s3-role'. There is a 'Create new IAM role' link next to the dropdown. At the bottom right, there are 'Cancel' and 'Update IAM role' buttons.

Run the instance using ssh key

```
PS C:\Users\Admin\Downloads> ssh -i "demo.pem" ec2-user@ec2-51-20-54-249.eu-north-1.compute.amazonaws.com
The authenticity of host 'ec2-51-20-54-249.eu-north-1.compute.amazonaws.com (51.20.54.249)' can't be established.
ED25519 key fingerprint is SHA256:dkAleIPxh45sQ0Z2xRydCld3vzl4V5y3AlhA2dDZdA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-51-20-54-249.eu-north-1.compute.amazonaws.com' (ED25519) to the list of known hosts
```



```
#
~\#####      Amazon Linux 2023
nn\#####\
nn\###|
nn\#/\_--
nnV~!'->
nn
nn _.-_-/_
nn _/m/'
```

```
[ec2-user@ip-172-31-42-172 ~]$ |
```

```
aws s3 ls
```

aws s3 ls Works

```
[ec2-user@ip-172-31-42-172 ~]$ aws s3 ls
2026-04-21 09:42:23 week-13-demo
[ec2-user@ip-172-31-42-172 ~]$
```